Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs are undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

Research Questions

1. Which factors in applications affect admission and success in graduate school?
    - Literature measures success with program completion and job placement ranking
2. Are admissions committees optimizing, or are they making mistakes?
    - Goal is to accept applicants with highest success probabilities who will matriculate
    - Predictors of admission only (success only) may be overweighted (underweighted)
3. If they are making mistakes, how can they make better admissions decisions?
    - In turn, by changing decision rules, how much better of a cohort can they admit?

★ This paper's methods generalize to admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - For some applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights several applicant characteristics
  - Counterfactual deviation gains of 6.5–7% points statistically significant at 5% level

Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant charateristics
  - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
  - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
  - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)

★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

1. **Administrative office receives data from university departments**
   - Algorithmically extracts variables from textual files, such as recommendation letters
2. **Synthetic data (Patki et al. 2016) used for model development**
   - Estimate joint distribution of variables, and simulate synthetic dataset from it
   - Rows do not correspond to real subjects, preventing identity disclosure
   - Cells with < 5 comparable observations binned to prevent attribute disclosure
3. **Administrative office executes final estimation code on actual data**
   - Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

▸▸ Synthetic Data Example

Outline

Data Description

- **Sample:** Applicants to R1 university's economics PhD program from 2015–2019
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Indicators for all stages of admissions process, including first-round filter
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables

Variables | Inference vs. Prediction | Sample Selection

Reduced-Form Summary

- **Method:** Logistic regression AMEs for acceptance, matriculation, and success
- **Results:** Different independent variables best predict each dependent variable
  - **Accept:** *Committee Score* is best predictor. Conditional on it, other effects (e.g. demographics, *GRE Quantitative Percentile*, *Work Experience*) are relatively smaller
  - **Matriculate:** Predictors of acceptance often negatively predict matriculation. However, best predictor of matriculation is *Open House* attendance
  - **Success:** *PhD Program Rank* is best predictor. Stronger programs place better, and conditional on that, other effects (e.g. *Work Experience*) are relatively smaller

Summary Statistics

Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	Top 100
Winsorized Age	-0.0001	-0.0223**	0.0053*
Female	0.0163**	-0.0490	-0.0032
U.S.	-0.0013	0.1472*	0.0192
International Asian	-0.0214***	0.1644**	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0409*
Work Experience	0.0140**	0.1591**	0.0358**
Committee Score	0.0788***	-0.0371	0.0007
Advanced Math	0.0246	-0.1950**	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0159
CV Coding	-0.0044*	-0.0414**	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0123*
Open House		0.2615***	
Program Rank			-0.0011***
Missing Program Rank			-0.3965***

Full Logistic Regression AMEs

Acceptance vs. Success AMEs

Dependent Variable	Accept	Top 100
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Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Elite U.S. University	-0.0034	-0.0192
Work Experience	0.0140**	0.0358**
#Professor Recommenders	-0.0033	-0.0069
Missing TOEFL/IELTS	0.0070	-0.0008
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
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- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

Insufficiency of Comparing AMEs

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Insufficiency of Comparing AMEs

Outline

Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
    - Per literature, committee should accept applicants with highest success probabilities
    - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
    - Quantify gains from deviating from actual to counterfactual decision rule

Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$, N = number of applicants, and N_A = maximum number of acceptances

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- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

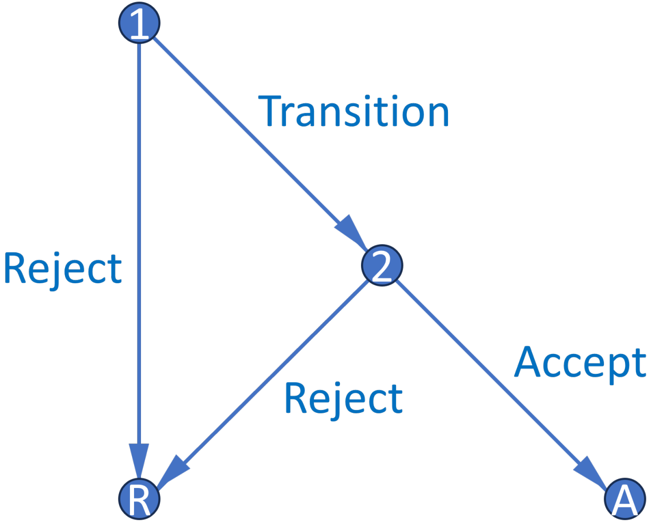
Portfolio Choice to Cutoff Rule

Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
  - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
  - **Round 2:** If chose to transition application file in Round 1, can accept or reject
  - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are additional models
  - Static model better handles high-dimensional subjective variables with missing values
  - Matriculation models account for both success and matriculation probabilities

Models of Admissions

Decision Graph



Dynamic Model

- **States:** r = round, x = objective variables, z = subjective variables, t = year, $\overline{k}$ = fixed program rank, ϵ_r = T1EV preference shock with scale parameter σ
- **Controls:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, y_r^* = marginal outcome

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Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S | x, z, t, \overline{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 2 Portfolio Choice

Dynamic Model

Round 1:

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{such that:}$$

$$\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] = \int_{\widetilde{z}} \sigma \log \left(e^{P(y=S|x,\widetilde{z},t,\overline{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\widetilde{z}|x, t)$$

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Conditional Choice Probabilities:

$$P(d_r|x, (z), t, \overline{k}) = \frac{\exp(u_r(x, (z), t, \overline{k}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x, (z), t, \overline{k}, \widetilde{d}_r)/\sigma)}$$

Round 1 Portfolio Choice
Dynamic Parameterization

Static Model

- **Motivation:** Subjective variables are high-dimensional and have missing values

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Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S | x, z, t, \overline{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

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- $P(y = S | x, t, \overline{k})$ estimates inconsistent, but fitted values account for z's effect on y

Round 1 Portfolio Choice

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where $(x, z) =$ variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, $N =$ number of applicants, and $N_M =$ maximum number of matriculants

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Theorem 1 (Proof)

For large N, this problem has a cutoff rule solution on **success** *probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- However, matriculation probabilities matter in Round 1 of a two-round model

Cutoff Rule with Matriculation

Matriculation-Dynamic Model

Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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Dynamic Portfolio Choice with Matriculation

Outline

Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{\left[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)} \left[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

Dynamic Likelihood | Matriculation Likelihood | Empirical Considerations

Cutoff Probability Identification

Theorem 2 (Proof)

The maximum likelihood cutoff probability estimate is such that the number of acceptances N_A equals the expected N_A. Also, for any cutoff probability and N_A, the likelihood is highest when the committee accepts the best N_A applicants.

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
    - If committee engages in sorting, Akaike Information Criterion more favorable
    - $\text{AIC} = 2\left[K - \log(\hat{\mathcal{L}}^R)\right]$, such that $K = \dim(\theta^R)$

Restricted Dynamic Estimates

	$L(\text{Cutoff})_1$	$P(\text{Cutoff})_2$	Sigma
2015	0.3327	32.63%	
	(0.2495, 0.4437)	(24.06%, 42.54%)	
2016	0.3902	38.35%	
	(0.2622, 0.5807)	(24.57%, 54.29%)	
2017	0.4615	45.25%	
	(0.3928, 0.5423)	(38.04%, 52.68%)	
2018	0.3473	34.14%	
	(0.2828, 0.4263)	(27.30%, 41.70%)	
2019	0.4173	41.06%	
	(0.3429, 0.5078)	(33.39%, 49.19%)	
Sigma			0.0018
			(0.0014)
Observations = 2329	Parameters = 528	LL = -13667.0	AIC = 28390.0

Full Dynamic Estimates
Full Myopic Estimates

Restricted Static Estimates

	$P(\text{Cutoff})_1$	$P(\text{Cutoff})_2$	Sigma
2015	29.33%	36.37%	
	(23.18%, 36.34%)	(28.46%, 45.10%)	
2016	29.17%	43.56%	
	(22.20%, 37.28%)	(34.01%, 53.62%)	
2017	37.04%	46.49%	
	(32.09%, 42.28%)	(37.65%, 55.56%)	
2018	38.19%	34.19%	
	(34.05%, 42.50%)	(26.12%, 43.29%)	
2019	34.04%	41.70%	
	(28.67%, 39.85%)	(33.20%, 50.72%)	
Sigma			0.0056
			(0.0035)
Observations = 2329	Parameters = 79	LL = -2528.4	AIC = 5214.8

Full Static Estimates

Restricted Matriculation-Dynamic Estimates

	$L(\text{Cutoff})_1$	$P(\text{Cutoff})_2$	Sigma
2015	0.0066	31.33%	
	(0.0023, 0.0190)	(21.85%, 42.69%)	
2016	0.0065	41.42%	
	(0.0024, 0.0180)	(30.36%, 53.41%)	
2017	0.0098	45.65%	
	(0.0036, 0.0267)	(39.08%, 52.37%)	
2018	0.0086	31.99%	
	(0.0032, 0.0231)	(24.21%, 40.91%)	
2019	0.0089	39.77%	
	(0.0031, 0.0253)	(31.52%, 48.65%)	
Sigma			0.0019**
			(0.0009)
Observations = 2329	Parameters = 567	LL = -13796.0	AIC = 28726.0

Full Matriculation-Dynamic Estimates

Outline

Deviation Gains Test

- Suppose committee adopts model's decision rule. Will average success probability of acceptance/matriculation set increase, and if so, by how much?
  - Solve model with $\sigma = 0$ in CCPs, compute average yearly $P(\text{Success})$ of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - $\hat{\theta}_S^R$ may be biased to rationalize suboptimal behavior, so plug in $\hat{\theta}_S^U$
- Committee may use success-predicting variable that I don't observe
  - Because the success probability measure used to evaluate gains cannot detect that variable, any edge the committee's private information provides goes unmeasured
  - Comprehensive data, especially committee score in z, protects against this limitation

Deviation Gains Interpretation

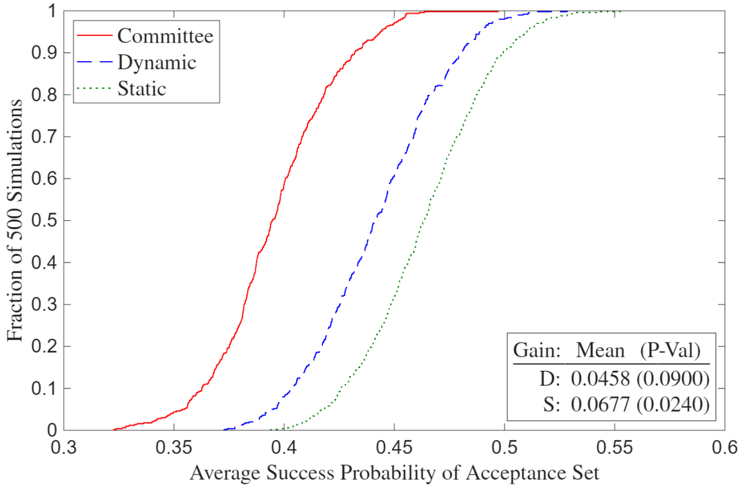
- Suppose there are meaningfully large deviation gains. Then committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Matriculation rate of model's acceptance set may be less than that of actual decision rule's acceptance set, making estimated gains an upper bound
  - In matriculation counterfactuals, compare success probabilities of matriculation sets by using matriculation probabilities to simulate accepted applicants' decisions

★ **Robustness:** Given too few observations for holdout set and infeasibility of large k-fold cross-validation due to runtime, implement randomization test

- Compute average success probabilities with perturbations of $\hat{\theta}_S^U$
- P-value is fraction of perturbations with committee's $\overline{P(\text{Success})} >$ model's

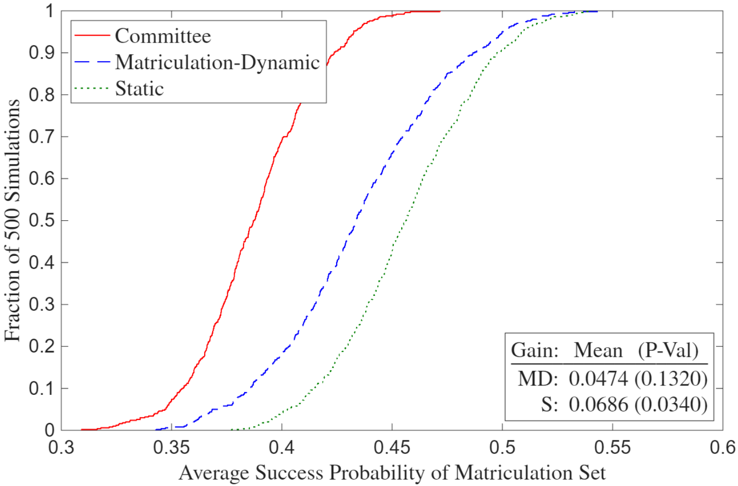
Deviation Gains Test

Deviation Gains



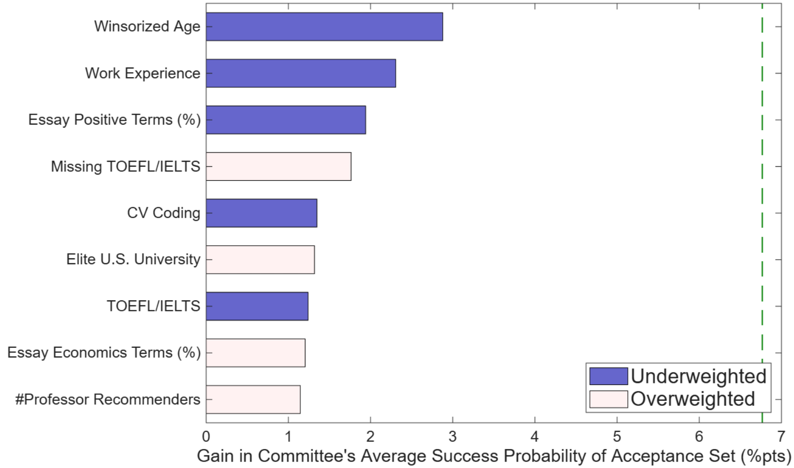
Full Deviation Gains

Matriculation Deviation Gains



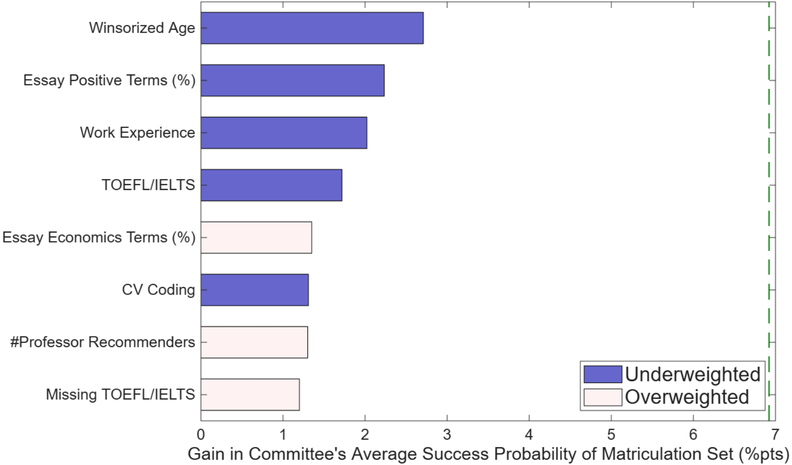
Full Matriculation Deviation Gains

Deviation Gains Attribution



▸▸ Full Deviation Gains Attribution ▸▸ Deviation Gains Attribution Table

Matriculation Deviation Gains Attribution



Full Matriculation Deviation Gains Attribution
Matriculation Deviation Gains Attribution Table

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
  1. For some applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights several applicant characteristics
  2. Counterfactual deviation gains of 6.5–7% points statistically significant at 5% level
- **Future Research:** Two-sided matching model across multiple universities

Simulated Results | Waitlist Models | Multiyear Models

Thank You!

A1: Synthetic Rows ≠ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	.	150
0	0	0	0	0	80	1	1	0	1	.	150
0	0	0	0	1	99	0	1	0	0	.	29
0	0	0	0	0	81	0	0	1	1	.	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	.	48
1	1	0	0	0	87	0	1	0	1	5	34

$\Downarrow$

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	.	30
0	0	0	0	1	89	0	0	0	0	.	25
0	0	0	0	0	85	1	0	0	0	.	150
0	0	0	0	0	76	0	1	0	1	.	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	.	35
0	0	0	0	0	70	1	0	0	0	.	150
1	1	0	0	1	93	0	1	0	0	5	57

Synthetic Data Protocol

A1: Dependent Variables

Table 1a: Dependent Variables

Admission	Success
$\mathbb{1}$**(Transition)**	$\mathbb{1}$**(Complete Program)**
$\mathbb{1}$(Accept)	$\mathbb{1}$**(Academic Placement)**
$\mathbb{1}$**(Waitlist)**	$\mathbb{1}$**(Government Placement)**
$\mathbb{1}$**(Open House)**	$\mathbb{1}$**(Consulting Placement)**
$\mathbb{1}$(Matriculate)	$\mathbb{1}$**(Tech Placement)**
$\mathbb{1}$**(Attend Program)**	$\mathbb{1}$**(Other Placement)**
Program Rank	$\mathbb{1}$**(No Placement)**
	Placement Rank
	$\mathbb{1}$**(Top 100 Placement)**

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

Data Description

A1: Objective Variables

Table 1b: Objective Independent Variables

Demographic	Performance
Year	GRE Verbal Percentile
$\mathbb{1}$*(Covid Year)*	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
$\mathbb{1}$(Female)	Undergraduate GPA
$\mathbb{1}$**(U.S.)**	TOEFL/IELTS
$\mathbb{1}$*(U.S. African/Hispanic/Native)*	$\mathbb{1}$(Elite U.S. University)
$\mathbb{1}$*(U.S. Asian)*	$\mathbb{1}$(Elite U.S. Liberal Arts College)
$\mathbb{1}$*(U.S. Other)*	$\mathbb{1}$(Other U.S. University/College)
$\mathbb{1}$(International Asian)	$\mathbb{1}$(Elite Foreign University)
$\mathbb{1}$(International Other)	$\mathbb{1}$(Other Foreign University)
	$\mathbb{1}$(Graduate Degree)
	$\mathbb{1}$(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	Concentration Dummy Variables

Data Description

A1: Subjective Variables

Table 1c: Subjective Independent Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	**#Math Courses**
Letters Length	**$\mathbb{1}$(Advanced Math)**
Positive Terms (%) (YS 2012)	**CV Length**
Negative Terms (%) (YS 2012)	**CV Coding**
Standout Terms (%) (BEMS 2022)	**CV $\mathbb{1}$(Publication/Working Paper)**
Ability Terms (%) (BEMS 2022)	**Essay Length**
Research Terms (%) (BEMS 2022)	**Essay $\mathbb{1}$(Specific Professor)**
Grindstone Terms (%) (BEMS 2022)	**Essay Economics Terms (%)**
Teaching Terms (%) (BEMS 2022)	**Essay Research Terms (%)**
Communal Terms (%) (BEMS 2022)	**Essay Positive Terms (%)**
Agentic Terms (%) (MHM 2009)	**Essay Negative Terms (%)**
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

Data Description

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Data Description

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

▸▸ Data Description

A1: Dictionaries

- **Recommendation Letters:**
  - **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
  - **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
  - **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
  - **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop
- **CVs and Essays:**
  - **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
  - **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
  - **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
  - **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

Data Description

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for "grindstone terms" (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
  - If GRE predicts success, committee should use it regardless of why

Data Description

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
  - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
  - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
  - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
  - **Instruments:** Variables that predict matriculation but not the outcome

Data Description

A1: Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Work experience has large effect on success, graduate degree smaller effect
  - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
  - **Acceptance Regression:** $\text{AME}_{\text{Experience}} = 1$, $\text{AME}_{\text{Degree}} = 0$
  - **Success Regression:** $1 > \text{AME}_{\text{Experience}} > \text{AME}_{\text{Degree}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

Acceptance vs. Success AMEs

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
  - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
  - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a "success" model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality

Acceptance vs. Success AMEs

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1+\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$, and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1+\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x;\theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} \left[1 - P(Y_n^E = 1|x_n^E; \theta^E)\right]^{1-Y_n^E}$$

$$\prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} \left[1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})\right]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E} \hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}} \hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. Then: $W = g(\hat{\theta})'[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

Acceptance vs. Success AMEs

A1: Linear Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\middle|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics)$\middle|$Predictors.

Let $f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$, and

$f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_S^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y\middle|x;\beta,\sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) \prod_{n=1}^{N_{NE}} f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
  Then $W = g(\hat{\beta}_1)'[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)\hat{V}(\hat{\beta},\hat{\sigma}^2)\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n,\ \sigma_n^2 = \sigma^2$)

▸▸ Acceptance vs. Success AMEs

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAships

Acceptance vs. Success AMEs

A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \ldots, d_N; \lambda) = \sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\},\ d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Model Framework

A2.1: Portfolio Choice to Cutoff Rule with Shocks

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left\{ \left[P(y_n = S|x_n, z_n) + \epsilon_n(A) \right] \mathbb{1}(d_n = A) + \epsilon_n(R)\mathbb{1}(d_n = R) \right\},$$

$$\text{such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \ldots, d_N; \lambda) =$$

$$\sum_{n=1}^{N} \left\{ \left[P(y_n = S|x_n, z_n) + \epsilon_n(A) \right] \mathbb{1}(d_n = A) + \left[\lambda + \epsilon_n(R) \right] \mathbb{1}(d_n = R) \right\} + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\}, d_n(x_n, z_n, \epsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 3 (Proof)

For large N, there is a cutoff rule solution with logistic CCPs. Model Framework

A2: Models of Admissions

1. **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
    1. **Static Model:** In Round 1, rank applicants based only on objective variables
    2. **Myopic Model:** Like dynamic, but distribution of z not conditional on x
2. **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)
3. **Waitlist-Dynamic/Hybrid Models:** In Round 1, rank applicants based on success probabilities. After Round 1, accept or waitlist applicants in Round 2. After acceptances matriculate or decline, accept or reject waitlisted applicants in Round 3. Favor likely matriculants in Round 2 to increase selectivity in Round 3

Cutoff Rule Problem

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \overline{k})$, where Success $= \mathbb{1}$(Complete Program) or $\mathbb{1}$(Good Placement). If reject applicant, get $P(y_2^* = S|t) = P$(Success) of year t's marginal accepted applicant

- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P$(Success) of year t's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ = expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
  - **Static:** If transition applicant, get $P(y = S|x, t, \overline{k})$, or P(Success) unconditional on z. It may differ from $P(y = S|x, z, t, \overline{k})$, though it still accounts for z's effect on y
  - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$, replace $F(z|x, t)$ with $F(z|t)$

Cutoff Rule Problem

A2: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \ldots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N_T\},\ d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Dynamic Model

A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n=S|x_n,\widetilde{z}_n),\overline{\lambda_2^*}\}dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n)>\lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^* \\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^*=S)\in\left[EV_{2,N_T+1},EV_{2,N_T}\right]$

Dynamic Model

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\big(d_1 = T \big| x, t; \theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1+\exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, t; \theta_A\big) = \frac{\exp(f_A(x,z,t)\theta_A)}{1+\exp(f_A(x,z,t)\theta_A)}$$

Success Probability:

$$P_y = P\big(y = S \big| x, z, t, k; \theta_S\big) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1+\exp(f_S(x,z,t,k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P\big(y_2^* = S \big| t; \theta_{S_2^*}\big) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1+\exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

Dynamic Model

A2: Dynamic Parameterization 2

$$\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\widetilde{d}_2)/\sigma}\right) dF(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z}_1}\cdots\int_{\widetilde{z}_J}\Big[\,\cdot\,\Big] dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z}_J,x,t)\cdots dF_J(\widetilde{z}_J|x,t)$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)},\ \ \ell = 1,\ldots,L_j-1$

Dynamic Model

A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$

Static Model

A2: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success often unobservable for applicants from fewer than six years ago

Maximum Likelihood Estimation

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice probabilities}}$$

$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^{N} \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{\neg c}|x,t}$$

$$\underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

Maximum Likelihood Estimation

A2: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in z as a proxy for this information

Maximum Likelihood Estimation

A2: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|w_n)\mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1,\dots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|w_n)\mathbb{1}(d_n = A)+\lambda\mathbb{1}(d_n = R)\right]P(m_n = M|w_n)+\lambda(N_M-\overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\dots,N\},\ d_n(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda^* \\ R & \text{if } P(y_n = S|w_n) < \lambda^* \end{cases}$$

Portfolio Choice with Matriculation

A2: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} \left[P(y = S|w) \delta(w) + \lambda(1 - \delta(w)) \right] P(m = M|w) f(w) dw + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

Portfolio Choice with Matriculation

A2: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2)=\sum_{n=1}^{N_T}\left[P(y_n=S|w_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]P(m_n=M|w_n)+\lambda_2(N_M-\overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^* \end{cases}$$

Matriculation-Dynamic Model

A2: Cutoff Rule with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n)=\int_{\widetilde{z}_n}\max\{P(y_n=S|x_n,\widetilde{z}_n)-\overline{\lambda_2^*},0\}P(m_n=M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1)=\sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases}T & \text{if } EV_{2,n}(x_n)>\lambda_1^*\\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^*\\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^*\end{cases}$$

Matriculation-Dynamic Model

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P\left(m = M \middle| x, z, t; \theta_M\right) = \frac{\exp(f_M(x,z,t)\theta_M)}{1 + \exp(f_M(x,z,t)\theta_M)}$$

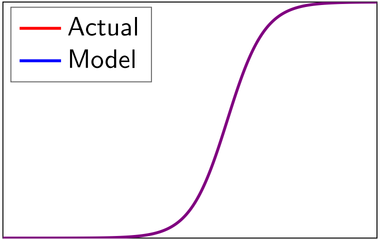
$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T)\left[P_{d_{2,n}}(\theta_A) P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}\right]^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice and matriculation probabilities}}$$

$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional Choice Probability}} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^{N} \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x,t} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional Choice Probability}}$$

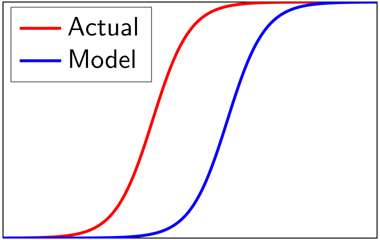
Maximum Likelihood Estimation

A2: How Deviation Gains Test Works

1. Write Bellman equations for committee's optimization problem
2. Estimate parameters. Rationality assumption distorts estimates
3. Re-estimate parameters without rationality, and use to solve model
4. Compare selections from model's decision rule to actual decision rule
5. To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

Deviation Gains Interpretation

A2: Deviation Gains Algorithm 1

Dynamic/Static/Myopic:

1. Rank each t's applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \overline{k})$ for static. Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$, and accept top N_A survivors. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

1. Rank each t's applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \overline{k})$ for static (best analogue). Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains Interpretation

A2: Deviation Gains Algorithm 2

Waitlist-Dynamic/Hybrid:

1. Rank each t's applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \ddot{m}_2, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \overline{k})$ for hybrid. Transition top N_T, and reject the rest
2. Rank surviving applicants, with $\sigma = 0$ and $\beta = 1$, by $P(y = S|x, z, t, \overline{k}) - \beta\mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \overline{k}, \ddot{m}_2]$. Accept top N_{2A}, each matriculating with probability $P(m_2 = M|x, z, t)$, and waitlist the rest
3. Rank waitlisted applicants by $P(y = S|x, z, t, \overline{k})$ and accept them in descending order, each matriculating with probability $P(m_3 = M|x, z, t)$, until N_M matriculate. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains Interpretation

A2: Randomization Test

Theorem 4 (Proof)

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$

$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- But for perturbation $\widetilde{\theta_{S,t}^U}$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$

$$< \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- For matriculation and waitlist models, compare matriculants, not acceptances

Deviation Gains Interpretation

A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
  - **Dynamic/Static:** Deviation gains significantly positive for simulated dataset where committee misweights applicant characteristics, but not for optimal simulated dataset
  - **Matriculation-Dynamic/Static:** Same as above
  - **Waitlist-Dynamic/Hybrid:** Same as above, and matriculation estimates as expected
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A
- By assuming h hours to read application file and w hourly wage, and calculating model's $\overline{P(\text{Success})}$ induced by all possible N_T, can put dollar value on success
  - Alternatively, what if committee uses AI to read every applicant's file?

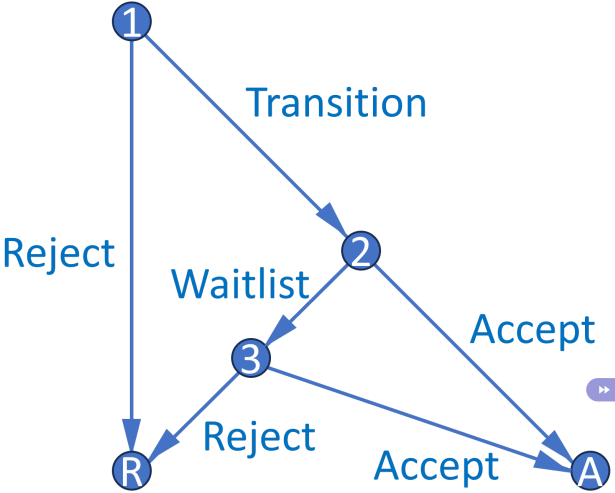
Conclusion

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
  - **Example:** Applicants with work experience more likely to succeed and matriculate
  - Can help committee find "people that we can afford" (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
  - Instead maximize combination of $P(\text{Success})$ and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields): $P(\text{Success})^{\alpha} \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see Appendix 3)
  - Three-round models, and two-round models that repeat over infinite horizon
  - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

Conclusion

A3.1: Waitlist Decision Graph



- Three-round model features waitlist and accounts for applicants' decisions
- Dynamic mechanism incentivizes giving early acceptances to likely matriculants
- In practice, need several years of matriculation data for identification

Conclusion

A3.1: Waitlist Model, Round 3

- Respecify Round 3 cutoff $P(y_3^* = S|t)$, where t = year, as $P(y_3^* = S|\overline{m_{2A}})$
  - $\overline{m_{2A}}$ = average early accepted applicant's matriculation probability
  - $P(y_3^* = S|\overline{m_{2A}})$ increases in $\overline{m_{2A}}$ because larger $\overline{m_{2A}}$ means fewer openings

---

$$v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x, z, t, \overline{k}, \overline{m_{2A}}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \overline{k}, \overline{m_{2A}}, d_3) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_3 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_3^* = S|\overline{m_{2A}})\mathbb{1}(d_3 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Conclusion

A3.1: Waitlist Model, Round 2

- What information available at start of Round 2 determines $\overline{m_{2A}}$?
  - $\ddot{m}_2$ is defined for all transitioned applicants as $\overline{m_2}$ with m_2 excluded, where $m_2 =$ applicant's own matriculation probability. Within a year, high m_2 implies low $\ddot{m}_2$
  - $\ddot{m}_2$ appears when applicant is waitlisted $\Rightarrow$ represents $\overline{m_2}$ of applicants committee *can* accept (vs. $\overline{m_{2A}}$, which represents $\overline{m_2}$ of applicants committee *does* accept)

$$v_2(x, z, t, \overline{k}, \ddot{m}_2, \epsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x, z, t, \overline{k}, \ddot{m}_2, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \overline{k}, \ddot{m}_2, d_2) = \underbrace{P(y = S | x, z, t, \overline{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{\beta \mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \overline{k}, \ddot{m}_2] \mathbb{1}(d_2 = W)}_{\text{If waitlist, get discounted } \mathbb{E}[\text{Social Surplus}]}$$

$$\mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \overline{k}, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left(\sum_{\widetilde{d}_3 \in \{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}} | \ddot{m}_2)$$

Conclusion

A3.1: Waitlist Dynamics

Theorem 5 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- Compared to low m_2 applicant, high m_2 applicant has low $\ddot{m}_2$, and therefore low $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, $\mathbb{E}[P(\text{Cutoff})_3]$, and $\mathbb{E}[v_3]$. This increases relative payoff of accepting them
- Why not accept low m_2 applicant and fill spot with waitlisted high m_2 applicant?
  1. m_2 declines between Rounds 2 and 3 and has farther to fall for high m_2 applicant
  2. In general, matriculation probabilities can change unpredictably between rounds

★ **Upshot:** Accepting applicants who want to matriculate benefits the program

Conclusion

A3.1: Waitlist Model, Round 1

Round 1:

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \overline{k}, \ddot{m}_2, \epsilon_2)|x, t, \overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{such that:}$$

$$\mathbb{E}[v_2(x, z, t, \overline{k}, \ddot{m}_2, \epsilon_2)|x, t, \overline{k}] = \int_{\widetilde{z}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,W\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t)$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Conclusion

A3.1: Waitlist Model, Round 1

Round 1 (Hybrid):

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{P(y = S|x, t, \overline{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Conclusion

A3.1: Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, \ddot{m}_2; \theta_{A_2}\right) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}{1 + \exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}$$

$$P_{d_3} = P\left(d_3 = A \middle| x, z, \overline{m_{2A}}; \theta_{A_3}\right) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}{1 + \exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P\left(y = S \middle| x, z, t, k; \theta_S\right) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1+\exp(f_S(x,z,t,k)\theta_S)}, P_{m_2} = P\left(m_2 = M \middle| x, z, t; \theta_{M_2}\right) = \frac{\exp(f_{M_2}(x,z,t)\theta_{M_2})}{1+\exp(f_{M_2}(x,z,t)\theta_{M_2})}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\left(y_3^* = S \middle| \overline{m_{2A}}; \theta_{S_3^*}\right) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}$$

Conclusion

A3.1: Waitlist-Dynamic Parameterization 2

$$
\begin{aligned}
\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,\epsilon_2)|x,t,\overline{k}] &= \int_{\widetilde{z}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x,t) \\
&= \int_{\widetilde{z_1}} \cdots \int_{\widetilde{z_{J_z}}} \Big[\cdot \Big] dF_{z|x,t,1}(\widetilde{z_1}|\widetilde{z_2},\ldots,\widetilde{z_{J_z}},x,t) \cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)
\end{aligned}
$$

$$
\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left(\sum_{\widetilde{d}_3 \in \{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)
$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}, \quad \ell = 1,\ldots,L_j-1$

Conclusion

A3.1: Waitlist-Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(\overline{m_{2A,n}} | \ddot{m}_{2,n}; \theta_{F_m}\right)}_{\text{Density of } \overline{m_{2A}}|\ddot{m}_2} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}}$$

$$\underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate θ_{M_2} with 1st-stage likelihood to get $m_{2,n}=$ matriculation probability $P_{m_{2,n}}$: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1 - P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

Conclusion

A3.1: Waitlist Notes

- **Estimation:** Estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; slope parameter's expected sign is positive from variation in $\overline{m_2}$ across years (e.g. due to Covid)
  - **Assumption:** Effect of $\ddot{m}_2$ on $\overline{m_{2A}}$ is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ from changes in d_2
- **Dynamics:** What if committee doesn't think dynamically?
  - If $\ddot{m}_2$ increasing increases $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which increases $\mathbb{E}[P(\text{Cutoff})_3]$, then it should
  - Don't need success data for all years to identify these two effects
- **Over/Undershooting:** Waitlist helps avoid both in reality, not just in expectation

Conclusion

A3.1: Symmetric Waitlist Model 1

Round 3:

$$v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\epsilon_3) = \max_{d_3\in\{A,R\}} u_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},d_3) = P(y=S|x,z,t,\overline{k})\mathbb{1}(d_3=A) + P(y_3^*=S|\overline{m_{2A}},\overline{m_3})\mathbb{1}(d_3=R)$$

Round 2:

$$v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\epsilon_2) = \max_{d_2\in\{A,W\}} u_2(x,z,t,\overline{k},\ddot{m}_2,m_3,d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x,z,t,\overline{k},\ddot{m}_2,m_3,d_2) =$$
$$P(y=S|x,z,t,\overline{k})\mathbb{1}(d_2=A) + \beta\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3]\mathbb{1}(d_2=W), \text{ such that:}$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\widetilde{\overline{m_3}},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3] =$$
$$\int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma\log\left(\sum_{\tilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\tilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

Conclusion

A3.1: Symmetric Waitlist Model 2

Round 1:

$$v_1(x,t,\overline{k}) = \max_{d_1 \in \{T,R\}} u_1(x,t,\overline{k},d_1) + \epsilon_1(d_1), \text{ such that:}$$

- **Hybrid:** $u_1(x,t,\overline{k},d_1) = P(y = S|x,t,\overline{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R)$
- **Dynamic:**
  $u_1(x,t,\overline{k},d_1) = \mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R)$, such that:

$$\mathbb{E}[v_2(x,z,t,\overline{k}\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}$$

Conclusion

A3.1: Symmetric Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, \ddot{m}_2, m_3; \theta_{A_2}\right) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}{1 + \exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}$$

$$P_{d_3} = P\left(d_3 = A \middle| x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3}\right) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}{1 + \exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P\left(y = S \middle| x, z, t, k; \theta_S\right) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1+\exp(f_S(x,z,t,k)\theta_S)}, \; P_{m_3} = P\left(m_3 = M \middle| x, z, t; \theta_{M_3}\right) = \frac{\exp(f_{M_3}(x,z,t)\theta_{M_3})}{1+\exp(f_{M_3}(x,z,t)\theta_{M_3})}$$

Cutoff Levels/Probabilities:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\left(y_3^* = S \middle| \overline{m_{2A}}, \overline{m_3}; \theta_{S_3^*}\right) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}$$

Conclusion

A3.1: Symmetric Waitlist-Dynamic Parameterization 2

★ Assume $(\overline{m_{2A}}, \overline{m_3})$ are conditionally independent:

$$\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k}\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z}_1}\cdots\int_{\widetilde{z_{J_z}}}\Big[\,\cdot\,\Big] dF_{z|x,t,1}(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3] = \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma\log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}}\exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)},\ \ell = 1,\ldots,L_j-1$

Conclusion

A3.1: Symmetric Waitlist-Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{f\left(\overline{m_{2A,n}} | \ddot{m}_{2,n}; \theta_{F_{m_2}}\right)}_{\text{Density of } \overline{m_{2A}} | \ddot{m}_2} \underbrace{f\left(\overline{m_{3,n}} | m_{3,n}; \theta_{F_{m_3}}\right)}_{\text{Density of } \overline{m_3} | m_3} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}}$$

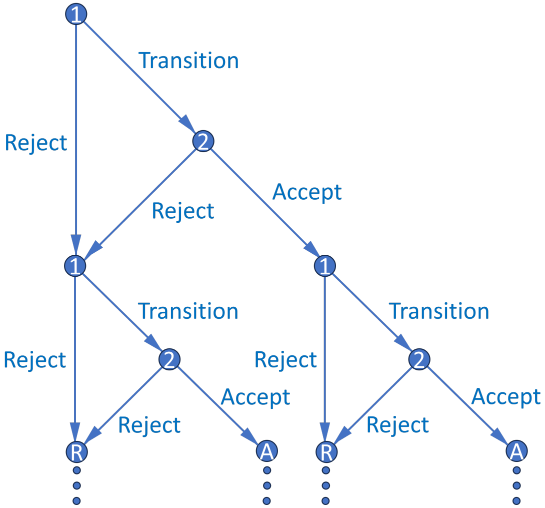
$$\underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate θ_{M_3} with 1st-stage likelihood to get $m_{3,n} =$ matriculation probability $P_{m_{3,n}}$: $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A, m_{3,n}=M)}(1 - P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A, m_{3,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

Conclusion

A3.2: Multiyear Decision Graph



Conclusion

A3.2: Multiyear Memory (Matriculation)

- Suppose committee maximizes discounted "lifetime" cohort success
- Respecify $P(y_2^* = S|t)$, where $t =$ year, as $L(y_2^* = S|c_{-1})$
  - $c_{-1} =$ previous year's cohort size. $L(\text{Cutoff}_2)$ increases in c_{-1} because committee must keep total number of students in program relatively constant across years
- What information available at start of Round 2 determines c?
  - **If and only if $\boldsymbol{d_2 = A}$:** $m =$ applicant's matriculation probability
  - For simplicity, suppose matriculation probabilities are exogenous, and success probabilities depend only on applicant characteristics (i.e. not program rank)

Conclusion

A3.2: Multiyear Dynamics (Matriculation)

Theorem 6 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- With memory, optimal dynamic decision may not equal optimal myopic one
  - High m means high $\overline{F}_{c|m}$, which means high $\mathbb{E}[L(\text{Cutoff}_2)']$ and future EV
  - Accepting likely matriculants above this year's margin means that next year in expectation, committee won't have to accept as many marginal applicants
- Harder to estimate since multiyear problem has no closed-form solution
  - Principal component analysis can help reduce dimensionality

Conclusion

$$v_2(x,z,t,m,c_{-1},\epsilon_2) = \max_{d_2\in\{A,R\}} u_2(x,z,t,m,c_{-1},d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x,z,t,m,c_{-1},d_2) = {} & \{P(y=S|x,z,t) + \beta\mathbb{E}[v_1(x',t',c,{\epsilon_1}')|m]\}\mathbb{1}(d_2=A) \\ & + \{L(y_2^*=S|c_{-1}) + \beta\mathbb{E}[v_1(x',t',c,{\epsilon_1}')]\}\mathbb{1}(d_2=R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x',t',c,{\epsilon_1}')|(m)] = \int_c\int_{t'}\int_{x'} \sigma\log\left(\sum_{{d_1}'\in\{T,R\}} e^{u_1(x',t',c,{d_1}')/\sigma}\right) dF_x(x')dF_t(t')dF_{c|(m)}(c|(m))$$

Conditional Choice Probabilities:

$$P(d_2|x,z,t,m,c_{-1}) = \frac{\exp(u_2(x,z,t,m,c_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,m,c_{-1},\widetilde{d_2})/\sigma)}$$

Conclusion

A3.2: Multiyear Model (Matriculation), Round 1

$$v_1(x, t, c_{-1}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, c_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$\begin{aligned} u_1(x, t, c_{-1}, d_1) = & \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}]\mathbb{1}(d_1 = T) \\ & + \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x', t', c, \epsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:} \end{aligned}$$

$$\begin{aligned} & \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}] = \\ & \int_{\widetilde{m}} \int_{\widetilde{z}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x, \widetilde{z}, t, \widetilde{m}, c_{-1}, \widetilde{d}_2)/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t) dF_{m|x,t}(\widetilde{m}|x, t) \end{aligned}$$

Conditional Choice Probabilities:

$$P(d_1|x, t, c_{-1}) = \frac{\exp(u_1(x, t, c_{-1}, d_1)/\sigma)}{\sum_{\widetilde{d}_1 \in \{T,R\}} \exp(u_1(x, t, c_{-1}, \widetilde{d}_1)/\sigma)}$$

Conclusion

A3.2: Multiyear (Matriculation) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t, c_{-1}; \theta_T\right) = \frac{\exp(f_T(x, t, c_{-1})\theta_T)}{1 + \exp(f_T(x, t, c_{-1})\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, m, c_{-1}; \theta_A\right) = \frac{\exp(f_A(x, z, m, c_{-1})\theta_A)}{1 + \exp(f_A(x, z, m, c_{-1})\theta_A)}$$

Success Probability:

$$P_y = P\left(y = S \middle| x, z, t; \theta_S\right) = \frac{\exp(f_S(x, z, t)\theta_S)}{1 + \exp(f_S(x, z, t)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L\left(y_2^* = S \middle| c_{-1}; \theta_{S_2^*}\right) = f_{S_2^*}(c_{-1})\theta_{S_2^*}$$

Conclusion

A3.2: Multiyear (Matriculation) Parameterization 2

- Assume (x, t) are independent, and next year is independent of current year:

$$\mathbb{E}[v_1(x', t', c, \epsilon_1{}')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1{}' \in \{T,R\}} e^{u_1(x',t',c,d_1{}')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m)) =$$

$$\int_c \int_{t_1{}'} \cdots \int_{t_{J_t}{}'} \int_{x_1{}'} \cdots \int_{x_{J_x}{}'} \Big[\cdot \Big] dF_{x,1}(x_1{}'|x_2{}', \ldots, x_{J_x}{}') \cdots dF_{x,J_x}(x_{J_x}{}') dF_{t,1}(t_1{}'|t_2{}', \ldots, t_{J_t}{}') \cdots dF_{t,J_t}(t_{J_t}{}') dF_{c|(m)}(c|(m))$$

- Assume (z, m) are conditionally independent:

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}] = \int_{\widetilde{m}} \int_{\widetilde{z}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d}_2)/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t) dF_{m|x}(\widetilde{m}|x, t) =$$

$$\int_{\widetilde{m}} \int_{\widetilde{z}_1} \cdots \int_{\widetilde{z_{J_z}}} \Big[\cdot \Big] dF_{z|x,t,1}(\widetilde{z}_1|\widetilde{z}_2, \ldots, \widetilde{z_{J_z}}, x, t) \cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x, t) dF_{m|x,t}(\widetilde{m}|x, t)$$

Conclusion

A3.2: Multiyear (Matriculation) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{f\left(c_n | m_n; \theta_{F_c}\right)}_{\text{Density of } c|m} \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(m_n | x_n, t_n; \theta_{F_m}\right)}_{\text{Density of } m|x, t}$$

$$\underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{f\left(x_n; \theta_{F_x}\right) f\left(t_n; \theta_{F_t}\right) f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Densities of } x,\ t, \text{ and } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $f_(c|m)$ on N_A accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

Conclusion

A3.2: Multiyear Memory and Dynamics (Diversity)

- Let additional state $a =$ any diversity variable affecting success
  - For example, $a = \mathbb{1}(\text{TheoryField})$; theory is less common than applied
  - Let $\bar{a}_{-1} =$ percentage of last year's acceptance set that's theory
  - If $|a - \bar{a}_{-1}|$ is small, $P(\text{Success})$ may decrease because future advisor may be too busy (effect may be unidentifiable if program never gets too imbalanced)

Theorem 7 (Proof)

Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.

- Optimal dynamic decision may not equal optimal myopic one. May accept diverse applicant with marginally lower $P(\text{Success})$ to increase $P(\text{Success})$ of next year's pool
  - Relative to $a = 0$, $a = 1 \Rightarrow$ higher $\overline{F}_{\bar{a}|a} \Rightarrow$ higher $\mathbb{E}[P(\text{Success})']$ and future EV

Conclusion

A3.2: Multiyear Model (Diversity), Round 2

$$v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, a, \bar{a}_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x, z, t, a, \bar{a}_{-1}, d_2) = &\{P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')|a]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')|a] = \int_{\widetilde{\bar{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{\bar{a}}, \widetilde{t'}, d_1')/\sigma}\right) dF_{x'}(\widetilde{x'}) dF_{t'}(\widetilde{t'}) dF_{\bar{a}|a}(\widetilde{\bar{a}}|a)$$

Conditional Choice Probabilities:

$$P(d_2|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(u_2(x, z, t, a, \bar{a}_{-1}, d_2)/\sigma)}{\sum_{\widetilde{d}_2 \in \{A,R\}} \exp(u_2(x, z, t, a, \bar{a}_{-1}, \widetilde{d}_2)/\sigma)}$$

Conclusion

A3.2: Multiyear Model (Diversity), Round 1

$$v_1(x, t, \bar{a}_{-1}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \bar{a}_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{a}_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}]\mathbb{1}(d_1 = T)$$
$$+ \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}] = \int_{\widetilde{a}} \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in A,R} e^{u_2(x,\widetilde{z},t,\widetilde{a},\bar{a}_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x, t)dF_a(\widetilde{a})$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')] = \int_{\widetilde{\bar{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\bar{a}},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\bar{a}}(\widetilde{\bar{a}})$$

Conditional Choice Probabilities:

$$P(d_1|x, t, \bar{a}_{-1}) = \frac{\exp(u_1(x, t, \bar{a}_{-1}, d_1)/\sigma)}{\sum_{\widetilde{d}_1 \in \{T,R\}} \exp(u_1(x, t, \bar{a}_{-1}, \widetilde{d}_1)/\sigma)}$$

Conclusion

A3.2: Multiyear (Diversity) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\big(d_1 = T \big| x, t, \bar{a}_{-1}; \theta_T\big) = \frac{\exp(f_T(x, t, \bar{a}_{-1})\theta_T)}{1 + \exp(f_T(x, t, \bar{a}_{-1})\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, t, a, \bar{a}_{-1}; \theta_A\big) = \frac{\exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}{1 + \exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}$$

Success Probability:

$$P_y = P\big(y = S \big| x, z, t, |a - \bar{a}_{-1}|; \theta_S\big) = \frac{\exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}{1 + \exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L\big(y_2^* = S \big| t; \theta_{S_2^*}\big) = f_{S_2^*}(t)\theta_{S_2^*}$$

Conclusion

A3.2: Multiyear (Diversity) Parameterization 2

- Assume $(x, t, \overline{a})$ mutually independent, such that $\overline{a}|a$:

$$\mathbb{E}[v_1(x', t', \overline{a}, \epsilon_1')|a] = \int_{\widetilde{\overline{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{t'}, \widetilde{\overline{a}}, d_1')/\sigma} \right) dF_{x'}(\widetilde{x'}) dF_{t'}(\widetilde{t'}) dF_{\overline{a}|a}(\widetilde{\overline{a}}|a) =$$

$$\int_{\widetilde{\overline{a}}} \int_{\widetilde{t_1'}} \cdots \int_{\widetilde{t_{J_t}'}} \int_{\widetilde{x_1'}} \cdots \int_{\widetilde{x_{J_x}'}} \Big[\cdot \Big] F_{x,1}({x_1}'|{x_2}', \ldots, {x_{J_x}}') \cdots dF_{x,J_x}({x_{J_x}}') dF_{t,1}({t_1}'|{t_2}', \ldots, {t_{J_t}}') \cdots dF_{t,J_t}({t_{J_t}}') dF_{\overline{a}|a}(\widetilde{\overline{a}}|a)$$

- Assume (z, a) independent, such that $z|x$:

$$\mathbb{E}[v_2(x, z, t, a, \overline{a}_{-1}, \epsilon_2)|x, t, \overline{a}_{-1}] = \int_{\widetilde{a}} \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d_2} \in A,R} e^{u_2(x, \widetilde{z}, t, \widetilde{a}, \overline{a}_{-1}, \widetilde{d_2})/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t) dF_a(\widetilde{a})$$

$$= \int_{\widetilde{a}} \int_{\widetilde{z_1}} \cdots \int_{\widetilde{z_J}} \Big[\cdot \Big] dF_1(\widetilde{z_1}|\widetilde{z_2}, \ldots \widetilde{z_J}, x, t) \cdots dF_J(\widetilde{z_J}|x, t) dF_a(\widetilde{a})$$

Conclusion

A3.2: Multiyear (Diversity) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{f\left(\bar{a}_n | a_n; \theta_{F_{\bar{a}}}\right)}_{\text{Density of } \bar{a}|a} \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(a_n; \theta_{F_a}\right)}_{\text{Density of } a}$$

$$\underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{f\left(x_n; \theta_{F_x}\right) f\left(t_n; \theta_{F_t}\right) f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Densities of } x,\ t, \text{ and } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $f(\bar{a}|a)$ on N_A accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

Conclusion

A4: Economics Department Summary Statistics

Table 2: Economics Department Summary Statistics

(a) Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

(b) Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Reduced-Form Summary

A4: Economics Department Correlation Matrix

Table 3: Economics Department Correlation Matrix

Columns 1–17:

	Trans	Wait	Acc	Mat	OH	Att	Com	T100	ProR	MProR	PlaR	MPlaR	2016	2017	2018	2019	Age
Transition	1																
Waitlist	0.8083	1															
Accept	0.4136	-0.1908	1														
Matriculate	0.2200	-0.0994	0.5320	1													
Open House	0.2192	-0.1104	0.5496	0.5400	1												
Attend Program	0.1266	0.0767	0.1209	0.1522	0.0833	1											
Complete Program	0.1085	0.0889	0.0763	0.0427	0.0154	0.9124	1										
Top 100 Placement	0.1073	0.0778	0.0977	0.0682	0.0498	0.4025	0.4412	1									
Program Rank	-0.1531	-0.0909	-0.1565	-0.1693	-0.0999	-0.8635	-0.7942	-0.4106	1								
Missing Program Rank	-0.1341	-0.0860	-0.1219	-0.1594	-0.0856	-0.9548	-0.8749	-0.4018	0.9044	1							
Placement Rank	-0.1153	-0.0802	-0.1057	-0.0553	-0.0666	-0.5176	-0.5673	-0.9004	0.4999	0.5032	1						
Missing Placement Rank	-0.1051	-0.0835	-0.0778	-0.0435	-0.0164	-0.9084	-0.9957	-0.4431	0.7924	0.8708	0.5698	1					
2016	-0.0752	-0.0883	-0.0229	-0.0013	-0.0126	-0.0194	0.0073	-0.0005	0.0077	0.0178	-0.0069	-0.0073	1				
2017	-0.0480	-0.0732	-0.0162	-0.0113	0.0162	0.0076	0.0138	0.0096	-0.0025	-0.0064	-0.0040	-0.0138	-0.2552	1			
2018	0.1637	0.1695	0.0120	0.0287	0.0009	0.0101	0.0078	0.0096	-0.0118	-0.0233	-0.0076	-0.0034	-0.2483	-0.2454	1		
2019	0.0121	-0.0050	0.0133	0.0058	-0.0045	0.0145	-0.0565	-0.0081	0.0123	0.0017	0.0166	0.0543	-0.2562	-0.2532	-0.2464	1	
Winsorized Age	0.0738	0.0157	0.0102	0.0201	0.0122	-0.0260	-0.0179	0.0398	0.0474	0.0304	-0.0148	0.0229	0.0102	-0.0091	-0.0128	0.0160	1
Female	0.0411	0.0255	0.0350	0.0102	0.0328	-0.0068	-0.0179	-0.0189	0.0074	0.0168	0.0007	0.0198	0.0066	-0.0163	0.0142	0.0358	-0.1468
U.S.	0.0046	-0.0393	0.0530	0.0301	0.0318	-0.1069	-0.1165	-0.0310	0.0900	0.1075	0.0538	0.1141	0.0083	0.0263	-0.0073	-0.0409	-0.1537
International Asian	0.0717	0.1739	-0.0976	-0.0189	-0.0210	0.1744	0.1736	0.0416	-0.1725	-0.1816	-0.0646	-0.1707	-0.0210	-0.0162	-0.0107	0.0407	-0.0407
GRE Verbal Percentile	0.3349	0.2346	0.1825	0.0746	0.1228	0.0524	0.0343	0.0479	-0.0681	-0.0536	-0.0548	-0.0358	-0.0272	-0.0158	0.0260	0.0343	-0.1056
GRE Quantitative Percentile	0.3448	0.2791	0.1559	0.0623	0.0605	0.1957	0.1930	0.0980	-0.2102	-0.1965	-0.1287	-0.1899	0.0061	0.0095	0.0116	-0.0255	-0.0622
GRE Analytic Percentile	0.1960	0.1198	0.1330	0.0505	0.0710	-0.0468	-0.0700	-0.0083	0.0359	0.0504	0.0058	0.0695	-0.0542	0.0113	0.0537	0.0379	-0.1533
Undergraduate GPA	0.0517	-0.0095	0.0747	0.0322	0.0440	0.0329	0.0179	-0.0166	-0.0382	-0.0336	0.0056	-0.0179	-0.0272	-0.0058	0.0260	0.0109	-0.0853
TOEFL/IELTS	0.1251	0.1060	0.0898	0.0017	0.0413	-0.0200	-0.0127	0.0342	-0.0053	0.0226	-0.0431	0.0108	-0.0018	0.0047	0.0176	-0.0092	-0.1177
Elite U.S. University	0.1203	0.0775	0.1048	0.0285	0.0769	-0.0151	-0.0196	-0.0167	-0.0171	0.0108	0.0124	0.0179	-0.0305	-0.0271	0.0070	0.0392	-0.0769
Elite U.S. Liberal Arts College	0.0256	-0.0020	0.0274	0.0372	0.0205	-0.0081	-0.0280	-0.0038	-0.0103	0.0104	-0.0012	0.0272	0.0178	0.0133	-0.0405	-0.0125	-0.0639
Elite Foreign University	0.0459	0.0691	-0.0066	-0.0130	-0.0062	0.0218	0.0305	0.0071	-0.0445	-0.0261	0.0027	-0.0314	0.0084	0.0215	-0.0246	-0.0290	-0.0129
Other Foreign University	0.0241	0.0431	-0.0309	0.0146	-0.0327	0.0571	0.0564	0.0311	-0.0260	-0.0496	-0.0454	-0.0512	-0.0228	-0.0085	-0.0772	0.1586	0.1709
Missing Undergraduate Institution	-0.0601	-0.0583	-0.0185	-0.0424	-0.0215	0.0169	0.0435	0.0180	-0.0124	-0.0198	-0.0227	-0.0458	0.0300	-0.0008	0.1946	-0.2576	-0.0011
Graduate Degree	0.1131	0.1254	0.0049	0.0046	0.0248	0.0529	0.0706	0.0226	-0.0399	-0.0548	-0.0328	-0.0698	0.0443	0.1379	-0.1556	-0.0729	0.3704
Work Experience	0.1103	0.0043	0.1074	0.0834	0.0484	-0.0723	-0.0789	0.0371	0.0722	0.0851	-0.0148	0.0795	-0.0017	-0.0002	-0.0056	0.0093	0.4141
#Professor Recommenders	0.0083	0.0374	-0.0114	-0.0136	-0.0257	0.0606	0.0466	-0.0073	-0.0472	-0.0547	0.0069	-0.0514	-0.0791	0.0186	0.0329	0.0390	-0.2398
#Prolific Recommenders	0.1044	0.0794	0.0733	0.0076	0.0374	0.1042	0.1084	0.0632	-0.1156	-0.1054	-0.0567	-0.1070	-0.0262	0.0330	0.0430	0.0062	0.0200
Missing GRE	-0.1102	-0.1318	-0.0579	-0.0217	-0.0358	-0.0731	-0.0649	-0.0322	0.0715	0.0786	0.0303	0.0641	0.0118	-0.0042	-0.0123	0.0071	0.0718
Missing Undergraduate GPA	-0.0759	-0.0361	-0.0929	-0.0464	-0.0501	0.0261	0.0292	0.0119	-0.0105	-0.0213	-0.0191	-0.0269	-0.0372	-0.0582	-0.0334	0.2064	0.1637
Missing TOEFL/IELTS	-0.0246	-0.0503	0.0389	0.0146	0.0438	-0.1211	-0.1296	-0.0660	0.1027	0.1244	0.0753	0.1260	-0.0248	-0.0372	-0.0241	0.0552	-0.1342
Missing Winsorized Age	-0.0250	-0.0231	-0.0103	-0.0055	-0.0063	-0.0361	-0.0329	-0.0145	0.0312	0.0345	0.0187	0.0328	0.0214	0.0218	-0.0143	-0.0148	0.0000
Committee Score	0.0000	-0.3453	0.5999	0.3028	0.3395	0.0873	0.0554	0.0493	-0.1117	-0.0842	-0.0463	-0.0594	-0.0390	-0.0215	0.0691	0.0682	-0.0598
#Math Courses	0.0107	0.0311	-0.0215	-0.0177	-0.0467	0.0368	0.0367	0.0132	-0.0467	-0.0422	-0.0177	-0.0375	0.0291	0.0473	0.0642	-0.1633	-0.0764
Advanced Math	0.2226	0.1304	0.2298	0.1071	0.1558	0.0463	0.0406	0.0031	-0.0742	-0.0510	-0.0116	-0.0420	-0.0182	-0.0752	0.0408	0.0398	-0.0349
Letters Length/100	0.0189	-0.1069	0.1781	0.0827	0.0792	0.0261	0.0189	0.0146	-0.0368	-0.0238	-0.0114	-0.0197	0.0075	-0.0049	-0.0155	0.0170	0.0376
Letters Positive Terms (%)	0.0161	-0.0760	0.0674	0.0780	0.0058	-0.0500	-0.0268	0.0708	0.0375	0.0280	-0.0454	0.0186	0.0464	-0.0001	-0.0522	0.0536	0.0432
Letters Negative Terms (%)	-0.0093	-0.0545	0.0077	0.0413	-0.0479	-0.0158	-0.0610	-0.0737	0.0245	0.0241	0.0592	0.0574	0.0186	0.0081	0.0824	-0.0272	0.0807
Letters Standout Terms (%)	0.0003	-0.1493	0.1464	0.0698	-0.0346	0.0222	0.0131	0.0878	-0.0031	-0.0252	-0.0422	-0.0153	0.0733	-0.0587	-0.0863	0.0900	0.0507
Letters Ability Terms (%)	0.0375	-0.0301	0.0487	0.0543	0.0006	-0.1785	-0.1573	-0.0137	0.1340	0.1765	0.0371	0.1502	0.0512	0.0014	0.0153	-0.0218	0.0866
Letters Research Terms (%)	0.0499	-0.0879	0.1800	0.0897	0.0944	0.0439	0.0375	0.0492	-0.0771	-0.0588	-0.0934	-0.0376	-0.0062	-0.0162	0.1051	0.0225	0.0258
Letters Grindstone Terms (%)	0.1125	-0.0320	0.1493	0.1481	0.0630	-0.1067	-0.1156	-0.0128	0.0728	0.1050	0.0347	0.1104	-0.0133	-0.0563	-0.0136	0.0294	0.1012
Letters Teaching Terms (%)	0.0258	-0.0022	-0.0087	0.0696	0.0224	-0.1522	-0.1563	0.0241	0.1552	0.1493	0.0234	0.1579	0.0595	-0.0270	-0.0398	0.0459	0.1921
Letters Communal Terms (%)	0.0338	0.0517	-0.0749	-0.0272	-0.0216	-0.0020	0.0357	-0.0500	0.0132	0.0135	0.0186	-0.0394	0.0324	0.0711	-0.0603	-0.0627	0.0382
Letters Agentic Terms (%)	0.0421	0.0589	0.0557	0.0884	0.0913	-0.0509	-0.0716	0.0030	0.0147	0.0225	0.0016	0.0686	0.0927	-0.0091	0.0022	-0.0789	-0.0172
Letters Terms (%) PC1	0.0130	-0.0283	0.0593	0.0870	0.0251	-0.0585	-0.0577	0.0109	0.0481	0.0525	0.0028	0.0547	0.0310	-0.0050	-0.0129	0.0043	0.0514
Letters Terms (%) PC2	0.0088	-0.0033	0.0396	0.0383	0.0466	-0.0062	-0.0170	-0.0190	-0.0053	0.0063	0.0007	0.0174	-0.0204	-0.0007	0.0628	-0.0096	0.0192
CV Length/100	0.0067	-0.0429	0.0959	0.0421	0.0772	0.0199	0.0202	0.0299	-0.0124	-0.0185	-0.0335	-0.0222	0.0320	-0.0248	-0.0129	-0.0118	0.0394
CV Coding	-0.0067	-0.0225	0.0102	-0.0320	-0.0333	0.0280	0.0338	0.0297	-0.0250	-0.0246	-0.0377	-0.0365	-0.0225	-0.0306	0.0089	0.0368	-0.0633
CV Publication/Working Paper	0.1587	0.0464	0.2207	0.1422	0.1135	0.0666	0.0560	0.0432	-0.0707	-0.0633	-0.0408	-0.0569	-0.0284	-0.0575	0.0197	0.0603	0.0894
Essay Length/100	0.0017	-0.0422	0.0591	0.0712	0.0195	0.0473	0.0287	0.0026	-0.0530	-0.0452	-0.0053	-0.0274	0.0204	-0.0137	-0.0110	0.0253	0.0388
Essay Specific Professor	0.2518	0.1247	0.2312	0.1775	0.1550	0.0489	0.0361	0.0306	-0.0633	-0.0470	-0.0396	-0.0374	-0.0440	-0.0641	0.0419	0.0752	0.0149
Essay Economics Terms (%)	-0.0009	0.0317	-0.0293	0.0156	-0.0042	-0.0358	-0.0452	-0.0431	0.0338	0.0270	0.0456	0.0426	-0.0102	0.0036	0.0191	-0.0257	-0.0088
Essay Research Terms (%)	-0.0101	0.0042	-0.0059	-0.0277	-0.0234	0.0838	0.0766	0.0144	-0.0690	-0.0720	-0.0305	-0.0802	-0.0296	0.0235	-0.0081	0.0445	-0.0767
Essay Positive Terms (%)	-0.0132	0.0211	-0.0442	-0.0405	-0.0379	-0.0035	0.0102	0.0332	0.0043	0.0060	-0.0127	-0.0110	-0.0123	-0.0010	0.0151	0.0029	-0.0450
Essay Negative Terms (%)	0.0074	0.0139	-0.0041	-0.0057	-0.0264	0.0107	0.0005	-0.0066	-0.0026	-0.0143	0.0085	-0.0030	0.0015	0.0056	-0.0294	0.0257	0.0215
Missing Committee Score	-0.9903	-0.7997	-0.4037	-0.1981	-0.2043	-0.1226	-0.1054	-0.1012	0.1477	0.1297	0.1106	0.1020	0.0748	0.0433	-0.1554	-0.0105	-0.0656
Missing Application Pdf	-0.3301	-0.2162	-0.2739	-0.1737	-0.1677	-0.0565	-0.0516	-0.0230	0.0840	0.0596	0.0243	0.0514	0.0528	0.0955	-0.0508	-0.1152	-0.0056
Missing #Math Courses	-0.3326	-0.2159	-0.2739	-0.1654	-0.1709	-0.0585	-0.0547	-0.0293	0.0872	0.0610	0.0309	0.0544	0.0498	0.0902	-0.0516	-0.1115	-0.0077
Missing Letters	-0.3273	-0.2105	-0.2771	-0.1755	-0.1700	-0.0545	-0.0491	-0.0230	0.0815	0.0574	0.0229	0.0489	0.0502	0.0986	-0.0509	-0.1130	-0.0081
Missing CV	-0.3261	-0.2095	-0.2752	-0.1661	-0.1568	-0.0590	-0.0528	-0.0248	0.0842	0.0615	0.0230	0.0525	0.0571	0.0893	-0.0528	-0.1156	-0.0142
Missing Essay	-0.3233	-0.2066	-0.2726	-0.1770	-0.1718	-0.0533	-0.0476	-0.0253	0.0819	0.0560	0.0253	0.0474	0.0508	0.0912	-0.0448	-0.1101	-0.0035
Observations	2329																

Columns 18–34:

	Fem	US	IntAsn	GREV	GREQ	GREA	GPA	T/I	EUni	ELA	EFor	OFor	MUg	Grad	Work	Prof	Pro
Female	1																
U.S.	-0.0886	1															
International Asian	0.1108	-0.5345	1														
GRE Verbal Percentile	-0.0155	0.2891	-0.1077	1													
GRE Quantitative Percentile	-0.0398	-0.1080	0.2269	0.2114	1												
GRE Analytic Percentile	0.0022	0.3950	-0.2554	0.5519	-0.0386	1											
Undergraduate GPA	0.0693	0.0418	-0.0425	0.0835	0.0793	0.0863	1										
TOEFL/IELTS	0.0969	0.0119	-0.0524	0.2772	0.0337	0.2465	0.0031	1									
Elite U.S. University	0.0095	0.2082	-0.0410	0.1155	0.0303	0.1320	-0.0917	0.0035	1								
Elite U.S. Liberal Arts College	0.0260	0.1177	-0.0567	0.0805	0.0260	0.1074	0.0216	0.0113	-0.0665	1							
Elite Foreign University	0.0447	-0.1008	0.1325	0.0335	0.0642	-0.0303	0.0047	0.0366	-0.0764	-0.0350	1						
Other Foreign University	-0.0105	-0.4231	0.2355	-0.1253	0.0709	-0.2141	-0.0002	-0.0057	-0.3141	-0.1441	-0.1654	1					
Missing Undergraduate Institution	0.0092	-0.0364	-0.0761	-0.0414	-0.0056	-0.0322	0.0301	-0.0164	-0.2024	-0.0929	-0.1066	-0.4384	1				
Graduate Degree	-0.0010	-0.2492	0.1402	-0.1043	0.0039	-0.1313	-0.0877	0.0107	-0.1458	-0.0808	0.0597	0.1649	0.0831	1			
Work Experience	-0.0229	0.0870	-0.2765	0.0369	-0.1144	0.1031	0.0121	0.0128	0.0083	0.0294	-0.0367	-0.0399	0.0163	0.1219	1		
#Professor Recommenders	-0.0114	-0.0330	0.1402	-0.0051	0.1028	-0.0399	0.0557	-0.0180	-0.0592	0.0069	0.0270	0.0136	-0.0023	-0.0631	-0.2477	1	
#Prolific Recommenders	-0.0017	-0.2072	0.1686	-0.0282	0.1814	-0.1224	-0.0547	0.0052	-0.0311	-0.0497	0.0492	0.1487	-0.0014	0.1173	-0.0724	0.2047	1
Missing GRE	0.0122	-0.0303	-0.0502	-0.0081	-0.0129	0.0026	-0.0117	-0.0034	0.0080	-0.0325	0.0122	-0.0028	-0.0183	-0.0073	0.0434	-0.0375	-0.0163
Missing Undergraduate GPA	-0.0095	-0.3206	0.1579	-0.2161	0.0599	-0.3214	0.0000	-0.0211	-0.2096	-0.1183	0.0940	0.3226	0.0180	0.1345	-0.0097	0.0036	0.1459
Missing TOEFL/IELTS	-0.0207	0.4256	-0.2472	0.0917	-0.1772	0.2285	0.0072	0.0000	0.2876	0.1352	-0.0396	-0.3778	-0.1160	-0.1265	0.0345	-0.0031	-0.2002
Missing Winsorized Age	-0.0235	-0.0163	-0.0282	-0.0442	-0.0172	0.0000	0.0000	0.0000	-0.0112	-0.0051	-0.0059	0.0057	0.0198	0.0289	0.0279	0.0150	-0.0071
Committee Score	-0.0014	-0.0064	-0.0812	0.0408	0.0378	0.0481	0.0800	0.0726	0.0496	-0.0133	-0.0126	-0.0104	0.0234	-0.0609	0.0246	0.0279	0.0849
#Math Courses	-0.0104	0.0306	0.0164	0.0307	-0.0093	0.0315	-0.0128	0.0409	0.0339	-0.0032	-0.0111	-0.0406	-0.0053	-0.0392	0.0041	-0.0020	-0.0002
Advanced Math	-0.0054	-0.0328	0.0827	0.0181	0.1157	-0.0207	0.0099	0.0072	0.0701	0.0337	0.0150	0.0042	-0.0763	-0.0478	-0.0561	0.0363	0.1345
Letters Length/100	0.0367	0.0330	-0.0790	0.0531	-0.0110	0.0579	-0.0050	0.0487	0.0012	0.0635	0.0250	-0.0216	-0.0030	-0.0020	0.1001	-0.0420	0.0333
Letters Positive Terms (%)	-0.0145	0.1376	-0.0883	0.0349	0.0533	0.1164	-0.0099	-0.0573	-0.0018	0.1339	-0.0416	-0.0931	-0.0256	-0.0638	0.0769	-0.1049	-0.1867
Letters Negative Terms (%)	-0.0036	0.1392	-0.0830	0.0764	-0.0665	0.0632	-0.0612	-0.1270	0.0379	0.0199	0.0261	-0.0506	-0.0449	0.0424	0.0943	-0.1391	0.0507
Letters Standout Terms (%)	-0.0141	0.0120	-0.0394	0.0100	0.0078	0.0742	0.0477	0.0066	-0.1116	0.0846	0.0073	0.0514	-0.0690	-0.0211	0.0198	0.0226	0.0132
Letters Ability Terms (%)	-0.0015	0.1424	-0.1930	0.0423	0.0487	0.1273	0.0808	-0.0476	-0.0047	0.0583	0.0035	-0.1331	0.0026	0.0081	0.1310	-0.1811	-0.2310
Letters Research Terms (%)	0.0245	0.1273	-0.0539	0.1004	-0.0220	0.1052	-0.0448	0.0235	0.0296	0.0482	-0.0391	-0.0767	0.0477	0.0037	0.1626	-0.2141	-0.0945
Letters Grindstone Terms (%)	0.0839	0.1233	-0.1591	0.0209	-0.0701	0.0946	0.0185	-0.0524	0.0140	0.1687	-0.0368	-0.1797	0.0350	0.0451	0.2148	-0.1778	-0.1497
Letters Teaching Terms (%)	0.0369	0.0349	-0.1064	-0.0334	-0.0692	0.0481	0.0218	-0.0055	-0.0565	0.0274	-0.0166	0.0065	-0.0288	0.0670	0.1777	-0.0771	-0.1383
Letters Communal Terms (%)	-0.0136	-0.1655	0.0499	-0.0011	0.0538	0.0001	0.0203	-0.1205	-0.0344	0.0260	0.0362	-0.0034	0.0305	0.0770	-0.0646	0.1095	-0.0050
Letters Agentic Terms (%)	0.1191	0.1148	-0.0943	0.0638	-0.0830	0.0517	-0.0681	0.0347	0.0740	0.1425	-0.0270	-0.1247	0.0009	-0.0336	0.0006	-0.1529	-0.0772
Letters Terms (%) PC1	0.0207	0.0601	-0.0803	0.0219	-0.0120	0.0601	0.0043	-0.0258	-0.0053	0.0708	-0.0127	-0.0645	-0.0019	0.0041	0.0764	-0.0732	-0.0963
Letters Terms (%) PC2	0.0150	0.0516	-0.0323	0.0285	-0.0245	0.0213	-0.0207	-0.0088	0.0366	-0.0004	-0.0110	-0.0449	0.0199	0.0192	0.0790	-0.0774	-0.0236
CV Length/100	0.0249	0.0341	-0.0336	0.0120	-0.0114	0.0448	0.0014	0.0399	-0.0027	0.0241	-0.0241	0.0074	-0.0123	0.0314	0.1054	-0.0738	0.0126
CV Coding	-0.0004	-0.0209	0.0151	-0.0125	0.0021	-0.0083	-0.0131	0.0023	0.0122	0.0125	-0.0072	0.0100	-0.0158	0.0065	-0.0100	-0.0106	0.0452
CV Publication/Working Paper	-0.0215	-0.0475	0.0269	-0.0141	0.0741	-0.0197	-0.0054	-0.0400	0.0292	0.0109	-0.0006	0.0358	-0.0256	0.0282	0.0825	-0.0292	0.0846
Essay Length/100	-0.0133	-0.0100	0.0078	-0.0314	-0.0020	-0.0472	-0.0304	-0.0084	-0.0051	-0.0053	0.0103	0.0145	-0.0109	0.0979	0.0273	0.0028	0.0398
Essay Specific Professor	-0.0044	-0.0712	0.0804	-0.0046	0.1188	-0.0416	0.0123	0.0222	0.0562	-0.0091	-0.0067	0.0521	-0.0577	0.0065	-0.0138	0.0279	0.1057
Essay Economics Terms (%)	-0.0026	-0.0373	0.0596	-0.0069	0.0022	-0.0280	0.0313	-0.0115	0.0065	-0.0007	0.0222	-0.0436	0.0058	0.0017	-0.0049	-0.0217	0.0205
Essay Research Terms (%)	0.0498	-0.0432	0.0665	-0.0242	0.0132	-0.0050	0.0206	-0.0017	0.0157	0.0134	0.0160	-0.0295	-0.0003	-0.0478	-0.0345	0.0073	0.0095
Essay Positive Terms (%)	0.0208	-0.0019	0.0002	-0.0149	0.0057	0.0096	0.0287	0.0103	0.0155	-0.0262	-0.0164	-0.0213	-0.0069	-0.0358	-0.0021	0.0141	-0.0244
Essay Negative Terms (%)	-0.0251	-0.0063	0.0323	-0.0046	0.0103	-0.0007	-0.0384	-0.0316	0.0534	0.0249	0.0123	-0.0148	-0.0414	-0.0002	0.0003	-0.0086	0.0500
Missing Committee Score	-0.0471	-0.0078	-0.0793	-0.3358	-0.3437	-0.1972	-0.0516	-0.1281	-0.1189	-0.0273	-0.0479	-0.0179	0.0551	-0.1108	-0.1049	-0.0094	-0.1045
Missing Application Pdf	-0.0041	0.1078	-0.1112	0.0270	-0.1797	0.0709	-0.0220	0.0179	-0.0650	-0.0082	-0.0079	-0.0674	0.0874	0.0263	0.0656	-0.0632	-0.1400
Missing #Math Courses	-0.0025	0.1021	-0.1108	0.0188	-0.1817	0.0657	-0.0200	0.0155	-0.0670	-0.0039	-0.0105	-0.0646	0.0844	0.0275	0.0646	-0.0636	-0.1423
Missing Letters	-0.0063	0.1052	-0.1060	0.0263	-0.1765	0.0686	-0.0209	0.0219	-0.0674	-0.0092	-0.0091	-0.0632	0.0848	0.0274	0.0645	-0.0622	-0.1461
Missing CV	-0.0020	0.1142	-0.1198	0.0255	-0.1806	0.0760	-0.0193	0.0171	-0.0579	-0.0109	-0.0110	-0.0757	0.0915	0.0297	0.0556	-0.0596	-0.1431
Missing Essay	-0.0099	0.1058	-0.1106	0.0263	-0.1746	0.0683	-0.0222	0.0160	-0.0660	-0.0101	-0.0042	-0.0649	0.0881	0.0230	0.0668	-0.0601	-0.1415

Columns 35–51:

	MGRE	MGPA	MT/I	MAge	Score	NMath	AMath	LetN	LPos	LNeg	LSta	LAbi	LRes	LGri	LTea	LCom	LAge
Missing GRE	1																
Missing Undergraduate GPA	0.0625	1															
Missing TOEFL/IELTS	0.0714	-0.2810	1														
Missing Winsorized Age	0.1575	0.0153	0.0236	1													
Committee Score	-0.0451	-0.0466	0.0188	0.0000	1												
#Math Courses	0.0170	-0.0632	0.0423	0.0004	-0.0232	1											
Advanced Math	-0.0417	-0.0261	-0.0030	-0.0092	0.1388	0.2255	1										
Letters Length/100	-0.0082	-0.0084	0.0034	0.0007	0.1631	0.0027	0.0502	1									
Letters Positive Terms (%)	0.1234	-0.0626	-0.0402		0.0587	-0.0070	-0.0916	0.0015	1								
Letters Negative Terms (%)	0.0151	-0.0109	0.1318		0.0330	0.0100	-0.0104	0.0546	0.1546	1							
Letters Standout Terms (%)	0.1382	0.0456	-0.0033		0.1475	0.0194	-0.1303	0.0427	0.4324	-0.0327	1						
Letters Ability Terms (%)	-0.0190	-0.0979	0.0358		0.0943	-0.0408	-0.1419	0.0162	0.5004	0.1896	0.2105	1					
Letters Research Terms (%)	-0.0446	0.0039	0.0552		0.1776	0.0127	0.0874	0.2071	0.0820	0.1569	-0.0520	0.0863	1				
Letters Grindstone Terms (%)	0.0905	-0.0876	0.0740		0.1127	-0.0523	-0.0447	0.0523	0.3502	0.1286	0.0678	0.3655	0.2281	1			
Letters Teaching Terms (%)	-0.0047	-0.0060	-0.0189		-0.0163	-0.0878	-0.1488	0.1290	0.3650	-0.0016	0.1177	0.3161	0.0976	0.2383	1		
Letters Communal Terms (%)	0.1483	-0.0116	-0.1435		-0.1044	-0.0440	-0.0754	-0.1683	0.2329	-0.0224	0.0070	0.0719	-0.1124	0.1381	0.0216	1	
Letters Agentic Terms (%)	0.0548	-0.1457	0.1325		0.0353	0.1087	0.0595	0.0830	0.3284	0.0143	0.1124	0.3027	0.0256	0.2276	0.2610	0.0207	1
Letters Terms (%) PC1	0.0261	-0.0426	0.0213	0.0004	0.0492	-0.0199	-0.1116	0.0803	0.8114	0.2386	0.3964	0.7486	0.2082	0.6157	0.5816	0.2083	0.5580
Letters Terms (%) PC2	-0.0271	-0.0068	0.0497	0.0003	0.0365	-0.0202	0.0655	0.1600	-0.2292	0.4519	-0.5572	-0.0082	0.7168	0.3168	0.0333	-0.2524	-0.1306
CV Length/100	-0.0228	-0.0163	-0.0019	0.0002	0.0962	0.0312	0.0677	0.2051	-0.0026	-0.0135	0.0985	0.0481	0.1497	0.0613	0.0839	-0.1494	0.1032
CV Coding	-0.0094	0.0346	0.0194	-0.0002	0.0740	0.0970	0.0863	0.1100	-0.0178	0.0284	0.0407	-0.0115	0.2123	0.0344	-0.0499	-0.0274	0.0278
CV Publication/Working Paper	-0.0289	0.0128	-0.0457	-0.0064	0.1599	-0.0381	0.2648	0.1565	0.1046	0.0451	0.1148	0.0639	0.2120	0.1178	0.1343	-0.0182	0.0086
Essay Length/100	-0.0057	0.0249	0.0041	0.0001	0.0474	-0.0164	0.0317	0.1083	-0.0279	0.0331	-0.0042	-0.0206	0.0524	-0.0092	-0.0672	-0.0119	-0.0283
Essay Specific Professor	-0.0406	0.0133	-0.0322	-0.0091	0.1428	0.0022	0.4462	0.0738	-0.0047	0.0772	-0.0022	-0.0215	0.0483	0.0270	0.0159	-0.0317	0.0294
Essay Economics Terms (%)	0.0164	-0.0006	0.0160	0.0000	0.0035	0.1185	0.0515	-0.0282	-0.0572	-0.0354	-0.0553	0.0057	0.1233	0.0995	0.0289	0.0579	0.0002
Essay Research Terms (%)	-0.0019	-0.0008	0.0049	-0.0003	0.0376	0.0664	0.0430	0.0499	-0.0016	0.0032	0.0595	-0.0279	0.2357	-0.0866	-0.0744	-0.0378	-0.0300
Essay Positive Terms (%)	0.0013	-0.0200	0.0169	-0.0005	-0.0277	0.0540	-0.0383	-0.1339	0.0887	-0.0558	-0.0026	-0.0028	-0.0436	-0.0223	0.0400	-0.0516	-0.0216
Essay Negative Terms (%)	-0.0132	0.0141	0.0353	0.0003	0.0270	0.0719	0.0827	-0.0128	0.0174	0.1960	0.0757	-0.0206	-0.0427	-0.0338	-0.0425	-0.0131	0.0363
Missing Committee Score	0.1134	0.0768	0.0216	0.0247	0.0000	-0.0153	-0.2077	-0.0137	-0.0089	0.0003	0.0162	-0.0200	-0.0288	-0.0786	0.0211	-0.0473	-0.0157
Missing Application Pdf	0.0578	-0.0143	0.0482	0.0140	-0.1544	0.0263	-0.6598	0.0467									
Missing #Math Courses	0.0623	-0.0108	0.0517	0.0137	-0.1618	0.0268	-0.6547	0.0402	-0.0241	-0.0133	0.0534	-0.0279	-0.0512	0.0813	0.0401	-0.0056	0.0053
Missing Letters	0.0571	-0.0142	0.0437	0.0139	-0.1570	0.0336	-0.6607	0.0470									
Missing CV	0.0558	-0.0205	0.0536	0.0137	-0.1568	0.0332	-0.6449	0.0394	-0.0109	-0.0393	-0.0461	0.0037	-0.0206	0.0649	0.0421	0.0770	0.1614
Missing Essay	0.0564	-0.0119	0.0429	0.0138	-0.1489	0.0146	-0.6606	0.0474	-0.0201	-0.0545	-0.0171	0.0164	-0.0382	0.0205	-0.0490	-0.0453	-0.0824

Columns 52–68:

	LPC1	LPC2	CVN	CVCod	CVPap	EssN	EssProf	EssEcon	EssRes	EssPos	EssNeg	MScore	MPDF	MNMath	ML	MCV	MEss
Letters Terms (%) PC1	1																
Letters Terms (%) PC2	-0.0204	1															
CV Length/100	0.0820	0.0721	1														
CV Coding	0.0147	0.1218	0.2274	1													
CV Publication/Working Paper	0.1158	0.0865	0.2482	0.1413	1												
Essay Length/100	-0.0332	0.0460	0.0706	0.1168	0.0772	1											
Essay Specific Professor	-0.0084	0.0360	0.0795	0.1038	0.2923	0.0701	1										
Essay Economics Terms (%)	0.0166	0.1110	0.0016	0.0578	0.0391	-0.0593	0.0049	1									
Essay Research Terms (%)	-0.0307	0.0886	0.0185	0.1414	0.0621	0.0651	-0.0097	0.1593	1								
Essay Positive Terms (%)	0.0132	-0.0494	-0.0413	-0.0248	-0.0291	-0.1956	-0.0535	-0.0689	-0.0250	1							
Essay Negative Terms (%)	0.0140	-0.0083	0.0279	0.0566	0.0102	0.0716	0.0666	-0.1217	-0.1206	-0.0714	1						
Missing Committee Score	-0.0015	-0.0054	-0.0070	0.0098	-0.1443	0.0100	-0.2336	-0.0045	0.0027	0.0134	-0.0106	1					
Missing Application Pdf	0.0322	0.0218	0.0165	-0.0164	-0.4566	0.0042	-0.6494	-0.0023	-0.0250	-0.0327	0.0183	0.3105	1				
Missing #Math Courses	0.0343	0.0164	0.0076	-0.0239	-0.4531	0.0007	-0.6399	0.0044	-0.0279	-0.0345	0.0283	0.3149	0.9843	1			
Missing Letters	0.0324	0.0219	0.0199	-0.0202	-0.4599	0.0066	-0.6462	0.0039	-0.0208	-0.0261	0.0113	0.3076	0.9929	0.9769	1		
Missing CV	0.0419	0.0187	0.0168	-0.0168	-0.4653	0.0026	-0.6299	-0.0129	-0.0385	-0.0266	0.0242	0.3061	0.9814	0.9650	0.9740	1	
Missing Essay	0.0253	0.0200	0.0134	-0.0246	-0.4464	0.0042	-0.6578	-0.0023	-0.0253	-0.0331	0.0185	0.3034	0.9871	0.9710	0.9798	0.9709	1

⏩ Reduced-Form Summary

A4: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

Reduced-Form Summary

A4: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	Top 100	Top 100	Top 100	Placement Rank	Placement Rank	Placement Rank
2016	-0.3826	-0.5049	-0.2577	0.0230	0.0301	0.0885	1.8038	1.7607	-0.6618
	(0.2365)	(0.4703)	(0.6853)	(0.1708)	(0.1711)	(0.1887)	(4.2583)	(4.2504)	(3.6266)
2017	-0.0954	-1.3247***	0.1357	0.0829	0.0711	0.1460	2.7746	3.0235	0.7473
	(0.2564)	(0.4881)	(0.6732)	(0.1707)	(0.1721)	(0.1934)	(4.2295)	(4.2188)	(3.7157)
2018	-0.2206	-3.7609***	0.9476	0.0624	-0.0392	0.0142	2.7265	5.2685	4.9770
	(0.2648)	(0.6518)	(0.6200)	(0.1761)	(0.1807)	(0.2011)	(4.4414)	(4.5011)	(3.8786)
2019	-0.1059	-3.1670***	0.5909	0.0594	0.0144	0.0366	5.7085	6.6750	5.2241
	(0.2653)	(0.7114)	(0.6855)	(0.1821)	(0.1830)	(0.2052)	(4.6400)	(4.6266)	(4.1424)
Winsorized Age	-0.0218	-0.0080	-0.1567**	0.0240	0.0242	0.0420*	0.3596	0.3632	-0.1277
	(0.0342)	(0.0592)	(0.0778)	(0.0207)	(0.0205)	(0.0236)	(0.5457)	(0.5410)	(0.4839)
Female	0.3759**	0.8784**	-0.3445	-0.0880	-0.1026	-0.0256	1.8085	2.0921	-1.6375
	(0.1657)	(0.3813)	(0.3794)	(0.1126)	(0.1130)	(0.1252)	(2.8776)	(2.8750)	(2.5117)
U.S.	-0.4982*	-0.0681	1.0348*	0.1167	0.1396	0.1520	-0.6372	-0.9759	-0.7059
	(0.2624)	(0.4538)	(0.6098)	(0.1903)	(0.1909)	(0.2236)	(4.7990)	(4.7913)	(4.3464)
International Asian	-1.1569***	-1.1507***	1.1559**	0.1695	0.1949	-0.1005	-7.8212**	-8.2008**	3.3007
	(0.2129)	(0.4146)	(0.4904)	(0.1379)	(0.1395)	(0.1546)	(3.5528)	(3.5708)	(3.1577)
GRE Verbal Percentile	0.0264***	0.0079	-0.0107	0.0049	0.0030	0.0007	-0.1958**	-0.1470*	-0.0330
	(0.0054)	(0.0130)	(0.0118)	(0.0031)	(0.0032)	(0.0036)	(0.0787)	(0.0806)	(0.0735)
GRE Quantitative Percentile	0.1329***	0.0589**	-0.1033**	0.0245***	0.0184**	0.0047	-0.8498***	-0.7196***	-0.2428*
	(0.0202)	(0.0275)	(0.0464)	(0.0077)	(0.0075)	(0.0084)	(0.1567)	(0.1578)	(0.1460)
GRE Analytic Percentile	0.0076**	0.0094	-0.0119	-0.0020	-0.0027	-0.0021	0.0656	0.0821	0.0160
	(0.0039)	(0.0085)	(0.0095)	(0.0025)	(0.0025)	(0.0029)	(0.0669)	(0.0671)	(0.0618)
Undergraduate GPA	1.2726*	1.0509	-0.7209	-0.3976	-0.4722	-0.8687*	4.8040	6.9311	17.2597**
	(0.6975)	(1.2453)	(1.2760)	(0.3912)	(0.3848)	(0.4442)	(10.1461)	(10.0615)	(7.7507)
TOEFL/IELTS	0.4236	-0.1165	-0.4434	0.1778	0.1437	0.3240*	-2.2504	-1.2960	-7.6837*
	(0.2924)	(0.5976)	(0.5525)	(0.1571)	(0.1559)	(0.1757)	(4.3852)	(4.3431)	(3.9285)
Elite U.S. University	0.7328***	-0.1822	-0.4980	-0.0159	-0.0853	-0.1523	-2.3790	-0.3552	-0.3450
	(0.2774)	(0.5918)	(0.5799)	(0.2133)	(0.2133)	(0.2344)	(5.2934)	(5.2869)	(4.4943)
Elite U.S. Liberal Arts College	0.0167	0.5446	0.9907	0.0731	0.0803	-0.1010	-3.9470	-3.8047	-3.4006
	(0.4568)	(0.6896)	(0.8447)	(0.3370)	(0.3361)	(0.3936)	(8.8571)	(8.7916)	(8.2332)
Elite Foreign University	0.1244	-0.5507	-0.0673	0.1210	0.0751	0.0969	-2.7705	-1.5414	-2.2142
	(0.5319)	(0.9725)	(1.1224)	(0.3239)	(0.3259)	(0.3526)	(7.4748)	(7.4679)	(6.1380)
Other Foreign University	0.5323*	0.6328	0.7373	0.1483	0.1091	0.1587	-6.7584	-5.7186	-9.1260**
	(0.3022)	(0.6505)	(0.6347)	(0.1964)	(0.1970)	(0.2216)	(4.9679)	(4.9620)	(4.3486)
Missing Undergraduate Institution	0.3260	0.2482	-0.6851	0.2292	0.2082	0.1770	-7.9862	-7.5382	-8.1758*
	(0.3111)	(0.6016)	(0.7260)	(0.2017)	(0.2022)	(0.2345)	(5.1133)	(5.1129)	(4.5085)
Graduate Degree	0.1726	-0.2058	-0.2954	-0.0406	-0.0921	-0.1835	-3.2803	-2.0242	2.6215
	(0.1788)	(0.3845)	(0.5017)	(0.1231)	(0.1273)	(0.1370)	(3.1662)	(3.2087)	(2.7452)
Work Experience	0.7785***	0.7519**	1.1183**	0.2298*	0.1772	0.2839**	-3.5942	-2.2559	-6.4380**
	(0.1915)	(0.3400)	(0.5291)	(0.1253)	(0.1270)	(0.1422)	(3.2005)	(3.2301)	(2.9093)
#Professor Recommenders	-0.0377	-0.1797	0.3587	-0.0472	-0.0513	-0.0547	0.8765	0.9972	0.7371
	(0.1079)	(0.2434)	(0.3115)	(0.0749)	(0.0754)	(0.0862)	(2.0492)	(2.0495)	(1.7160)
#Prolific Recommenders	0.2208***	0.2081	-0.2993	0.1011*	0.0899*	0.0600	-1.4623	-1.1967	0.1726
	(0.0811)	(0.1721)	(0.2060)	(0.0528)	(0.0533)	(0.0585)	(1.3680)	(1.3670)	(1.1704)
Missing GRE	-1.5399	1.0141		-0.4515	-0.3258	0.0174	11.5202	8.5434	-3.7106
	(0.9821)	(1.1356)		(0.3510)	(0.3544)	(0.3797)	(8.6385)	(8.6602)	(6.1802)
Missing Undergraduate GPA	-0.3432	-0.2798	0.0798	-0.1010	-0.0639	-0.0513	2.4296	1.5256	-0.8710
	(0.2210)	(0.4413)	(0.4943)	(0.1565)	(0.1585)	(0.1756)	(3.9808)	(3.9842)	(3.3572)
Missing TOEFL/IELTS	0.4719**	0.3743	-0.0734	-0.1261	-0.1447	-0.0065	4.0029	4.3290	-2.3792
	(0.2062)	(0.4223)	(0.5891)	(0.1267)	(0.1282)	(0.1443)	(3.3209)	(3.3244)	(2.9657)
Committee Score		4.2419***	-0.2608		0.0943**	0.0059		-2.5088**	0.6884
		(0.4534)	(0.2250)		(0.0456)	(0.0488)		(1.1860)	(1.0132)
#Math Courses	-0.0949	-0.2706	0.0755	0.0626	0.0662	0.0008	-2.3828	-2.4550	0.9116
	(0.0773)	(0.2829)	(0.1680)	(0.0759)	(0.0766)	(0.0885)	(1.8033)	(1.8149)	(1.5082)
Advanced Math	0.6333**	1.3229	-1.3706**	-0.1932	-0.2078	-0.2275	3.0400	3.4570	1.9957
	(0.3036)	(0.9203)	(0.5621)	(0.2605)	(0.2639)	(0.2949)	(6.4446)	(6.4377)	(5.5819)
Letters Length/100	0.1358***	0.1084	-0.0910	0.0067	-0.0018	-0.0150	-0.2593	-0.0078	0.7206
	(0.0518)	(0.1424)	(0.0712)	(0.0335)	(0.0339)	(0.0388)	(0.8254)	(0.8325)	(0.7084)
Letters Terms (%) PC1	0.1507	0.2939	0.4716**	0.0362	0.0251	0.1260	1.6082	1.8374	-0.5612
	(0.0923)	(0.2912)	(0.2051)	(0.0732)	(0.0748)	(0.0915)	(2.0561)	(2.0540)	(1.8186)
Letters Terms (%) PC2	-0.0913	-0.4980	0.2724	-0.1226	-0.1199	-0.1420	0.7118	0.5616	0.1155
	(0.1464)	(0.4143)	(0.2604)	(0.1162)	(0.1177)	(0.1272)	(2.7992)	(2.8081)	(2.3450)
CV Length/100	0.0361	0.0559	-0.1986	0.0301	0.0274	0.0352	-1.0456	-1.0120	-0.8208
	(0.0617)	(0.1186)	(0.1337)	(0.0562)	(0.0563)	(0.0593)	(1.3224)	(1.3130)	(1.0252)
CV Coding	-0.0439	-0.2369*	-0.2911**	0.0716	0.0710	0.0992	-1.9601	-1.8941	-2.0142
	(0.0765)	(0.1381)	(0.1390)	(0.0532)	(0.0532)	(0.0654)	(1.5459)	(1.5358)	(1.3384)
CV Publication/Working Paper	0.6162*	0.8309	-0.1668	0.2940	0.2834	0.1310	-11.5166	-10.8849	-4.6204
	(0.3421)	(0.9326)	(0.5756)	(0.2998)	(0.3036)	(0.3383)	(7.2873)	(7.3131)	(6.7001)
Essay Length/100	0.1747*	0.2225	0.1776	-0.0008	-0.0008	-0.0711	-0.4758	-0.5351	1.8425
	(0.0933)	(0.3920)	(0.1536)	(0.0751)	(0.0764)	(0.0821)	(1.8648)	(1.8721)	(1.6240)
Essay Specific Professor	0.3842	0.4170	0.6845	0.2144	0.1774	0.2872	-6.8688	-5.9104	-7.5240
	(0.2908)	(0.7014)	(0.5119)	(0.2407)	(0.2410)	(0.2789)	(6.0223)	(5.9982)	(5.2718)
Essay Economics Terms (%)	-0.0169	0.0070	0.2576**	-0.1126**	-0.1150**	-0.0822	3.6540***	3.7277***	2.1702*
	(0.0547)	(0.0866)	(0.1159)	(0.0533)	(0.0531)	(0.0618)	(1.2540)	(1.2496)	(1.1179)
Essay Research Terms (%)	-0.0018	-0.0459	-0.1868	0.0697	0.0642	0.0150	-3.7084**	-3.6795**	-1.0575
	(0.0736)	(0.1690)	(0.1451)	(0.0633)	(0.0639)	(0.0674)	(1.6220)	(1.6209)	(1.3903)
Essay Positive Terms (%)	-0.0202	-0.0700	-0.0504	0.0724	0.0772*	0.0977*	-0.9256	-0.9929	-0.7266
	(0.0512)	(0.1044)	(0.1331)	(0.0453)	(0.0458)	(0.0542)	(1.1634)	(1.1496)	(0.9752)
Essay Negative Terms (%)	-0.0151	-0.0886	0.0019	-0.0386	-0.0454	-0.0472	0.8164	0.9436	0.9507
	(0.0770)	(0.1557)	(0.1071)	(0.0663)	(0.0663)	(0.0748)	(1.6317)	(1.6259)	(1.3985)
Missing Committee Score		-0.5168			-0.3458**	-0.2199		9.2234***	2.8589
		(0.6383)			(0.1397)	(0.1519)		(3.4582)	(3.0258)
Missing Application Pdf	-1.0107***	0.1926	-0.7407	0.1461	0.2686	0.2416	-5.1611	-8.4497	-6.8690
	(0.3188)	(0.8088)	(0.6084)	(0.2442)	(0.2479)	(0.2763)	(6.0328)	(6.1404)	(5.4325)
Open House			1.8384***						
			(0.4115)						
Program Rank						-0.0087***			0.1813***
						(0.0021)			(0.0404)
Missing Program Rank						-3.1432***			89.4721***
						(0.4077)			(5.6947)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2957	0.8108	0.2877	0.0308	0.0353	0.2430	0.0089	0.0097	0.1065
CV Pseudo-R^2	0.2047	0.7160	-0.2841	-0.0107	-0.0069	0.1999	0.0014	0.0022	0.0992

Robust standard errors in parentheses: *$p<0.10$, **$p<0.05$, ***$p<0.01$.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A4: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

Dependent Variable	Accept	Accept	Matriculate	Top 100	Top 100	Top 100	Placement Rank	Placement Rank	Placement Rank
2016	-0.0274	-0.0094	-0.0367	0.0035	0.0046	0.0112	0.8694	0.8485	-0.3038
	(0.0171)	(0.0085)	(0.0976)	(0.0262)	(0.0261)	(0.0238)	(2.0523)	(2.0482)	(1.6650)
2017	-0.0068	-0.0246***	0.0193	0.0127	0.0108	0.0184	1.3374	1.4571	0.3431
	(0.0184)	(0.0090)	(0.0958)	(0.0262)	(0.0263)	(0.0244)	(2.0386)	(2.0331)	(1.7054)
2018	-0.0158	-0.0699***	0.1348	0.0096	-0.0060	0.0018	1.3142	2.5391	2.2847
	(0.0189)	(0.0086)	(0.0857)	(0.0270)	(0.0276)	(0.0254)	(2.1405)	(2.1686)	(1.7791)
2019	-0.0076	-0.0589***	0.0841	0.0091	0.0022	0.0046	2.7516	3.2169	2.3982
	(0.0190)	(0.0109)	(0.0970)	(0.0279)	(0.0279)	(0.0259)	(2.2355)	(2.2284)	(1.8977)
Winsorized Age	-0.0016	-0.0001	-0.0223**	0.0037	0.0037	0.0053*	0.1734	0.1751	-0.0586
	(0.0024)	(0.0011)	(0.0108)	(0.0032)	(0.0031)	(0.0030)	(0.2629)	(0.2606)	(0.2222)
Female	0.0269**	0.0163**	-0.0490	-0.0135	-0.0157	-0.0032	0.8717	1.0082	-0.7517
	(0.0118)	(0.0068)	(0.0534)	(0.0173)	(0.0172)	(0.0158)	(1.3867)	(1.3851)	(1.1534)
U.S.	-0.0356*	-0.0013	0.1472*	0.0179	0.0213	0.0192	-0.3071	-0.4703	-0.3241
	(0.0188)	(0.0084)	(0.0864)	(0.0292)	(0.0291)	(0.0282)	(2.3132)	(2.3090)	(1.9951)
International Asian	-0.0828***	-0.0214***	0.1644**	0.0260	0.0298	-0.0127	-3.7699**	-3.9523**	1.5152
	(0.0148)	(0.0073)	(0.0691)	(0.0212)	(0.0213)	(0.0195)	(1.7114)	(1.7195)	(1.4490)
GRE Verbal Percentile	0.0019***	0.0001	-0.0015	0.0008	0.0005	0.0001	-0.0944**	-0.0708*	-0.0151
	(0.0004)	(0.0002)	(0.0017)	(0.0005)	(0.0005)	(0.0005)	(0.0379)	(0.0388)	(0.0337)
GRE Quantitative Percentile	0.0095***	0.0011**	-0.0147**	0.0038***	0.0028**	0.0006	-0.4096***	-0.3468***	-0.1114*
	(0.0014)	(0.0005)	(0.0063)	(0.0012)	(0.0011)	(0.0011)	(0.0757)	(0.0762)	(0.0670)
GRE Analytic Percentile	0.0005**	0.0002	-0.0017	-0.0003	-0.0004	-0.0003	0.0316	0.0395	0.0073
	(0.0003)	(0.0002)	(0.0013)	(0.0004)	(0.0004)	(0.0004)	(0.0322)	(0.0323)	(0.0283)
Undergraduate GPA	0.0910*	0.0195	-0.1026	-0.0610	-0.0721	-0.1096**	2.3156	3.3404	7.9232**
	(0.0499)	(0.0235)	(0.1820)	(0.0600)	(0.0587)	(0.0558)	(4.8911)	(4.8499)	(3.5564)
TOEFL/IELTS	0.0303	-0.0022	-0.0631	0.0273	0.0219	0.0409*	-1.0847	-0.6246	-3.5273*
	(0.0209)	(0.0111)	(0.0776)	(0.0241)	(0.0238)	(0.0221)	(2.1145)	(2.0935)	(1.8033)
Elite U.S. University	0.0524***	-0.0034	-0.0708	-0.0024	-0.0130	-0.0192	-1.1467	-0.1712	-0.1584
	(0.0198)	(0.0111)	(0.0816)	(0.0327)	(0.0326)	(0.0296)	(2.5514)	(2.5479)	(2.0632)
Elite U.S. Liberal Arts College	0.0012	0.0101	0.1409	0.0112	0.0123	-0.0127	-1.9025	-1.8336	-1.5611
	(0.0327)	(0.0126)	(0.1197)	(0.0517)	(0.0513)	(0.0496)	(4.2692)	(4.2369)	(3.7802)
Elite Foreign University	0.0089	-0.0102	-0.0096	0.0186	0.0115	0.0122	-1.3354	-0.7428	-1.0164
	(0.0381)	(0.0182)	(0.1597)	(0.0497)	(0.0497)	(0.0445)	(3.6022)	(3.5986)	(2.8173)
Other Foreign University	0.0381*	0.0118	0.1049	0.0228	0.0167	0.0200	-3.2576	-2.7560	-4.1894**
	(0.0216)	(0.0119)	(0.0890)	(0.0301)	(0.0301)	(0.0279)	(2.3945)	(2.3912)	(1.9955)
Missing Undergraduate Institution	0.0233	0.0046	-0.0975	0.0352	0.0318	0.0223	-3.8495	-3.6329	-3.7531*
	(0.0222)	(0.0111)	(0.1014)	(0.0309)	(0.0308)	(0.0296)	(2.4639)	(2.4635)	(2.0698)
Graduate Degree	0.0123	-0.0038	-0.0420	-0.0062	-0.0141	-0.0231	-1.5811	-0.9755	1.2034
	(0.0128)	(0.0071)	(0.0712)	(0.0189)	(0.0194)	(0.0172)	(1.5257)	(1.5461)	(1.2606)
Work Experience	0.0557***	0.0140**	0.1591**	0.0353*	0.0271	0.0358**	-1.7325	-1.0872	-2.9554**
	(0.0135)	(0.0064)	(0.0700)	(0.0192)	(0.0194)	(0.0178)	(1.5432)	(1.5570)	(1.3344)
#Professor Recommenders	-0.0027	-0.0033	0.0510	-0.0072	-0.0078	-0.0069	0.4225	0.4806	0.3384
	(0.0077)	(0.0045)	(0.0437)	(0.0115)	(0.0115)	(0.0109)	(0.9881)	(0.9881)	(0.7878)
#Prolific Recommenders	0.0158***	0.0039	-0.0426	0.0155*	0.0137*	0.0076	-0.7048	-0.5767	0.0792
	(0.0058)	(0.0031)	(0.0285)	(0.0081)	(0.0081)	(0.0074)	(0.6595)	(0.6589)	(0.5373)
Missing GRE	-0.1102	0.0188		-0.0693	-0.0497	0.0022	5.5529	4.1174	-1.7034
	(0.0706)	(0.0212)		(0.0538)	(0.0541)	(0.0479)	(4.1627)	(4.1727)	(2.8358)
Missing Undergraduate GPA	-0.0246	-0.0052	0.0114	-0.0155	-0.0098	-0.0065	1.1711	0.7352	-0.3999
	(0.0158)	(0.0082)	(0.0703)	(0.0240)	(0.0242)	(0.0221)	(1.9187)	(1.9201)	(1.5412)
Missing TOEFL/IELTS	0.0338**	0.0070	-0.0104	-0.0194	-0.0221	-0.0008	1.9295	2.0863	-1.0922
	(0.0147)	(0.0078)	(0.0839)	(0.0194)	(0.0195)	(0.0182)	(1.6007)	(1.6020)	(1.3613)
Committee Score		0.0788***	-0.0371		0.0144**	0.0007		-1.2091**	0.3160
		(0.0045)	(0.0322)		(0.0069)	(0.0062)		(0.5716)	(0.4649)
#Math Courses	-0.0068	-0.0050	0.0107	0.0096	0.0101	0.0001	-1.1485	-1.1832	0.4185
	(0.0055)	(0.0053)	(0.0239)	(0.0116)	(0.0117)	(0.0112)	(0.8692)	(0.8747)	(0.6922)
Advanced Math	0.0453**	0.0246	-0.1950**	-0.0296	-0.0317	-0.0287	1.4653	1.6660	0.9161
	(0.0216)	(0.0168)	(0.0793)	(0.0399)	(0.0402)	(0.0372)	(3.1061)	(3.1022)	(2.5621)
Letters Length/100	0.0097***	0.0020	-0.0129	0.0010	-0.0003	-0.0019	-0.1250	-0.0038	0.3308
	(0.0037)	(0.0026)	(0.0100)	(0.0051)	(0.0052)	(0.0049)	(0.3978)	(0.4012)	(0.3253)
Letters Terms (%) PC1	0.0108	0.0055	0.0671**	0.0056	0.0038	0.0159	0.7752	0.8855	-0.2576
	(0.0066)	(0.0054)	(0.0283)	(0.0112)	(0.0114)	(0.0115)	(0.9905)	(0.9892)	(0.8348)
Letters Terms (%) PC2	-0.0065	-0.0093	0.0387	-0.0188	-0.0183	-0.0179	0.3431	0.2706	0.0530
	(0.0105)	(0.0077)	(0.0371)	(0.0178)	(0.0179)	(0.0160)	(1.3495)	(1.3535)	(1.0765)
CV Length/100	0.0026	0.0010	-0.0282	0.0046	0.0042	0.0044	-0.5040	-0.4877	-0.3768
	(0.0044)	(0.0022)	(0.0190)	(0.0086)	(0.0086)	(0.0075)	(0.6374)	(0.6328)	(0.4706)
CV Coding	-0.0031	-0.0044*	-0.0414**	0.0110	0.0108	0.0125	-0.9448	-0.9129	-0.9246
	(0.0055)	(0.0026)	(0.0193)	(0.0082)	(0.0081)	(0.0082)	(0.7454)	(0.7403)	(0.6139)
CV Publication/Working Paper	0.0441*	0.0154	-0.0237	0.0451	0.0433	0.0165	-5.5511	-5.2459	-2.1210
	(0.0244)	(0.0173)	(0.0820)	(0.0460)	(0.0463)	(0.0427)	(3.5094)	(3.5215)	(3.0760)
Essay Length/100	0.0125*	0.0041	0.0253	-0.0001	-0.0001	-0.0090	-0.2293	-0.2579	0.8458
	(0.0067)	(0.0073)	(0.0219)	(0.0115)	(0.0117)	(0.0103)	(0.8988)	(0.9022)	(0.7451)
Essay Specific Professor	0.0275	0.0077	0.0974	0.0329	0.0271	0.0362	-3.3108	-2.8484	-3.4539
	(0.0208)	(0.0130)	(0.0724)	(0.0369)	(0.0368)	(0.0351)	(2.9044)	(2.8921)	(2.4207)
Essay Economics Terms (%)	-0.0012	0.0001	0.0366**	-0.0173**	-0.0175**	-0.0104	1.7613***	1.7965***	0.9962*
	(0.0039)	(0.0016)	(0.0162)	(0.0082)	(0.0081)	(0.0078)	(0.6048)	(0.6027)	(0.5134)
Essay Research Terms (%)	-0.0001	-0.0009	-0.0266	0.0107	0.0098	0.0019	-1.7875**	-1.7733**	-0.4855
	(0.0053)	(0.0031)	(0.0206)	(0.0097)	(0.0098)	(0.0085)	(0.7813)	(0.7808)	(0.6382)
Essay Positive Terms (%)	-0.0014	-0.0013	-0.0072	0.0111	0.0118*	0.0123*	-0.4461	-0.4785	-0.3335
	(0.0037)	(0.0019)	(0.0189)	(0.0069)	(0.0070)	(0.0068)	(0.5609)	(0.5541)	(0.4475)
Essay Negative Terms (%)	-0.0011	-0.0016	0.0003	-0.0059	-0.0069	-0.0060	0.3935	0.4548	0.4364
	(0.0055)	(0.0029)	(0.0152)	(0.0102)	(0.0101)	(0.0094)	(0.7865)	(0.7837)	(0.6420)
Missing Committee Score		-0.0096			-0.0528**	-0.0277		4.4451***	1.3124
		(0.0120)			(0.0212)	(0.0191)		(1.6677)	(1.3892)
Missing Application Pdf	-0.0723***	0.0036	-0.1054	0.0224	0.0410	0.0305	-2.4877	-4.0722	-3.1533
	(0.0227)	(0.0149)	(0.0858)	(0.0375)	(0.0379)	(0.0348)	(2.9090)	(2.9614)	(2.4962)
Open House			0.2615***						
			(0.0512)						
Program Rank						-0.0011***			0.0832***
						(0.0003)			(0.0185)
Missing Program Rank						-0.3965***			41.0729***
						(0.0512)			(2.6010)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A4: Economics Department Logistic Regression Results (Synthetic)

Table 6a: Economics Department Logistic Regression Results (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	Top 100	Top 100	Top 100	Placement Rank	Placement Rank	Placement Rank
2016	-0.2091	-0.1311	0.1291	0.0997	0.0892	0.1020	-0.8042	-0.5831	-1.3927
	(0.2288)	(0.2577)	(0.6478)	(0.1812)	(0.1814)	(0.1970)	(4.5726)	(4.5702)	(4.1436)
2017	-0.3355	-0.5461**	0.6767	0.2827	0.2903	0.1565	-12.4495***	-12.2002***	-6.3639
	(0.2390)	(0.2627)	(0.6543)	(0.1761)	(0.1767)	(0.1925)	(4.4305)	(4.4347)	(3.9262)
2018	0.2889	-0.2993	0.2947	0.1627	0.2192	0.1272	-9.3709**	-9.2465**	-6.0098
	(0.2056)	(0.2212)	(0.5517)	(0.1806)	(0.1827)	(0.1972)	(4.4278)	(4.4749)	(3.9975)
2019	-0.1933	-0.3208	-0.0662	0.0116	0.0023	-0.1488	-5.3768	-4.7378	1.0400
	(0.2227)	(0.2490)	(0.6069)	(0.1927)	(0.1933)	(0.2061)	(4.4927)	(4.5026)	(3.9171)
Winsorized Age	-0.0508**	-0.0414*	0.0065	-0.0141	-0.0150	-0.0094	0.4466	0.4383	0.2496
	(0.0220)	(0.0243)	(0.0600)	(0.0170)	(0.0170)	(0.0185)	(0.4147)	(0.4141)	(0.3766)
Female	-0.1826	-0.1535	0.0691	0.1351	0.1379	0.2109*	-1.5097	-1.6305	-4.7144*
	(0.1446)	(0.1590)	(0.4006)	(0.1167)	(0.1168)	(0.1259)	(2.9357)	(2.9318)	(2.6789)
U.S.	0.0002	-0.0549	0.0998	0.0198	0.0241	-0.0123	2.8625	2.5697	4.2450
	(0.2240)	(0.2468)	(0.6663)	(0.1794)	(0.1801)	(0.1997)	(4.4691)	(4.4669)	(4.1868)
International Asian	-0.1230	-0.0646	-0.1205	0.2014	0.1972	0.1700	-0.7475	-0.9433	1.1783
	(0.1688)	(0.1838)	(0.5028)	(0.1394)	(0.1392)	(0.1541)	(3.3754)	(3.3732)	(3.1231)
GRE Verbal Percentile	-0.0025	-0.0001	0.0008	0.0004	0.0001	-0.0021	-0.0668	-0.0623	0.0181
	(0.0034)	(0.0036)	(0.0089)	(0.0026)	(0.0026)	(0.0029)	(0.0661)	(0.0663)	(0.0595)
GRE Quantitative Percentile	-0.0028	-0.0005	-0.0125	0.0145**	0.0143**	-0.0003	-0.5982***	-0.5916***	-0.0184
	(0.0075)	(0.0080)	(0.0195)	(0.0065)	(0.0065)	(0.0069)	(0.1570)	(0.1569)	(0.1376)
GRE Analytic Percentile	0.0041	0.0019	-0.0010	0.0009	0.0011	0.0019	0.0181	0.0156	-0.0008
	(0.0031)	(0.0033)	(0.0074)	(0.0024)	(0.0024)	(0.0026)	(0.0587)	(0.0587)	(0.0529)
Undergraduate GPA	-0.4230	-0.3157	0.9077	0.4075	0.3986	0.4627	-2.7870	-2.7400	-5.8113
	(0.3862)	(0.4536)	(1.8572)	(0.3984)	(0.3990)	(0.4326)	(9.4800)	(9.4624)	(8.8952)
TOEFL/IELTS	0.2329	0.2776	-0.2073	-0.0682	-0.0625	-0.0142	2.4237	2.2404	-0.4376
	(0.2154)	(0.2202)	(0.4100)	(0.1509)	(0.1507)	(0.1670)	(3.7070)	(3.7021)	(3.1595)
Elite U.S. University	-0.0473	-0.1157	-0.2546	0.3145	0.3119	0.3760*	-6.7562	-6.6352	-7.1987
	(0.2618)	(0.2943)	(0.7160)	(0.2054)	(0.2062)	(0.2230)	(5.3790)	(5.3875)	(4.8870)
Elite U.S. Liberal Arts College	-0.6642	-0.6053		0.0069	-0.0044	-0.0958	-2.2643	-2.2283	3.3486
	(0.5448)	(0.5744)		(0.3656)	(0.3659)	(0.3964)	(8.9098)	(8.8713)	(7.5578)
Elite Foreign University	-0.1797	-0.1798	0.5156	-0.0538	-0.0692	-0.0298	-1.0302	-0.9315	-0.2765
	(0.3900)	(0.4341)	(1.2147)	(0.3273)	(0.3294)	(0.3495)	(8.2453)	(8.2793)	(7.1254)
Other Foreign University	-0.0406	-0.1089	0.6613	0.1055	0.1125	0.0046	-6.0523	-6.2650	-0.6409
	(0.2078)	(0.2262)	(0.5235)	(0.1688)	(0.1689)	(0.1840)	(4.2123)	(4.2076)	(3.8417)
Missing Undergraduate Institution	-0.0012	0.0377	-0.6367	0.1037	0.0966	0.0297	-5.9887	-5.9936	-2.3404
	(0.2217)	(0.2413)	(0.5829)	(0.1844)	(0.1842)	(0.1977)	(4.5947)	(4.5865)	(4.1622)
Graduate Degree	-0.0846	-0.1354	-0.0100	0.1515	0.1611	0.1906	-1.4756	-1.6341	-2.0725
	(0.1458)	(0.1626)	(0.3834)	(0.1166)	(0.1165)	(0.1252)	(2.8643)	(2.8619)	(2.5933)
Work Experience	0.2547*	0.2149	-0.2181	0.0750	0.0805	0.1443	0.4084	0.3542	-2.7442
	(0.1463)	(0.1588)	(0.3573)	(0.1161)	(0.1160)	(0.1250)	(2.8682)	(2.8651)	(2.6268)
#Professor Recommenders	0.0194	-0.0565	0.1806	0.0038	0.0074	-0.0302	-0.2230	-0.1761	1.2584
	(0.0950)	(0.1021)	(0.2725)	(0.0759)	(0.0761)	(0.0794)	(1.9087)	(1.9104)	(1.7006)
#Prolific Recommenders	-0.0388	-0.0071	0.0168	0.0027	-0.0000	0.0325	0.3327	0.3170	-1.1079
	(0.0671)	(0.0713)	(0.1685)	(0.0521)	(0.0522)	(0.0575)	(1.3245)	(1.3242)	(1.2261)
Missing GRE	0.3002	0.0736		0.3136	0.3314	0.2674	-12.0035	-11.9570	-11.5860
	(0.3757)	(0.3872)		(0.2858)	(0.2857)	(0.3297)	(8.1036)	(8.0998)	(8.0641)
Missing Undergraduate GPA	0.1997	0.1720	0.2910	-0.1321	-0.1290	-0.0816	5.2305	5.1674	4.3351
	(0.1863)	(0.2016)	(0.5443)	(0.1391)	(0.1389)	(0.1534)	(3.6167)	(3.6124)	(3.3094)
Missing TOEFL/IELTS	-0.1740	-0.0792	-0.3009	0.1948	0.1815	0.2286*	-1.5099	-1.4121	-3.1183
	(0.1609)	(0.1741)	(0.4256)	(0.1280)	(0.1281)	(0.1380)	(3.1162)	(3.1192)	(2.7616)
Committee Score		0.0372	-0.1004		0.0465	0.0330		-1.6627	-0.4906
		(0.0423)	(0.1060)		(0.0505)	(0.0558)		(1.1655)	(1.1645)
#Math Courses	0.0632	0.0756	-0.1683	-0.1269	-0.1305*	-0.1764**	1.1826	1.2073	3.0509**
	(0.0687)	(0.0733)	(0.1601)	(0.0786)	(0.0780)	(0.0863)	(1.7287)	(1.7209)	(1.5010)
Advanced Math	0.0222	-0.0336	0.9873	-0.1639	-0.1630	-0.2842	1.1644	1.2290	6.1671
	(0.2590)	(0.2747)	(0.7209)	(0.3004)	(0.3013)	(0.3256)	(6.4828)	(6.4875)	(5.6351)
Letters Length/100	-0.0254	-0.0128	-0.0575	-0.1049***	-0.1057***	-0.1218***	1.6052**	1.6308**	2.1010***
	(0.0333)	(0.0361)	(0.0885)	(0.0381)	(0.0376)	(0.0428)	(0.7873)	(0.7869)	(0.6934)
Letters Terms (%) PC1	0.0280	-0.0073	0.0381	0.0574	0.0643	0.0566	1.1364	1.1816	1.0356
	(0.0779)	(0.0821)	(0.2097)	(0.0828)	(0.0834)	(0.0930)	(1.8933)	(1.9000)	(1.6894)
Letters Terms (%) PC2	-0.2328*	-0.2369*	-0.4921	0.0802	0.0718	0.1318	-2.8589	-2.7586	-5.0669**
	(0.1226)	(0.1255)	(0.3028)	(0.1205)	(0.1211)	(0.1250)	(2.6915)	(2.7005)	(2.1872)
CV Length/100	0.0971**	0.0795	0.1022	-0.0358	-0.0336	-0.0095	2.4879**	2.5172**	0.5354
	(0.0492)	(0.0508)	(0.1340)	(0.0548)	(0.0551)	(0.0641)	(1.2406)	(1.2458)	(1.0685)
CV Coding	-0.0198	-0.0473	-0.0834	0.0688	0.0720*	0.0795*	-1.6529	-1.6778	-1.0250
	(0.0411)	(0.0448)	(0.1069)	(0.0434)	(0.0434)	(0.0474)	(1.0435)	(1.0451)	(0.8904)
CV Publication/Working Paper	0.3079	0.3005	1.3491*	0.1111	0.1183	0.1111	2.7524	2.4246	5.5955
	(0.2852)	(0.2962)	(0.7475)	(0.3264)	(0.3272)	(0.3482)	(7.3505)	(7.3749)	(6.2211)
Essay Length/100	0.0679	0.0790	-0.0558	-0.0667	-0.0636	-0.0694	0.5523	0.4398	0.3231
	(0.0770)	(0.0831)	(0.1489)	(0.0780)	(0.0794)	(0.0871)	(1.7254)	(1.7321)	(1.3973)
Essay Specific Professor	-0.1756	-0.2494	-0.1115	-0.0547	-0.0524	0.0009	1.2639	1.2219	-3.1507
	(0.2564)	(0.2714)	(0.6302)	(0.2774)	(0.2798)	(0.3022)	(6.3588)	(6.3580)	(5.5747)
Essay Economics Terms (%)	-0.0123	-0.0094	-0.0406	-0.0091	-0.0082	-0.0222	0.2276	0.2960	0.6769
	(0.0542)	(0.0572)	(0.1334)	(0.0539)	(0.0547)	(0.0624)	(1.3179)	(1.3189)	(1.1799)
Essay Research Terms (%)	0.0303	0.0066	-0.0464	0.1042	0.1084	0.1176	-1.8679	-1.9069	-1.8170
	(0.0656)	(0.0702)	(0.1578)	(0.0703)	(0.0705)	(0.0723)	(1.5218)	(1.5245)	(1.2596)
Essay Positive Terms (%)	0.0180	0.0074	0.3699***	0.0871*	0.0868*	0.0947	-1.5542	-1.4775	-1.1629
	(0.0474)	(0.0484)	(0.1213)	(0.0511)	(0.0517)	(0.0609)	(1.1518)	(1.1568)	(1.1006)
Essay Negative Terms (%)	0.0333	0.0312	-0.0731	-0.0304	-0.0280	-0.0168	0.5011	0.4716	0.2911
	(0.0314)	(0.0351)	(0.0769)	(0.0359)	(0.0359)	(0.0382)	(0.8608)	(0.8614)	(0.7100)
Missing Committee Score		-4.3141***			0.2635**	0.4129***		-0.9440	-6.1957**
		(0.4312)			(0.1267)	(0.1388)		(3.0258)	(2.8364)
Missing Application Pdf	-0.9321***	-0.0588	0.9473	0.0574	-0.0560	-0.1100	4.5136	4.8399	5.6614
	(0.2534)	(0.2753)	(0.6793)	(0.2628)	(0.2678)	(0.2943)	(5.9238)	(6.0503)	(5.4160)
Open House			1.0643						
			(0.8268)						
Program Rank						-0.0049**			0.1176***
						(0.0019)			(0.0386)
Missing Program Rank						-2.5925***			94.5255***
						(0.3273)			(5.6183)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.0552	0.2804	0.1402	0.0221	0.0245	0.1935	0.0040	0.0042	0.0953
CV Pseudo-R^2	0.0028	0.2137	-0.6295	-0.0225	-0.0212	0.1432	-0.0033	-0.0033	0.0890

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

⏩ Acceptance, Matriculation, and Success AMEs

A4: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	Top 100	Top 100	Top 100	Placement Rank	Placement Rank	Placement Rank
2016	-0.0185	-0.0100	0.0201	0.0136	0.0122	0.0120	-0.3764	-0.2729	-0.6112
	(0.0203)	(0.0197)	(0.1005)	(0.0248)	(0.0248)	(0.0233)	(2.1401)	(2.1390)	(1.8189)
2017	-0.0297	-0.0417**	0.1052	0.0387	0.0396	0.0185	-5.8265***	-5.7099***	-2.7929
	(0.0212)	(0.0199)	(0.1005)	(0.0241)	(0.0241)	(0.0227)	(2.0748)	(2.0770)	(1.7228)
2018	0.0256	-0.0229	0.0458	0.0223	0.0299	0.0150	-4.3857**	-4.3275**	-2.6375
	(0.0182)	(0.0168)	(0.0856)	(0.0247)	(0.0250)	(0.0233)	(2.0732)	(2.0958)	(1.7555)
2019	-0.0171	-0.0245	-0.0103	0.0016	0.0003	-0.0176	-2.5164	-2.2174	0.4564
	(0.0197)	(0.0189)	(0.0944)	(0.0264)	(0.0264)	(0.0243)	(2.1013)	(2.1061)	(1.7192)
Winsorized Age	-0.0045**	-0.0032*	0.0010	-0.0019	-0.0020	-0.0011	0.2090	0.2051	0.1095
	(0.0020)	(0.0019)	(0.0093)	(0.0023)	(0.0023)	(0.0022)	(0.1941)	(0.1938)	(0.1654)
Female	-0.0162	-0.0117	0.0107	0.0185	0.0188	0.0249*	-0.7066	-0.7631	-2.0690*
	(0.0128)	(0.0121)	(0.0621)	(0.0160)	(0.0159)	(0.0149)	(1.3745)	(1.3728)	(1.1767)
U.S.	0.0000	-0.0042	0.0155	0.0027	0.0033	-0.0015	1.3397	1.2026	1.8630
	(0.0198)	(0.0189)	(0.1035)	(0.0246)	(0.0246)	(0.0236)	(2.0914)	(2.0905)	(1.8379)
International Asian	-0.0109	-0.0049	-0.0187	0.0276	0.0269	0.0201	-0.3498	-0.4415	0.5171
	(0.0150)	(0.0140)	(0.0782)	(0.0191)	(0.0190)	(0.0182)	(1.5798)	(1.5789)	(1.3706)
GRE Verbal Percentile	-0.0002	-0.0000	0.0001	0.0001	0.0000	-0.0002	-0.0313	-0.0291	0.0080
	(0.0003)	(0.0003)	(0.0014)	(0.0004)	(0.0004)	(0.0003)	(0.0309)	(0.0310)	(0.0261)
GRE Quantitative Percentile	-0.0002	-0.0000	-0.0019	0.0020**	0.0020**	-0.0000	-0.2800***	-0.2769***	-0.0081
	(0.0007)	(0.0006)	(0.0030)	(0.0009)	(0.0009)	(0.0008)	(0.0734)	(0.0734)	(0.0604)
GRE Analytic Percentile	0.0004	0.0001	-0.0002	0.0001	0.0002	0.0002	0.0085	0.0073	-0.0004
	(0.0003)	(0.0003)	(0.0011)	(0.0003)	(0.0003)	(0.0003)	(0.0275)	(0.0275)	(0.0232)
Undergraduate GPA	-0.0375	-0.0241	0.1411	0.0558	0.0544	0.0546	-1.3043	-1.2824	-2.5504
	(0.0342)	(0.0346)	(0.2875)	(0.0545)	(0.0545)	(0.0510)	(4.4373)	(4.4292)	(3.9043)
TOEFL/IELTS	0.0206	0.0212	-0.0322	-0.0093	-0.0085	-0.0017	1.1343	1.0486	-0.1921
	(0.0191)	(0.0168)	(0.0637)	(0.0207)	(0.0206)	(0.0197)	(1.7354)	(1.7331)	(1.3866)
Elite U.S. University	-0.0042	-0.0088	-0.0396	0.0431	0.0426	0.0444*	-3.1620	-3.1054	-3.1593
	(0.0232)	(0.0225)	(0.1109)	(0.0281)	(0.0282)	(0.0263)	(2.5179)	(2.5220)	(2.1436)
Elite U.S. Liberal Arts College	-0.0588	-0.0462		0.0009	-0.0006	-0.0113	-1.0597	-1.0429	1.4696
	(0.0483)	(0.0439)		(0.0500)	(0.0500)	(0.0468)	(4.1696)	(4.1517)	(3.3177)
Elite Foreign University	-0.0159	-0.0137	0.0801	-0.0074	-0.0095	-0.0035	-0.4822	-0.4360	-0.1214
	(0.0346)	(0.0332)	(0.1891)	(0.0448)	(0.0450)	(0.0412)	(3.8590)	(3.8749)	(3.1271)
Other Foreign University	-0.0036	-0.0083	0.1028	0.0144	0.0154	0.0005	-2.8325	-2.9321	-0.2813
	(0.0184)	(0.0173)	(0.0813)	(0.0231)	(0.0231)	(0.0217)	(1.9702)	(1.9680)	(1.6858)
Missing Undergraduate Institution	-0.0001	0.0029	-0.0989	0.0142	0.0132	0.0035	-2.8028	-2.8051	-1.0271
	(0.0196)	(0.0184)	(0.0902)	(0.0252)	(0.0251)	(0.0233)	(2.1499)	(2.1460)	(1.8264)
Graduate Degree	-0.0075	-0.0103	-0.0015	0.0207	0.0220	0.0225	-0.6906	-0.7648	-0.9095
	(0.0129)	(0.0124)	(0.0596)	(0.0160)	(0.0159)	(0.0147)	(1.3406)	(1.3395)	(1.1381)
Work Experience	0.0226*	0.0164	-0.0339	0.0103	0.0110	0.0170	0.1911	0.1658	-1.2044
	(0.0129)	(0.0121)	(0.0551)	(0.0159)	(0.0158)	(0.0147)	(1.3422)	(1.3408)	(1.1533)
#Professor Recommenders	0.0017	-0.0043	0.0281	0.0005	0.0010	-0.0036	-0.1044	-0.0824	0.5523
	(0.0084)	(0.0078)	(0.0423)	(0.0104)	(0.0104)	(0.0094)	(0.8933)	(0.8941)	(0.7465)
#Prolific Recommenders	-0.0034	-0.0005	0.0026	0.0004	-0.0000	0.0038	0.1557	0.1484	-0.4862
	(0.0059)	(0.0054)	(0.0262)	(0.0071)	(0.0071)	(0.0068)	(0.6197)	(0.6196)	(0.5384)
Missing GRE	0.0266	0.0056		0.0429	0.0453	0.0316	-5.6177	-5.5961	-5.0848
	(0.0333)	(0.0296)		(0.0391)	(0.0390)	(0.0389)	(3.7950)	(3.7934)	(3.5412)
Missing Undergraduate GPA	0.0177	0.0131	0.0452	-0.0181	-0.0176	-0.0096	2.4479	2.4184	1.9026
	(0.0165)	(0.0154)	(0.0847)	(0.0190)	(0.0190)	(0.0181)	(1.6942)	(1.6921)	(1.4526)
Missing TOEFL/IELTS	-0.0154	-0.0060	-0.0468	0.0267	0.0248	0.0270*	-0.7067	-0.6609	-1.3685
	(0.0143)	(0.0133)	(0.0659)	(0.0175)	(0.0175)	(0.0162)	(1.4590)	(1.4604)	(1.2133)
Committee Score		0.0028	-0.0156		0.0064	0.0039		-0.7782	-0.2153
		(0.0032)	(0.0165)		(0.0069)	(0.0066)		(0.5455)	(0.5111)
#Math Courses	0.0056	0.0058	-0.0262	-0.0174	-0.0178*	-0.0208**	0.5535	0.5650	1.3390**
	(0.0061)	(0.0056)	(0.0248)	(0.0107)	(0.0106)	(0.0101)	(0.8095)	(0.8059)	(0.6591)
Advanced Math	0.0020	-0.0026	0.1534	-0.0224	-0.0223	-0.0335	0.5450	0.5752	2.7066
	(0.0229)	(0.0210)	(0.1122)	(0.0411)	(0.0411)	(0.0384)	(3.0339)	(3.0362)	(2.4720)
Letters Length/100	-0.0022	-0.0010	-0.0089	-0.0144***	-0.0144***	-0.0144***	0.7512**	0.7632**	0.9221***
	(0.0029)	(0.0028)	(0.0137)	(0.0052)	(0.0051)	(0.0050)	(0.3694)	(0.3693)	(0.3050)
Letters Terms (%) PC1	0.0025	-0.0006	0.0059	0.0079	0.0088	0.0067	0.5318	0.5530	0.4545
	(0.0069)	(0.0063)	(0.0325)	(0.0113)	(0.0114)	(0.0110)	(0.8858)	(0.8889)	(0.7413)
Letters Terms (%) PC2	-0.0206*	-0.0181*	-0.0765*	0.0110	0.0098	0.0156	-1.3380	-1.2911	-2.2237**
	(0.0108)	(0.0095)	(0.0462)	(0.0165)	(0.0165)	(0.0147)	(1.2609)	(1.2651)	(0.9607)
CV Length/100	0.0086**	0.0061	0.0159	-0.0049	-0.0046	-0.0011	1.1643**	1.1781**	0.2350
	(0.0043)	(0.0039)	(0.0210)	(0.0075)	(0.0075)	(0.0076)	(0.5802)	(0.5825)	(0.4688)
CV Coding	-0.0018	-0.0036	-0.0130	0.0094	0.0098*	0.0094*	-0.7736	-0.7852	-0.4499
	(0.0036)	(0.0034)	(0.0167)	(0.0059)	(0.0059)	(0.0056)	(0.4887)	(0.4894)	(0.3909)
CV Publication/Working Paper	0.0273	0.0230	0.2097*	0.0152	0.0162	0.0131	1.2881	1.1348	2.4557
	(0.0252)	(0.0226)	(0.1128)	(0.0447)	(0.0447)	(0.0411)	(3.4400)	(3.4515)	(2.7307)
Essay Length/100	0.0060	0.0060	-0.0087	-0.0091	-0.0087	-0.0082	0.2585	0.2058	0.1418
	(0.0068)	(0.0063)	(0.0231)	(0.0107)	(0.0108)	(0.0103)	(0.8075)	(0.8107)	(0.6133)
Essay Specific Professor	-0.0156	-0.0190	-0.0173	-0.0075	-0.0072	0.0001	0.5915	0.5719	-1.3828
	(0.0227)	(0.0207)	(0.0978)	(0.0380)	(0.0382)	(0.0357)	(2.9760)	(2.9757)	(2.4463)
Essay Economics Terms (%)	-0.0011	-0.0007	-0.0063	-0.0012	-0.0011	-0.0026	0.1065	0.1385	0.2971
	(0.0048)	(0.0044)	(0.0207)	(0.0074)	(0.0075)	(0.0074)	(0.6168)	(0.6173)	(0.5179)
Essay Research Terms (%)	0.0027	0.0005	-0.0072	0.0143	0.0148	0.0139	-0.8742	-0.8925	-0.7975
	(0.0058)	(0.0054)	(0.0245)	(0.0096)	(0.0096)	(0.0085)	(0.7127)	(0.7141)	(0.5530)
Essay Positive Terms (%)	0.0016	0.0006	0.0575***	0.0119*	0.0118*	0.0112	-0.7274	-0.6915	-0.5104
	(0.0042)	(0.0037)	(0.0182)	(0.0070)	(0.0071)	(0.0072)	(0.5394)	(0.5418)	(0.4832)
Essay Negative Terms (%)	0.0030	0.0024	-0.0114	-0.0042	-0.0038	-0.0020	0.2345	0.2207	0.1278
	(0.0028)	(0.0027)	(0.0120)	(0.0049)	(0.0049)	(0.0045)	(0.4029)	(0.4032)	(0.3117)
Missing Committee Score		-0.3296***			0.0360**	0.0487***		-0.4418	-2.7191**
		(0.0330)			(0.0173)	(0.0162)		(1.4166)	(1.2447)
Missing Application Pdf	-0.0826***	-0.0045	0.1472	0.0079	-0.0076	-0.0130	2.1124	2.2652	2.4846
	(0.0225)	(0.0210)	(0.1064)	(0.0360)	(0.0366)	(0.0347)	(2.7720)	(2.8317)	(2.3765)
Open House			0.1654						
			(0.1278)						
Program Rank						-0.0006**			0.0516***
						(0.0002)			(0.0169)
Missing Program Rank						-0.3060***			41.4849***
						(0.0381)			(2.4608)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A4: Economics Department Dynamic Estimates

Table 7: Economics Department Dynamic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0602***	-3.4656***	-28.5276***	-1.2765***	-3.7142	-0.5347
	(2.1751)	(2.0492)	(0.2371)	(6.9081)	(0.2933)	(3.3294)	(0.4783)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.5121	0.0737
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.3233)	(0.0460)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.6368*	0.0917**
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.3294)	(0.0466)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0600	0.0086
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.3133)	(0.0451)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.3498	0.0504
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.3214)	(0.0459)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0799**	0.0115**
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0351)	(0.0050)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	0.1872	0.0270
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1856)	(0.0267)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.6453*	0.0929*
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.3470)	(0.0494)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0233	-0.0033
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.2426)	(0.0349)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	-0.0005	-0.0001
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0062)	(0.0009)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	-0.0064	-0.0009
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0238)	(0.0034)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0064	-0.0009
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0043)	(0.0006)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3738	0.0538
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.2358)	(0.0338)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	0.1629	0.0234
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.3360)	(0.0484)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	0.1414	0.0204
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.5521)	(0.0795)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.2454	0.0353
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.4927)	(0.0709)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.2550	0.0367
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.3455)	(0.0497)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1146	0.0165
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.3847)	(0.0554)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.3618*	-0.0521*
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.2120)	(0.0302)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.0458	0.0066
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.2252)	(0.0324)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0215	-0.0031
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.1331)	(0.0192)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0961	0.0138
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0843)	(0.0121)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	-0.3074	-0.0443
	(0.9422)	(0.5268)	(0.0772)	(1.8091)	(0.0811)	(0.9426)	(0.1357)
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.2725	-0.0392
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.2221)	(0.0317)
Committee Score				3.7089***	0.1660***	0.0176	0.0025
				(0.3984)	(0.0065)	(0.0510)	(0.0073)
#Math Courses	-0.0790			-0.2494	-0.0112	-0.0676	-0.0097
	(0.0519)			(0.2386)	(0.0108)	(0.1076)	(0.0155)
Advanced Math	0.2979			1.1003	0.0492	-0.1678	-0.0242
	(0.1889)			(0.7916)	(0.0347)	(0.3465)	(0.0498)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0252	-0.0036
	(0.0245)			(0.1080)	(0.0048)	(0.0431)	(0.0062)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1496	0.0215
	(0.0551)			(0.2489)	(0.0112)	(0.1082)	(0.0155)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.0395	-0.0057
	(0.0823)			(0.3450)	(0.0153)	(0.1470)	(0.0212)
CV Length/100	0.0513			0.0391	0.0017	0.0778	0.0112
	(0.0329)			(0.1060)	(0.0047)	(0.0700)	(0.0100)
CV Coding	0.0381			-0.2228*	-0.0100*	0.0863	0.0124
	(0.0424)			(0.1301)	(0.0060)	(0.0788)	(0.0113)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1054	0.0152
	(0.1912)			(0.8128)	(0.0364)	(0.3992)	(0.0575)
Essay Length/100	0.0927			0.2047	0.0092	-0.0987	-0.0142
	(0.0597)			(0.3483)	(0.0157)	(0.0948)	(0.0136)
Essay Specific Professor	0.1411			0.4125	0.0185	0.0869	0.0125
	(0.1744)			(0.6138)	(0.0275)	(0.3148)	(0.0453)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.1091	-0.0157
	(0.0312)			(0.0814)	(0.0036)	(0.0736)	(0.0105)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0033	0.0005
	(0.0426)			(0.1492)	(0.0067)	(0.0771)	(0.0111)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0948	0.0137
	(0.0345)			(0.0968)	(0.0043)	(0.0589)	(0.0084)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.1146	-0.0165
	(0.0443)			(0.1368)	(0.0061)	(0.0779)	(0.0111)
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.3292	0.0474
	(0.1841)			(0.7005)	(0.0311)	(0.3321)	(0.0477)
Program Rank						-0.0093***	-0.0013***
						(0.0033)	(0.0005)
Missing Program Rank						-3.3225***	-0.4783***
						(0.6860)	(0.0991)
$\sqrt{\text{MSE}}$	1.6467***						
	(0.0370)						
Obs | Param | LL | AIC	2329	581	-13672.6	28507.3			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-5.6290**	-1.1004***	0.3327	-0.7250***	32.63%	-0.8104**	
	(2.3609)	(0.1469)	(0.2495, 0.4437)	(0.2165)	(24.06%, 42.54%)	(0.4033)	
2016	-0.0236	0.1593	0.3902	0.2502	38.35%	0.2397	
	(0.1869)	(0.2367)	(0.2622, 0.5807)	(0.3792)	(24.57%, 54.29%)	(0.3786)	
2017	0.4614**	0.3272**	0.4615	0.5346**	45.25%	0.5202*	
	(0.2164)	(0.1661)	(0.3928, 0.5423)	(0.2664)	(38.04%, 52.68%)	(0.2671)	
2018	0.9169***	0.0427	0.3473	0.0677	34.14%	0.0247	
	(0.1990)	(0.1826)	(0.2828, 0.4263)	(0.2765)	(27.30%, 41.70%)	(0.2854)	
2019	0.5987***	0.2264	0.4173	0.3636	41.06%	0.3243	
	(0.1972)	(0.1689)	(0.3429, 0.5078)	(0.2663)	(33.39%, 49.19%)	(0.2681)	
Winsorized Age	-0.0459*					-0.0004	
	(0.0247)					(0.0009)	
Female	-0.1023					0.0103	
	(0.1052)					(0.0091)	
U.S.	-0.3174					0.0033	
	(0.2109)					(0.0076)	
International Asian	-0.3625**					-0.0044	
	(0.1586)					(0.0064)	
GRE Verbal Percentile	0.0110***					0.0002	
	(0.0036)					(0.0002)	
GRE Quantitative Percentile	0.0578***					0.0006	
	(0.0108)					(0.0007)	
GRE Analytic Percentile	0.0026					0.0001	
	(0.0028)					(0.0001)	
TOEFL/IELTS	0.3186*					0.0003	
	(0.1714)					(0.0067)	
Elite U.S. University	0.5964***					-0.0097	
	(0.1777)					(0.0105)	
Elite U.S. Liberal Arts College	-0.4482					0.0038	
	(0.3240)					(0.0137)	
Elite Foreign University	0.2282					-0.0076	
	(0.2741)					(0.0126)	
Other Foreign University	0.4355**					0.0043	
	(0.1947)					(0.0101)	
Missing Undergraduate Institution	0.3910*					-0.0001	
	(0.2050)					(0.0089)	
Graduate Degree	-0.0558					-0.0018	
	(0.1512)					(0.0058)	
Work Experience	0.0776					0.0077	
	(0.1419)					(0.0073)	
#Professor Recommenders	0.1021					-0.0002	
	(0.0784)					(0.0035)	
#Prolific Recommenders	0.0557					-0.0005	
	(0.0601)					(0.0024)	
Missing GRE	-1.2131					0.0197	
	(0.9076)					(0.0219)	
Missing TOEFL/IELTS	0.2158*					0.0029	
	(0.1255)					(0.0054)	
Committee Score						0.0462	
						(0.0359)	
#Math Courses	-0.1437					-0.0026	
	(0.0926)					(0.0044)	
Advanced Math	0.4116					-0.0019	
	(0.3021)					(0.0100)	
Letters Length/100	0.0902**					0.0004	
	(0.0439)					(0.0018)	
Letters Terms (%) PC1	0.2113**					0.0049	
	(0.0952)					(0.0054)	
Letters Terms (%) PC2	-0.1512					-0.0038	
	(0.1994)					(0.0060)	
CV Length/100	0.0276					0.0012	
	(0.0720)					(0.0019)	
CV Coding	0.0310					-0.0041	
	(0.0978)					(0.0041)	
CV Publication/Working Paper	1.1626***					0.0011	
	(0.4332)					(0.0123)	
Essay Length/100	0.2680**					0.0072	
	(0.1053)					(0.0086)	
Essay Specific Professor	0.1400					-0.0055	
	(0.2233)					(0.0128)	
Essay Economics Terms (%)	0.0664					0.0003	
	(0.0551)					(0.0014)	
Essay Research Terms (%)	0.1357*					-0.0002	
	(0.0821)					(0.0028)	
Essay Positive Terms (%)	0.0413					-0.0004	
	(0.0725)					(0.0027)	
Essay Negative Terms (%)	0.1022					0.0009	
	(0.0774)					(0.0032)	
Missing Application Pdf	-0.6972**					-0.0242	
	(0.2751)					(0.0184)	
Program Rank						-0.0087***	
						(0.0030)	
Missing Program Rank						-3.1650***	
						(0.6793)	
$\sqrt{\text{MSE}}$	1.7301***						
	(0.0480)						
Sigma							0.0018
							(0.0014)
Obs | Param | LL | AIC	2329	528	-13667.0	28390.0			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

A4: Economics Department Static Estimates

Table 8: Economics Department Static Estimates

(a) Unrestricted

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-23.0601***	-3.4656***	-28.5276***	-1.2765***	-4.6847***	-0.5998***	-3.7142	-0.5347
	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(1.6180)	(0.2054)	(3.3294)	(0.4783)
2016	-0.2896	-0.0435	-0.5294	-0.0237	0.0683	0.0087	0.5121	0.0737
	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1852)	(0.0237)	(0.3233)	(0.0460)
2017	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1166	0.0149	0.6368*	0.0917**
	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1877)	(0.0240)	(0.3294)	(0.0466)
2018	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0112	0.0014	0.0600	0.0086
	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.1920)	(0.0246)	(0.3133)	(0.0451)
2019	0.2905	0.0437	-3.0765***	-0.1377***	0.0177	0.0023	0.3498	0.0504
	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1913)	(0.0245)	(0.3214)	(0.0459)
Winsorized Age	0.0628***	0.0094***	-0.0133	-0.0006	0.0439*	0.0056*	0.0799**	0.0115**
	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0229)	(0.0029)	(0.0351)	(0.0050)
Female	0.3919***	0.0589***	0.8909**	0.0399***	-0.0342	-0.0044	0.1872	0.0270
	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1232)	(0.0158)	(0.1856)	(0.0267)
U.S.	-0.1124	-0.0169	0.0293	0.0013	0.1634	0.0209	0.6453*	0.0929*
	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2198)	(0.0281)	(0.3470)	(0.0494)
International Asian	0.0087	0.0013	-1.1484***	-0.0514***	-0.0894	-0.0114	-0.0233	-0.0033
	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1489)	(0.0191)	(0.2426)	(0.0349)
GRE Verbal Percentile	0.0326***	0.0049***	0.0039	0.0002	0.0015	0.0002	-0.0005	-0.0001
	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0062)	(0.0009)
GRE Quantitative Percentile	0.1829***	0.0275***	0.0402	0.0018	0.0065	0.0008	-0.0064	-0.0009
	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0076)	(0.0010)	(0.0238)	(0.0034)
GRE Analytic Percentile	0.0152***	0.0023***	0.0078	0.0004	-0.0011	-0.0001	-0.0064	-0.0009
	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0028)	(0.0004)	(0.0043)	(0.0006)
TOEFL/IELTS	-0.0097	-0.0015	-0.1286	-0.0058	0.3267*	0.0418*	0.3738	0.0538
	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1721)	(0.0220)	(0.2358)	(0.0338)
Elite U.S. University	0.9066***	0.1363***	-0.5317	-0.0238	-0.0506	-0.0065	0.1629	0.0234
	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2283)	(0.0292)	(0.3360)	(0.0484)
Elite U.S. Liberal Arts College	0.3268	0.0491	0.3253	0.0146	-0.0812	-0.0104	0.1414	0.0204
	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3863)	(0.0495)	(0.5521)	(0.0795)
Elite Foreign University	0.0518	0.0078	-0.9107	-0.0408	0.0910	0.0117	0.2454	0.0353
	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3413)	(0.0437)	(0.4927)	(0.0709)
Other Foreign University	0.1161	0.0174	0.2293	0.0103	0.2023	0.0259	0.2550	0.0367
	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2140)	(0.0274)	(0.3455)	(0.0497)
Missing Undergraduate Institution	-0.4337**	-0.0652**	0.0505	0.0023	0.1941	0.0249	0.1146	0.0165
	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2292)	(0.0293)	(0.3847)	(0.0554)
Graduate Degree	1.0229***	0.1537***	-0.3419	-0.0153	-0.1377	-0.0176	-0.3618*	-0.0521*
	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1312)	(0.0168)	(0.2120)	(0.0302)
Work Experience	0.7870***	0.1183***	0.5914*	0.0265*	0.3144**	0.0403**	0.0458	0.0066
	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1361)	(0.0173)	(0.2252)	(0.0324)
#Professor Recommenders	-0.0211	-0.0032	-0.1896	-0.0085	-0.0570	-0.0073	-0.0215	-0.0031
	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0846)	(0.0108)	(0.1331)	(0.0192)
#Prolific Recommenders	0.1325***	0.0199***	0.1903	0.0085	0.0559	0.0072	0.0961	0.0138
	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0574)	(0.0073)	(0.0843)	(0.0121)
Missing GRE	-1.8142***	-0.2727***	1.6533	0.0740	-0.0087	-0.0011	-0.3074	-0.0443
	(0.5268)	(0.0772)	(1.8091)	(0.0811)	(0.3716)	(0.0476)	(0.9426)	(0.1357)
Missing TOEFL/IELTS	0.0947	0.0142	0.4511	0.0202	-0.0080	-0.0010	-0.2725	-0.0392
	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1419)	(0.0182)	(0.2221)	(0.0317)
Committee Score			3.7089***	0.1660***			0.0176	0.0025
			(0.3984)	(0.0065)			(0.0510)	(0.0073)
#Math Courses			-0.2494	-0.0112			-0.0676	-0.0097
			(0.2386)	(0.0108)			(0.1076)	(0.0155)
Advanced Math			1.1003	0.0492			-0.1678	-0.0242
			(0.7916)	(0.0347)			(0.3465)	(0.0498)
Letters Length/100			0.0770	0.0034			-0.0252	-0.0036
			(0.1080)	(0.0048)			(0.0431)	(0.0062)
Letters Terms (%) PC1			0.2451	0.0110			0.1496	0.0215
			(0.2489)	(0.0112)			(0.1082)	(0.0155)
Letters Terms (%) PC2			-0.3767	-0.0169			-0.0395	-0.0057
			(0.3450)	(0.0153)			(0.1470)	(0.0212)
CV Length/100			0.0391	0.0017			0.0778	0.0112
			(0.1060)	(0.0047)			(0.0700)	(0.0100)
CV Coding			-0.2228*	-0.0100*			0.0863	0.0124
			(0.1301)	(0.0060)			(0.0788)	(0.0113)
CV Publication/Working Paper			0.7611	0.0341			0.1054	0.0152
			(0.8128)	(0.0364)			(0.3992)	(0.0575)
Essay Length/100			0.2047	0.0092			-0.0987	-0.0142
			(0.3483)	(0.0157)			(0.0948)	(0.0136)
Essay Specific Professor			0.4125	0.0185			0.0869	0.0125
			(0.6138)	(0.0275)			(0.3148)	(0.0453)
Essay Economics Terms (%)			0.0154	0.0007			-0.1091	-0.0157
			(0.0814)	(0.0036)			(0.0736)	(0.0105)
Essay Research Terms (%)			-0.0751	-0.0034			0.0033	0.0005
			(0.1492)	(0.0067)			(0.0771)	(0.0111)
Essay Positive Terms (%)			-0.0689	-0.0031			0.0948	0.0137
			(0.0968)	(0.0043)			(0.0589)	(0.0084)
Essay Negative Terms (%)			-0.0863	-0.0039			-0.1146	-0.0165
			(0.1368)	(0.0061)			(0.0779)	(0.0111)
Missing Application Pdf			0.2614	0.0117			0.3292	0.0474
			(0.7005)	(0.0311)			(0.3321)	(0.0477)
Program Rank					-0.0084***	-0.0011***	-0.0093***	-0.0013***
					(0.0020)	(0.0003)	(0.0033)	(0.0005)
Missing Program Rank					-3.1385***	-0.4018***	-3.3225***	-0.4784***
					(0.4051)	(0.0517)	(0.6860)	(0.0991)
Obs | Param | LL | AIC	2329	132	-2501.6	5267.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	-0.8795***	29.33%	-0.5593***	36.37%	-1.1693***	-0.9173**	
	(0.1627)	(23.18%, 36.34%)	(0.1850)	(28.46%, 45.10%)	(0.4372)	(0.4528)	
2016	-0.0075	29.17%	0.3004	43.56%	-0.0151	0.2884	
	(0.2468)	(22.20%, 37.28%)	(0.2694)	(34.01%, 53.62%)	(0.2479)	(0.2705)	
2017	0.3491*	37.04%	0.4187*	46.49%	0.3460*	0.3888	
	(0.1962)	(32.09%, 42.28%)	(0.2507)	(37.65%, 55.56%)	(0.1963)	(0.2507)	
2018	0.3979**	38.19%	-0.0958	34.19%	0.4378**	-0.1814	
	(0.1853)	(34.05%, 42.50%)	(0.2524)	(26.12%, 43.29%)	(0.1859)	(0.2502)	
2019	0.2181	34.04%	0.2241	41.70%	0.2264	0.1532	
	(0.2016)	(28.67%, 39.85%)	(0.2499)	(33.20%, 50.72%)	(0.2017)	(0.2511)	
Winsorized Age					0.0016	0.0001	
					(0.0011)	(0.0014)	
Female					0.0104	0.0216	
					(0.0070)	(0.0149)	
U.S.					-0.0030	0.0026	
					(0.0052)	(0.0119)	
International Asian					-0.0005	-0.0264	
					(0.0036)	(0.0175)	
GRE Verbal Percentile					0.0008	0.0001	
					(0.0005)	(0.0003)	
GRE Quantitative Percentile					0.0049	0.0009	
					(0.0031)	(0.0009)	
GRE Analytic Percentile					0.0004	0.0001	
					(0.0003)	(0.0002)	
TOEFL/IELTS					-0.0008	-0.0025	
					(0.0041)	(0.0129)	
Elite U.S. University					0.0222	-0.0116	
					(0.0150)	(0.0165)	
Elite U.S. Liberal Arts College					0.0077	0.0082	
					(0.0102)	(0.0197)	
Elite Foreign University					0.0011	-0.0195	
					(0.0076)	(0.0248)	
Other Foreign University					0.0033	0.0064	
					(0.0055)	(0.0165)	
Missing Undergraduate Institution					-0.0105	0.0020	
					(0.0078)	(0.0158)	
Graduate Degree					0.0270	-0.0091	
					(0.0170)	(0.0106)	
Work Experience					0.0207	0.0138	
					(0.0136)	(0.0111)	
#Professor Recommenders					-0.0004	-0.0047	
					(0.0022)	(0.0065)	
#Prolific Recommenders					0.0034	0.0041	
					(0.0025)	(0.0048)	
Missing GRE					-0.0497	0.0385	
					(0.0346)	(0.0471)	
Missing TOEFL/IELTS					0.0026	0.0100	
					(0.0037)	(0.0111)	
Committee Score						0.0859*	
						(0.0492)	
#Math Courses						-0.0061	
						(0.0064)	
Advanced Math						0.0257	
						(0.0225)	
Letters Length/100						0.0017	
						(0.0026)	
Letters Terms (%) PC1						0.0059	
						(0.0067)	
Letters Terms (%) PC2						-0.0088	
						(0.0095)	
CV Length/100						0.0010	
						(0.0024)	
CV Coding						-0.0050	
						(0.0042)	
CV Publication/Working Paper						0.0182	
						(0.0223)	
Essay Length/100						0.0046	
						(0.0081)	
Essay Specific Professor						0.0088	
						(0.0154)	
Essay Economics Terms (%)						0.0001	
						(0.0019)	
Essay Research Terms (%)						-0.0017	
						(0.0036)	
Essay Positive Terms (%)						-0.0014	
						(0.0023)	
Essay Negative Terms (%)						-0.0020	
						(0.0035)	
Missing Application Pdf						0.0063	
						(0.0167)	
Program Rank					-0.0085***	-0.0083***	
					(0.0021)	(0.0030)	
Missing Program Rank					-3.0033***	-3.2457***	
					(0.4079)	(0.6738)	
Sigma							0.0056
							(0.0035)
Obs | Param | LL | AIC	2329	79	-2528.4	5214.8			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates

A4: Economics Department Myopic Estimates

Table 9: Economics Department Myopic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	4.0405***	-23.0601***	-3.4656***	-28.5204***	-1.2762***	-3.7142	-0.5347
	(0.9990)	(2.0491)	(0.2371)	(6.9074)	(0.2933)	(3.3294)	(0.4783)
2016	0.1701	-0.2896	-0.0435	-0.5295	-0.0237	0.5121	0.0737
	(0.1966)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.3233)	(0.0460)
2017	0.4355**	-0.1533	-0.0230	-1.3366**	-0.0598**	0.6368*	0.0917**
	(0.2214)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.3294)	(0.0466)
2018	0.8410***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0600	0.0086
	(0.1873)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.3133)	(0.0451)
2019	0.7560***	0.2905	0.0437	-3.0764***	-0.1377***	0.3498	0.0504
	(0.1987)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.3214)	(0.0459)
Winsorized Age		0.0628***	0.0094***	-0.0134	-0.0006	0.0799**	0.0115**
		(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0351)	(0.0050)
Female		0.3920***	0.0589***	0.8909**	0.0399***	0.1872	0.0270
		(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1856)	(0.0267)
U.S.		-0.1124	-0.0169	0.0292	0.0013	0.6453*	0.0929*
		(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.3470)	(0.0494)
International Asian		0.0087	0.0013	-1.1484***	-0.0514***	-0.0233	-0.0033
		(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.2426)	(0.0349)
GRE Verbal Percentile		0.0326***	0.0049***	0.0039	0.0002	-0.0005	-0.0001
		(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0062)	(0.0009)
GRE Quantitative Percentile		0.1829***	0.0275***	0.0402	0.0018	-0.0064	-0.0009
		(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0238)	(0.0034)
GRE Analytic Percentile		0.0152***	0.0023***	0.0078	0.0004	-0.0064	-0.0009
		(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0043)	(0.0006)
TOEFL/IELTS		-0.0097	-0.0015	-0.1290	-0.0058	0.3738	0.0538
		(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.2358)	(0.0338)
Elite U.S. University		0.9066***	0.1363***	-0.5318	-0.0238	0.1629	0.0234
		(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.3360)	(0.0484)
Elite U.S. Liberal Arts College		0.3268	0.0491	0.3252	0.0146	0.1414	0.0204
		(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.5521)	(0.0795)
Elite Foreign University		0.0518	0.0078	-0.9106	-0.0407	0.2454	0.0353
		(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.4927)	(0.0709)
Other Foreign University		0.1161	0.0174	0.2293	0.0103	0.2550	0.0367
		(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.3455)	(0.0497)
Missing Undergraduate Institution		-0.4337**	-0.0652**	0.0504	0.0023	0.1146	0.0165
		(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.3847)	(0.0554)
Graduate Degree		1.0229***	0.1537***	-0.3418	-0.0153	-0.3618*	-0.0521*
		(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.2120)	(0.0302)
Work Experience		0.7870***	0.1183***	0.5914*	0.0265*	0.0458	0.0066
		(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.2252)	(0.0324)
#Professor Recommenders		-0.0211	-0.0032	-0.1896	-0.0085	-0.0215	-0.0031
		(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.1331)	(0.0192)
#Prolific Recommenders		0.1325***	0.0199***	0.1903	0.0085	0.0961	0.0138
		(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0843)	(0.0121)
Missing GRE		-1.8142***	-0.2727***	1.6530	0.0740	-0.3074	-0.0443
		(0.5268)	(0.0772)	(1.8090)	(0.0811)	(0.9426)	(0.1357)
Missing TOEFL/IELTS		0.0947	0.0142	0.4510	0.0202	-0.2725	-0.0392
		(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.2221)	(0.0317)
Committee Score				3.7088***	0.1660***	0.0176	0.0025
				(0.3984)	(0.0065)	(0.0510)	(0.0073)
#Math Courses	-0.0804			-0.2494	-0.0112	-0.0676	-0.0097
	(0.0520)			(0.2386)	(0.0108)	(0.1076)	(0.0155)
Advanced Math	0.3560*			1.1003	0.0492	-0.1678	-0.0242
	(0.1909)			(0.7915)	(0.0347)	(0.3465)	(0.0498)
Letters Length/100	0.1225***			0.0770	0.0034	-0.0252	-0.0036
	(0.0242)			(0.1080)	(0.0048)	(0.0431)	(0.0062)
Letters Terms (%) PC1	0.0974*			0.2451	0.0110	0.1496	0.0215
	(0.0536)			(0.2488)	(0.0112)	(0.1082)	(0.0155)
Letters Terms (%) PC2	-0.0925			-0.3767	-0.0169	-0.0395	-0.0057
	(0.0856)			(0.3450)	(0.0153)	(0.1470)	(0.0212)
CV Length/100	0.0588*			0.0391	0.0017	0.0778	0.0112
	(0.0323)			(0.1060)	(0.0047)	(0.0700)	(0.0100)
CV Coding	0.0551			-0.2228*	-0.0100*	0.0863	0.0124
	(0.0417)			(0.1301)	(0.0060)	(0.0788)	(0.0113)
CV Publication/Working Paper	0.5324***			0.7610	0.0341	0.1054	0.0152
	(0.1975)			(0.8127)	(0.0364)	(0.3992)	(0.0575)
Essay Length/100	0.0763			0.2047	0.0092	-0.0987	-0.0142
	(0.0567)			(0.3483)	(0.0157)	(0.0948)	(0.0136)
Essay Specific Professor	0.1442			0.4124	0.0185	0.0869	0.0125
	(0.1767)			(0.6138)	(0.0275)	(0.3148)	(0.0453)
Essay Economics Terms (%)	0.0042			0.0154	0.0007	-0.1091	-0.0157
	(0.0314)			(0.0814)	(0.0036)	(0.0736)	(0.0105)
Essay Research Terms (%)	0.0304			-0.0752	-0.0034	0.0033	0.0005
	(0.0446)			(0.1492)	(0.0067)	(0.0771)	(0.0111)
Essay Positive Terms (%)	0.0175			-0.0689	-0.0031	0.0948	0.0137
	(0.0373)			(0.0968)	(0.0043)	(0.0589)	(0.0084)
Essay Negative Terms (%)	0.0429			-0.0863	-0.0039	-0.1146	-0.0165
	(0.0460)			(0.1368)	(0.0061)	(0.0779)	(0.0111)
Missing Application Pdf	-0.4006**			0.2614	0.0117	0.3292	0.0474
	(0.1850)			(0.7005)	(0.0311)	(0.3321)	(0.0477)
Program Rank						-0.0093***	-0.0013***
						(0.0033)	(0.0005)
Missing Program Rank						-3.3225***	-0.4784***
						(0.6860)	(0.0991)
$\sqrt{MSE}$	1.7525***						
	(0.0403)						
Obs | Param | LL | AIC	2329	296	-14007.8	28607.5			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	3.5032***	-1.0719***	0.3424	-0.6968***	33.25%	-1.2920**	
	(1.2668)	(0.1515)	(0.2544, 0.4607)	(0.2303)	(24.08%, 43.90%)	(0.5375)	
2016	0.0794	0.1267	0.3886	0.1977	37.78%	0.1867	
	(0.2040)	(0.2580)	(0.2616, 0.5772)	(0.4150)	(23.97%, 53.90%)	(0.4143)	
2017	0.6881***	0.3114*	0.4674	0.5092*	45.32%	0.4878*	
	(0.2283)	(0.1671)	(0.4104, 0.5324)	(0.2687)	(39.09%, 51.71%)	(0.2707)	
2018	1.0665***	0.0059	0.3444	0.0142	33.57%	-0.0389	
	(0.1782)	(0.1860)	(0.2772, 0.4279)	(0.2844)	(26.49%, 41.47%)	(0.2888)	
2019	0.7839***	0.1487	0.3972	0.2375	38.72%	0.1780	
	(0.2074)	(0.2024)	(0.3060, 0.5157)	(0.3235)	(29.02%, 49.40%)	(0.3262)	
Winsorized Age						0.0009	
						(0.0006)	
Female						0.0119*	
						(0.0065)	
U.S.						0.0010	
						(0.0044)	
International Asian						-0.0067	
						(0.0048)	
GRE Verbal Percentile						0.0007*	
						(0.0004)	
GRE Quantitative Percentile						0.0035*	
						(0.0018)	
GRE Analytic Percentile						0.0003*	
						(0.0002)	
TOEFL/IELTS						0.0005	
						(0.0037)	
Elite U.S. University						0.0097	
						(0.0069)	
Elite U.S. Liberal Arts College						0.0055	
						(0.0077)	
Elite Foreign University						-0.0011	
						(0.0069)	
Other Foreign University						0.0059	
						(0.0055)	
Missing Undergraduate Institution						-0.0032	
						(0.0049)	
Graduate Degree						0.0177*	
						(0.0092)	
Work Experience						0.0167*	
						(0.0089)	
#Professor Recommenders						-0.0012	
						(0.0020)	
#Prolific Recommenders						0.0013	
						(0.0015)	
Missing GRE						-0.0395	
						(0.0277)	
Missing TOEFL/IELTS						0.0009	
						(0.0031)	
Committee Score						0.0614**	
						(0.0311)	
#Math Courses	-0.0602					-0.0010	
	(0.0688)					(0.0066)	
Advanced Math	0.3634					0.0026	
	(0.2725)					(0.0149)	
Letters Length/100	0.1461***					0.0020	
	(0.0317)					(0.0027)	
Letters Terms (%) PC1	0.1129					0.0016	
	(0.0786)					(0.0084)	
Letters Terms (%) PC2	-0.1002					-0.0061	
	(0.1155)					(0.0085)	
CV Length/100	0.0620					0.0018	
	(0.0408)					(0.0025)	
CV Coding	0.0448					-0.0047	
	(0.0577)					(0.0035)	
CV Publication/Working Paper	0.7302**					-0.0012	
	(0.2876)					(0.0144)	
Essay Length/100	0.1332					0.0074	
	(0.0855)					(0.0119)	
Essay Specific Professor	0.1790					0.0039	
	(0.3042)					(0.0199)	
Essay Economics Terms (%)	-0.0022					-0.0017	
	(0.0395)					(0.0020)	
Essay Research Terms (%)	0.0632					0.0020	
	(0.0598)					(0.0041)	
Essay Positive Terms (%)	0.0394					0.0008	
	(0.0475)					(0.0031)	
Essay Negative Terms (%)	0.0458					-0.0007	
	(0.0575)					(0.0046)	
Missing Application Pdf	-0.9256***					-0.0324**	
	(0.2410)					(0.0163)	
Program Rank						-0.0084***	
						(0.0031)	
Missing Program Rank						-3.1555***	
						(0.6820)	
$\sqrt{MSE}$	1.8063***						
	(0.0435)						
Sigma							0.0024**
							(0.0012)
Obs | Param | LL | AIC	2329	243	-14072.6	28631.2			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

▸▸ Restricted Dynamic Estimates

A4: Economics Department Matriculation-Dynamic Estimates

Table 10: Economics Department Matriculation-Dynamic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5277***	-1.2765***	-3.7142	-0.5347	21.0916***	3.3374***
	(2.1751)	(2.0492)	(0.2371)	(6.9081)	(0.2933)	(3.3294)	(0.4783)	(6.9868)	(1.0163)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.5121	0.0737	0.3206	0.0507
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.3233)	(0.0460)	(0.6530)	(0.1030)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.6368*	0.0917**	0.9971	0.1578
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.3294)	(0.0466)	(0.6410)	(0.1008)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0600	0.0086	1.6496***	0.2610***
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.3133)	(0.0451)	(0.5920)	(0.0886)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.3498	0.0504	1.0916*	0.1727*
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.3214)	(0.0459)	(0.6190)	(0.0971)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0799**	0.0115**	-0.1396*	-0.0221*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0351)	(0.0050)	(0.0738)	(0.0113)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	0.1872	0.0270	-0.4562	-0.0722
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1856)	(0.0267)	(0.3550)	(0.0556)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.6453*	0.0929*	0.9160	0.1449
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.3470)	(0.0494)	(0.5700)	(0.0896)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0233	-0.0033	1.1667**	0.1846***
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.2426)	(0.0349)	(0.4531)	(0.0689)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	-0.0005	-0.0001	-0.0043	-0.0007
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0062)	(0.0009)	(0.0115)	(0.0018)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	-0.0064	-0.0009	-0.1358***	-0.0215***
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0238)	(0.0034)	(0.0444)	(0.0064)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0064	-0.0009	-0.0185**	-0.0029**
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0043)	(0.0006)	(0.0087)	(0.0013)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3738	0.0538	-0.5140	-0.0813
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.2358)	(0.0338)	(0.5175)	(0.0807)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	0.1629	0.0234	-0.4342	-0.0687
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.3360)	(0.0484)	(0.5475)	(0.0858)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	0.1414	0.0204	1.0020	0.1586
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.5521)	(0.0795)	(0.7518)	(0.1185)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.2454	0.0353	-0.2759	-0.0437
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.4927)	(0.0709)	(0.9777)	(0.1548)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.2550	0.0367	0.4536	0.0718
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.3455)	(0.0497)	(0.6285)	(0.0992)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1146	0.0165	-0.8208	-0.1299
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.3847)	(0.0554)	(0.6805)	(0.1058)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.3618*	-0.0521*	-0.0653	-0.0103
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.2120)	(0.0302)	(0.4719)	(0.0746)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.0458	0.0066	0.9166**	0.1450**
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.2252)	(0.0324)	(0.4411)	(0.0667)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0215	-0.0031	0.3588	0.0568
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.1331)	(0.0192)	(0.2789)	(0.0433)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0961	0.0138	-0.2845	-0.0450
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0843)	(0.0121)	(0.1875)	(0.0288)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	-0.3074	-0.0443		
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.9426)	(0.1357)		
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.2725	-0.0392	-0.0091	-0.0014
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.2221)	(0.0317)	(0.5591)	(0.0885)
Committee Score				3.7089***	0.1660***	0.0176	0.0025	-0.1637	-0.0259
				(0.3984)	(0.0065)	(0.0510)	(0.0073)	(0.2099)	(0.0336)
#Math Courses	-0.0790			-0.2494	-0.0112	-0.0676	-0.0097	-0.0182	-0.0029
	(0.0519)			(0.2386)	(0.0108)	(0.1076)	(0.0155)	(0.1671)	(0.0265)
Advanced Math	0.2979			1.1003	0.0492	-0.1678	-0.0242	-1.1143**	-0.1763**
	(0.1889)			(0.7916)	(0.0347)	(0.3465)	(0.0498)	(0.5578)	(0.0877)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0252	-0.0036	-0.0744	-0.0118
	(0.0245)			(0.1080)	(0.0048)	(0.0431)	(0.0062)	(0.0711)	(0.0113)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1496	0.0215	0.3997**	0.0632**
	(0.0551)			(0.2489)	(0.0112)	(0.1082)	(0.0155)	(0.1903)	(0.0287)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.0395	-0.0057	0.2897	0.0458
	(0.0823)			(0.3450)	(0.0153)	(0.1470)	(0.0212)	(0.2460)	(0.0388)
CV Length/100	0.0513			0.0391	0.0017	0.0778	0.0112	-0.1109	-0.0175
	(0.0329)			(0.1060)	(0.0047)	(0.0700)	(0.0100)	(0.1105)	(0.0175)
CV Coding	0.0381			-0.2228*	-0.0100*	0.0863	0.0124	-0.2901**	-0.0459**
	(0.0424)			(0.1301)	(0.0060)	(0.0788)	(0.0113)	(0.1176)	(0.0181)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1054	0.0152	-0.3451	-0.0546
	(0.1912)			(0.8128)	(0.0364)	(0.3992)	(0.0575)	(0.5102)	(0.0812)
Essay Length/100	0.0927			0.2047	0.0092	-0.0987	-0.0142	0.0882	0.0140
	(0.0597)			(0.3483)	(0.0157)	(0.0948)	(0.0136)	(0.1386)	(0.0220)
Essay Specific Professor	0.1411			0.4125	0.0185	0.0869	0.0125	0.6812	0.1078
	(0.1744)			(0.6138)	(0.0275)	(0.3148)	(0.0453)	(0.4976)	(0.0773)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.1091	-0.0157	0.2244**	0.0355**
	(0.0312)			(0.0814)	(0.0036)	(0.0736)	(0.0105)	(0.1075)	(0.0169)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0033	0.0005	-0.1904	-0.0301
	(0.0426)			(0.1492)	(0.0067)	(0.0771)	(0.0111)	(0.1440)	(0.0226)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0948	0.0137	-0.0170	-0.0027
	(0.0345)			(0.0968)	(0.0043)	(0.0589)	(0.0084)	(0.1151)	(0.0182)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.1146	-0.0165	-0.0176	-0.0028
	(0.0443)			(0.1368)	(0.0061)	(0.0779)	(0.0111)	(0.1043)	(0.0165)
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.3292	0.0474	-0.7440	-0.1177
	(0.1841)			(0.7005)	(0.0311)	(0.3321)	(0.0477)	(0.5803)	(0.0919)
Program Rank						-0.0093***	-0.0013***		
						(0.0033)	(0.0005)		
Missing Program Rank						-3.3225***	-0.4784***		
						(0.6860)	(0.0991)		
$\sqrt{MSE}$	1.6467***								
	(0.0370)								
Obs | Param | LL | AIC	2329	620	-13796.1	28832.3					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

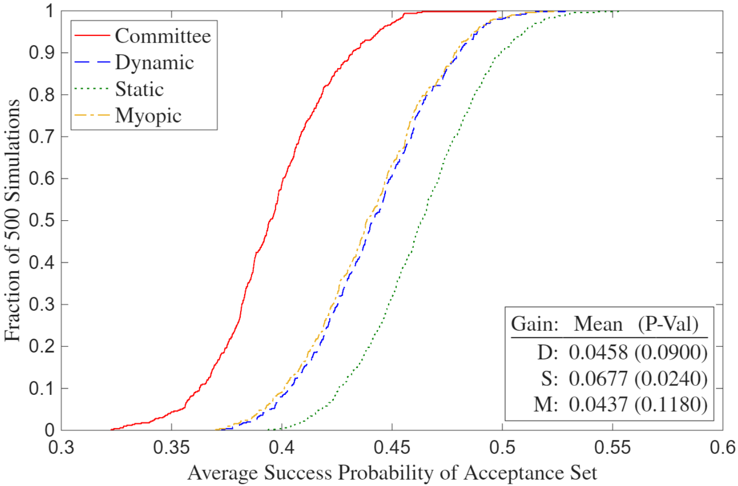
(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.5744***	-5.0258***	0.0066	-0.7845***	31.33%	-0.9724**	2.0177	
	(2.5443)	(0.5433)	(0.0023, 0.0190)	(0.2501)	(21.85%, 42.69%)	(0.4154)	(5.1883)	
2016	0.2797	-0.0027	0.0065	0.4378	41.42%	0.4344	-1.3526*	
	(0.2028)	(0.1805)	(0.0024, 0.0180)	(0.3469)	(30.36%, 53.41%)	(0.3480)	(0.7087)	
2017	0.6354***	0.3991**	0.0098	0.6101**	45.65%	0.5944**	-0.3904	
	(0.2429)	(0.1775)	(0.0036, 0.0267)	(0.2914)	(39.08%, 52.37%)	(0.2914)	(0.8385)	
2018	0.8480***	0.2676	0.0086	0.0302	31.99%	-0.0271	1.2580*	
	(0.1893)	(0.1910)	(0.0032, 0.0231)	(0.3068)	(24.21%, 40.91%)	(0.3129)	(0.7179)	
2019	0.5712***	0.3053	0.0089	0.3696	39.77%	0.3215	0.7374	
	(0.2115)	(0.2124)	(0.0031, 0.0253)	(0.3145)	(31.52%, 48.65%)	(0.3164)	(0.8671)	
Winsorized Age	-0.0572**					0.0000	-0.0328	
	(0.0251)					(0.0010)	(0.0536)	
Female	-0.0789					0.0109	0.0997	
	(0.1146)					(0.0082)	(0.2090)	
U.S.	-0.3691*					-0.0026	0.4429	
	(0.2091)					(0.0078)	(0.4392)	
International Asian	-0.5014***					-0.0092	0.5847*	
	(0.1609)					(0.0078)	(0.3315)	
GRE Verbal Percentile	0.0082**					0.0000	0.0093	
	(0.0037)					(0.0002)	(0.0072)	
GRE Quantitative Percentile	0.0707***					0.0008	-0.0640**	
	(0.0106)					(0.0008)	(0.0321)	
GRE Analytic Percentile	0.0015					0.0001	0.0025	
	(0.0029)					(0.0002)	(0.0063)	
TOEFL/IELTS	0.2866					-0.0022	-0.0613	
	(0.1752)					(0.0098)	(0.2699)	
Elite U.S. University	0.6413***					-0.0090	-0.0833	
	(0.1891)					(0.0138)	(0.3856)	
Elite U.S. Liberal Arts College	-0.3529					0.0121	-0.2907	
	(0.3495)					(0.0166)	(0.6651)	
Elite Foreign University	0.1882					-0.0171	0.4510	
	(0.2890)					(0.0186)	(0.5277)	
Other Foreign University	0.5558***					0.0038	0.2319	
	(0.2053)					(0.0141)	(0.3737)	
Missing Undergraduate Institution	0.5487***					0.0031	-0.3893	
	(0.2110)					(0.0131)	(0.4032)	
Graduate Degree	-0.1898					-0.0061	0.5020	
	(0.1552)					(0.0078)	(0.3403)	
Work Experience	0.0771					0.0074	0.5609	
	(0.1492)					(0.0068)	(0.3613)	
#Professor Recommenders	0.0815					-0.0019	0.2138	
	(0.0822)					(0.0052)	(0.1566)	
#Prolific Recommenders	0.0932					-0.0007	-0.1801*	
	(0.0590)					(0.0032)	(0.1040)	
Missing GRE	-1.3773					0.0129		
	(0.8760)					(0.0183)		
Missing TOEFL/IELTS	0.2646**					0.0019	0.0865	
	(0.1307)					(0.0060)	(0.2435)	
Committee Score						0.0604**	0.5114***	
						(0.0302)	(0.1474)	
#Math Courses	-0.1654					-0.0045	-0.1737	
	(0.1024)					(0.0054)	(0.1598)	
Advanced Math	0.7448*					0.0244	0.1226	
	(0.4190)					(0.0194)	(0.5210)	
Letters Length/100	0.1083**					0.0010	-0.0124	
	(0.0473)					(0.0021)	(0.0644)	
Letters Terms (%) PC1	0.2634***					0.0084	0.4294***	
	(0.0994)					(0.0070)	(0.1635)	
Letters Terms (%) PC2	-0.3064*					-0.0100	-0.2706	
	(0.1751)					(0.0086)	(0.2153)	
CV Length/100	0.0248					0.0003	-0.0592	
	(0.0758)					(0.0022)	(0.0814)	
CV Coding	0.0414					-0.0046	-0.0645	
	(0.1090)					(0.0043)	(0.1016)	
CV Publication/Working Paper	1.3563***					0.0185	-0.2243	
	(0.3391)					(0.0187)	(0.5277)	
Essay Length/100	0.2764*					0.0061	0.1399	
	(0.1467)					(0.0092)	(0.1527)	
Essay Specific Professor	0.3968					0.0116	0.6573	
	(0.3698)					(0.0163)	(0.4045)	
Essay Economics Terms (%)	0.0876					0.0008	0.0836	
	(0.0634)					(0.0021)	(0.0804)	
Essay Research Terms (%)	0.1309					0.0004	-0.1743	
	(0.0843)					(0.0027)	(0.1268)	
Essay Positive Terms (%)	0.0530					-0.0007	0.0065	
	(0.0763)					(0.0023)	(0.0960)	
Essay Negative Terms (%)	0.0804					-0.0017	-0.0464	
	(0.0842)					(0.0035)	(0.1030)	
Missing Application Pdf	-0.3982					0.0113	-1.5898***	
	(0.2791)					(0.0117)	(0.2207)	
Program Rank						-0.0084***		
						(0.0031)		
Missing Program Rank						-3.1674***		
						(0.6814)		
$\sqrt{MSE}$	1.7639***							
	(0.0466)							
Sigma								0.0019**
								(0.0009)
Obs | Param | LL | AIC	2329	567	-13796.0	28726.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

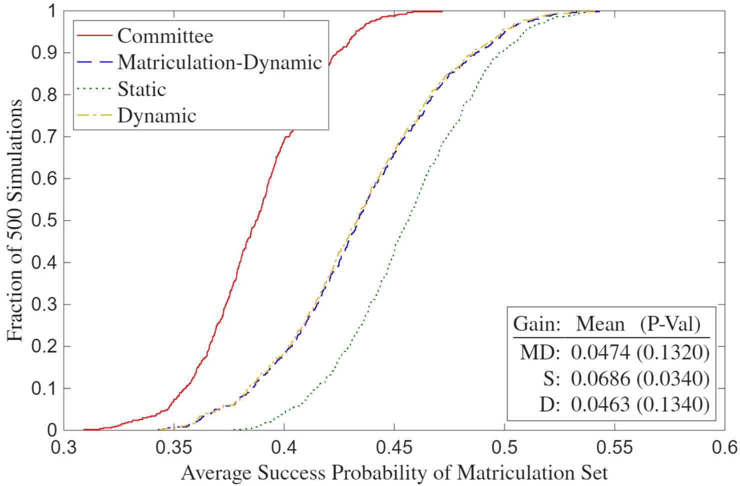
Restricted Matriculation-Dynamic Estimates

A4: Economics Department Deviation Gains



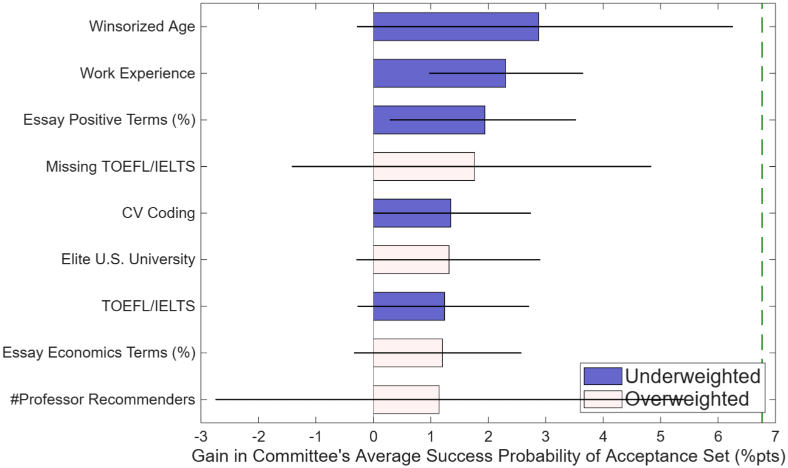
Deviation Gains

A4: Economics Department Matriculation Deviation Gains



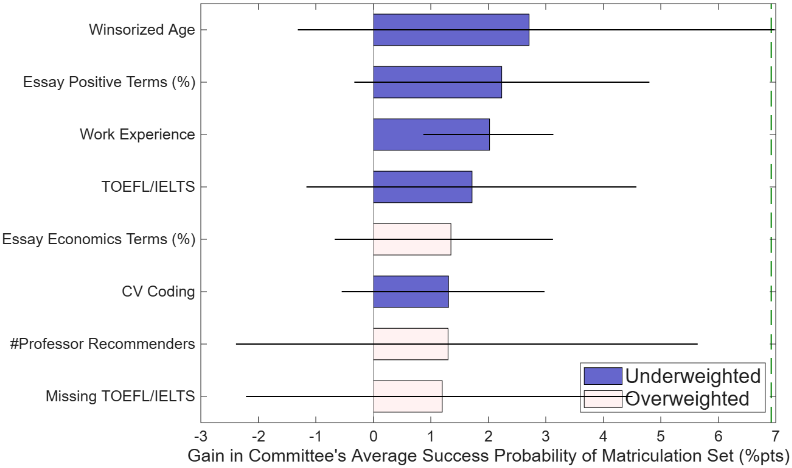
Matriculation Deviation Gains

A4: Economics Department Gains Attribution



Deviation Gains Attribution

A4: Economics Department Matriculation Gains Attribution



Matriculation Deviation Gains Attribution

A4: Economics Department Static Attribution

Table 11: Economics Department Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.8788	25.9910	26.2840	29.9572	29.2918	1.1674	0.9559
Work Experience	2.3065	0.5247	0.7276	0.9844	0.8833	0.5141	0.3116
Essay Positive Terms (%)	1.9410	11.7499	11.6521	12.5376	12.0896	0.7858	0.3882
Missing TOEFL/IELTS	1.7635	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.3478	2.8001	2.7820	3.3399	2.8488	0.6244	0.0748
Elite U.S. University	1.3179	0.1267	0.2335	0.0078	0.1323	-0.6784	-0.3041
TOEFL/IELTS	1.2400	7.4376	7.5675	7.7759	7.5990	0.5957	0.0903
Essay Economics Terms (%)	1.2043	9.7228	9.7423	9.0284	9.5511	-0.6411	-0.1717
#Professor Recommenders	1.1455	2.6230	2.6148	1.3852	2.3191	-1.6680	-0.4012

Deviation Gains Attribution

A4: Economics Department Matriculation-Static Attribution

Table 12: Economics Department Matriculation-Static Attribution

	Gain (% Points)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.7099	25.9910	26.3284	30.9159	29.6161	1.4580	1.0449
Essay Positive Terms (%)	2.2339	11.7499	11.6107	12.9136	12.0979	1.1534	0.4323
Work Experience	2.0220	0.5247	0.7938	0.9949	0.9391	0.4077	0.2910
TOEFL/IELTS	1.7179	7.4376	7.4939	7.8501	7.5082	1.0238	0.0407
Essay Economics Terms (%)	1.3503	9.7228	9.9994	9.0856	9.5668	-0.8271	-0.3885
CV Coding	1.3099	2.8001	2.6661	3.3395	2.7049	0.7618	0.0434
#Professor Recommenders	1.3005	2.6230	2.6171	1.4173	2.3702	-1.6321	-0.3350
Missing TOEFL/IELTS	1.1999	0.6076	0.6858	0.0442	0.4801	-1.3075	-0.4212

Matriculation Deviation Gains Attribution

A4: Graduate School Means by Program

Table 13a: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

A4: Graduate School Means by Program, Accept = 1

Table 13b: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

A4: Graduate School Correlation Matrix

Table 14: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Acceptance, Matriculation, and Success AMEs

A4: Graduate School Logistic Regression Results

Table 15a: Graduate School Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	-0.4332*	-0.5752*	-0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Winsorized Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2971)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5394	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7201)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0199**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7219)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3943	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts	1.1173***	0.6276*	1.0698**	-0.6088	2.3151**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4759	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4972	0.1495	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4781*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1790)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
MissingTOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2677	0.1525	0.1982	0.1871	0.2475
CV Pseudo-R^2	0.2102	0.0720	0.0604	-0.0265	-0.2787

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

Acceptance, Matriculation, and Success AMEs

A4: Graduate School Logistic Regression AMEs

Table 15b: Graduate School Logistic Regression AMEs

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Winsorized Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0334	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0393)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0006**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008**	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0208)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0987**	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0221**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0259*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

Acceptance, Matriculation, and Success AMEs

A4: Graduate School Logistic Regression Results (Synthetic)

Table 16a: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.2028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Winsorized Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.6866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	-0.2650	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230
CV Pseudo-R^2	0.0139	-0.0189	-0.0887	0.0055	-0.0014

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

▸▸ Acceptance, Matriculation, and Success AMEs

A4: Graduate School Logistic Regression AMEs (Synthetic)

Table 16b: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0192	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Winsorized Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0268	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0006)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1281**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0221)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1771***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0692**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0286***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

A5: Simulated Wald Test Results (Same)

Table 17a: Simulated Wald Test Results (Same)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848***	-1.5822***	-28.5524***	-1.3677***	-10.4586***	-1.9105***	-10.2610***	-1.8514***
	(1.1315)	(0.0956)	(1.4963)	(0.0500)	(0.7236)	(0.1070)	(0.4815)	(0.0628)
GRE Quantitative Percentile	0.1509***	0.0146***	0.3245***	0.0155***	0.1058***	0.0193***	0.1154***	0.0208***
	(0.0131)	(0.0012)	(0.0179)	(0.0007)	(0.0088)	(0.0014)	(0.0070)	(0.0011)
Gender	0.4714***	0.0455***	0.7880***	0.0377***	-0.1384	-0.0253	-0.0497	-0.0090
	(0.1475)	(0.0140)	(0.1696)	(0.0081)	(0.1095)	(0.0200)	(0.0895)	(0.0161)
Demographic 1	-0.6287***	-0.0607***	-1.8488***	-0.0886***	-0.3486**	-0.0637**	-0.3079***	-0.0556***
	(0.2088)	(0.0200)	(0.2445)	(0.0115)	(0.1467)	(0.0267)	(0.1123)	(0.0202)
Demographic 2	0.1531	0.0148	0.3703*	0.0177*	0.1756	0.0321	0.0282	0.0051
	(0.2072)	(0.0200)	(0.1929)	(0.0091)	(0.1558)	(0.0284)	(0.1069)	(0.0193)
Advanced Math	0.2889*	0.0279*			0.3267***	0.0597***		
	(0.1612)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0216***	0.3994***	0.0191***	0.1026***	0.0187***	0.0757***	0.0137***
	(0.0413)	(0.0039)	(0.0506)	(0.0023)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			0.8376***	0.0401***			1.9483***	0.3515***
			(0.2297)	(0.0107)			(0.1236)	(0.0205)
Observations | Pseudo-R^2	5000	0.3251			5000	0.1431		
Parameters | LL | AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic | DF | P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *$p<0.10$, **$p<0.05$, ***$p<0.01$.

Acceptance, Matriculation, and Success AMEs

A5: Simulated Wald Test Results (Different)

Table 17b: Simulated Wald Test Results (Different)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291***	-1.5784***	-8.4753***	-0.3642***	-10.4581***	-1.9104***	-3.2236***	-0.5997***
	(1.1274)	(0.0955)	(0.8673)	(0.0412)	(0.7237)	(0.1070)	(0.4018)	(0.0728)
GRE Quantitative Percentile	0.1502***	0.0145***	0.0402***	0.0017***	0.1058***	0.0193***	0.0034	0.0006
	(0.0131)	(0.0012)	(0.0128)	(0.0006)	(0.0088)	(0.0014)	(0.0062)	(0.0012)
Gender	0.4702***	0.0454***	0.5942***	0.0255***	-0.1372	-0.0251	0.0375	0.0070
	(0.1473)	(0.0140)	(0.1820)	(0.0079)	(0.1095)	(0.0200)	(0.0868)	(0.0161)
Demographic 1	-0.6246***	-0.0604***	-0.2541	-0.0109	-0.3544**	-0.0647**	-0.1152	-0.0214
	(0.2085)	(0.0200)	(0.2393)	(0.0103)	(0.1467)	(0.0267)	(0.1103)	(0.0205)
Demographic 2	0.1555	0.0150	0.0081	0.0003	0.1713	0.0313	-0.0562	-0.0105
	(0.2070)	(0.0200)	(0.2109)	(0.0091)	(0.1557)	(0.0284)	(0.1030)	(0.0192)
Advanced Math	0.2886*	0.0279*			0.3284***	0.0600***		
	(0.1611)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0217***	0.3817***	0.0164***	0.1026***	0.0187***	0.1808***	0.0336***
	(0.0413)	(0.0039)	(0.0528)	(0.0025)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			-0.0452	-0.0019			1.1393***	0.2120***
			(0.1995)	(0.0086)			(0.1219)	(0.0219)
Observations | Pseudo-R^2	5000	0.1488			5000	0.0758		
Parameters | LL | AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic | DF | P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance, Matriculation, and Success AMEs

A5: Simulated Unrestricted Dynamic Estimates (Optimal)

Table 18a: Simulated Unrestricted Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9001**	-0.4657**	4.8636***	-11.5885***	-2.2558***	-13.1542***	-2.3735***	-9.5556***	-1.4872***
	(0.7563)	(0.1830)	(1.3238)	(1.0232)	(0.1454)	(2.0077)	(0.2886)	(2.0721)	(0.2688)
GRE Quantitative Percentile	0.0207**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1161***	0.0181***
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0227)	(0.0029)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1289	0.0233	0.1263	0.0197
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.2699)	(0.0419)
Demographic 1	0.3692**	0.0905**	-0.3431	-0.3013	-0.0587	-0.2638	-0.0476	-0.4605	-0.0717
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3743)	(0.0672)	(0.3933)	(0.0608)
Demographic 2	0.0925	0.0227	-0.1806	0.1538	0.0299	0.1708	0.0308	-0.2340	-0.0364
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.4322)	(0.0672)
Advanced Math			0.6870***			0.7736***	0.1396***	0.6276**	0.0977**
			(0.2024)			(0.2535)	(0.0443)	(0.2749)	(0.0417)
Committee Score						0.0783	0.0141	0.0464	0.0072
						(0.0602)	(0.0108)	(0.0655)	(0.0102)
Year Transition More	0.0663	0.0163	0.0932	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0237	-0.0037
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.2763)	(0.0430)
Program Rank								-0.0282***	-0.0044***
								(0.0027)	(0.0003)
$\sqrt{\text{MSE}}$			1.9465***						
			(0.0673)						
Obs | Param | LL | AIC	1000	37	-2408.7	4891.4					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 18b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.8086**	-0.4435**	5.3561***	-11.3342***	-2.1952***	-6.7719***	-1.3374***	-9.1414***	-1.4363***
	(0.7548)	(0.1830)	(1.2694)	(1.0218)	(0.1474)	(1.7405)	(0.3203)	(1.9229)	(0.2518)
GRE Quantitative Percentile	0.0197**	0.0048**	-0.0026	0.1238***	0.0240***	0.0555***	0.0110***	0.1113***	0.0175***
	(0.0090)	(0.0022)	(0.0144)	(0.0118)	(0.0018)	(0.0191)	(0.0036)	(0.0212)	(0.0027)
Gender	-0.2332*	-0.0572*	-0.1025	0.2601*	0.0504*	0.0119	0.0024	0.0519	0.0082
	(0.1309)	(0.0319)	(0.2051)	(0.1472)	(0.0283)	(0.2369)	(0.0468)	(0.2652)	(0.0416)
Demographic 1	0.3717**	0.0911**	-0.4863*	-0.8309***	-0.1609***	-0.3354	-0.0662	-0.5014	-0.0788
	(0.1731)	(0.0421)	(0.2701)	(0.1972)	(0.0371)	(0.3396)	(0.0669)	(0.3621)	(0.0564)
Demographic 2	0.0905	0.0222	-0.2750	0.0815	0.0158	0.1755	0.0347	-0.4742	-0.0745
	(0.1909)	(0.0468)	(0.2740)	(0.2141)	(0.0415)	(0.3347)	(0.0659)	(0.3825)	(0.0597)
Advanced Math			0.7219***			0.3555	0.0702	0.5330**	0.0837**
			(0.1966)			(0.2376)	(0.0465)	(0.2687)	(0.0416)
Committee Score						0.2969***	0.0586***	0.0597	0.0094
						(0.0593)	(0.0106)	(0.0639)	(0.0100)
Year Transition More	0.0642	0.0158	0.0002	1.1142***	0.2158***	-0.7164***	-0.1415***	0.1143	0.0180
	(0.1280)	(0.0314)	(0.2010)	(0.1484)	(0.0257)	(0.2398)	(0.0455)	(0.2774)	(0.0434)
Program Rank								-0.0280***	-0.0044***
								(0.0028)	(0.0003)
$\sqrt{\text{MSE}}$			1.9186***						
			(0.0682)						
Obs | Param | LL | AIC	1000	37	-2452.7	4979.4					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

▸▸ Conclusion

A5: Simulated Restricted Dynamic Estimates (Optimal)

Table 19a: Simulated Restricted Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-2.1226***	4.7314***	-0.3330***	0.7168	-0.0067	49.83%	-9.5110***	
	(0.7573)	(1.3235)	(0.0802)	(0.6125, 0.8387)	(0.2115)	(39.62%, 60.06%)	(1.9433)	
GRE Quantitative Percentile	0.0235***	0.0024					0.1134***	
	(0.0090)	(0.0150)					(0.0214)	
Gender	-0.2151*	-0.0652					0.2238*	
	(0.1305)	(0.2113)					(0.1229)	
Demographic 1	0.3587**	-0.3509					-0.3323*	
	(0.1725)	(0.2903)					(0.1836)	
Demographic 2	0.0940	-0.1813					0.0172	
	(0.1905)	(0.3010)					(0.1773)	
Advanced Math		0.6919***					0.5658***	
		(0.2026)					(0.1714)	
Committee Score							0.0496	
							(0.0377)	
Year Transition More	0.0730	0.0954	-0.0244	0.6995	0.5755*	63.85%	0.0619	
	(0.1277)	(0.2069)	(0.0931)	(0.5948, 0.8227)	(0.3212)	(52.94%, 73.50%)	(0.2670)	
Program Rank							-0.0282***	
							(0.0027)	
$\sqrt{\text{MSE}}$		1.9468***						
		(0.0673)						
Sigma								0.1367***
								(0.0247)
Obs | Param | LL | AIC	1000	28	-2435.7	4927.5				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 19b: Simulated Restricted Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	L(Cutoff)$_1$	Cutoff 2	P(Cutoff)$_2$	Success	Sigma
Intercept	-2.1619***	4.4213***	-0.3434***	0.7094	-0.0905	47.74%	-7.5239***	
	(0.7594)	(1.2675)	(0.1028)	(0.5800, 0.8676)	(0.2232)	(37.10%, 58.59%)	(2.2918)	
GRE Quantitative Percentile	0.0240***	0.0086					0.0852***	
	(0.0090)	(0.0144)					(0.0245)	
Gender	-0.2194*	-0.0603					0.1567	
	(0.1307)	(0.2029)					(0.1225)	
Demographic 1	0.3468**	-0.5636**					-0.4967**	
	(0.1730)	(0.2718)					(0.2081)	
Demographic 2	0.1009	-0.2620					-0.0131	
	(0.1908)	(0.2740)					(0.1789)	
Advanced Math		0.7335***					0.3527**	
		(0.1973)					(0.1705)	
Committee Score							0.1726***	
							(0.0366)	
Year Transition More	0.0693	0.0080	-0.0615	0.6670	0.5395	61.04%	0.1139	
	(0.1278)	(0.2011)	(0.0978)	(0.5507, 0.8080)	(0.3422)	(49.37%, 71.57%)	(0.2774)	
Program Rank							-0.0279***	
							(0.0027)	
$\sqrt{\text{MSE}}$		1.9212***						
		(0.0688)						
Sigma								0.1458***
								(0.0395)
Obs | Param | LL | AIC	1000	28	-2497.3	5050.5				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Unrestricted Static Estimates (Optimal)

Table 20a: Simulated Unrestricted Static Estimates (Optimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5886***	-2.2559***	-13.1494***	-2.3728***	-8.1224***	-1.1660***	-9.5306***	-1.4839***
	(1.0232)	(0.1454)	(2.0072)	(0.2886)	(1.1371)	(0.1344)	(2.0699)	(0.2689)
GRE Quantitative Percentile	0.1235***	0.0240***	0.1371***	0.0247***	0.1078***	0.0155***	0.1158***	0.0180***
	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0133)	(0.0015)	(0.0227)	(0.0029)
Gender	0.2262	0.0440	0.1290	0.0233	0.1396	0.0200	0.1262	0.0197
	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1738)	(0.0249)	(0.2698)	(0.0419)
Demographic 1	-0.2995	-0.0583	-0.2639	-0.0476	-0.2529	-0.0363	-0.4577	-0.0713
	(0.1988)	(0.0386)	(0.3743)	(0.0672)	(0.2358)	(0.0338)	(0.3931)	(0.0608)
Demographic 2	0.1552	0.0302	0.1711	0.0309	-0.1180	-0.0169	-0.2318	-0.0361
	(0.2215)	(0.0431)	(0.3843)	(0.0693)	(0.2580)	(0.0371)	(0.4320)	(0.0672)
Advanced Math			0.7733***	0.1396***			0.6278**	0.0978**
			(0.2534)	(0.0443)			(0.2748)	(0.0417)
Committee Score			0.0781	0.0141			0.0462	0.0072
			(0.0602)	(0.0108)			(0.0655)	(0.0102)
Year Transition More	0.9837***	0.1915***	-0.9252***	-0.1670***	-0.0592	-0.0085	-0.0239	-0.0037
	(0.1472)	(0.0263)	(0.2512)	(0.0427)	(0.1689)	(0.0242)	(0.2762)	(0.0430)
Program Rank					-0.0310***	-0.0045***	-0.0282***	-0.0044***
					(0.0019)	(0.0002)	(0.0027)	(0.0003)
Obs | Param | LL | AIC	1000	30	-1375.9	2811.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Unrestricted Static Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Static Estimates (Suboptimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2297***	-2.1801***	-6.7297***	-1.3296***	-8.2833***	-1.1914***	-9.6070***	-1.4986***
	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.1393)	(0.1341)	(1.9611)	(0.2508)
GRE Quantitative Percentile	0.1226***	0.0238***	0.0550***	0.0109***	0.1097***	0.0158***	0.1163***	0.0181***
	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0133)	(0.0015)	(0.0216)	(0.0027)
Gender	0.2576*	0.0500*	0.0097	0.0019	0.1355	0.0195	0.0584	0.0091
	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1741)	(0.0250)	(0.2669)	(0.0416)
Demographic 1	-0.8291***	-0.1610***	-0.3339	-0.0660	-0.2777	-0.0399	-0.5157	-0.0804
	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2344)	(0.0337)	(0.3648)	(0.0563)
Demographic 2	0.0798	0.0155	0.1762	0.0348	-0.1808	-0.0260	-0.4742	-0.0740
	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2566)	(0.0369)	(0.3857)	(0.0598)
Advanced Math			0.3581	0.0708			0.5369**	0.0837**
			(0.2375)	(0.0465)			(0.2704)	(0.0416)
Committee Score			0.2962***	0.0585***			0.0616	0.0096
			(0.0592)	(0.0106)			(0.0643)	(0.0100)
Year Transition More	1.1109***	0.2157***	-0.7185***	-0.1419***	-0.0766	-0.0110	0.1314	0.0205
	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1688)	(0.0243)	(0.2795)	(0.0434)
Program Rank					-0.0304***	-0.0044***	-0.0279***	-0.0044***
					(0.0019)	(0.0002)	(0.0028)	(0.0003)
Obs | Param | LL | AIC	1000	30	-1404.1	2868.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Restricted Static Estimates (Optimal)

Table 21a: Simulated Restricted Static Estimates (Optimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.6884***	66.56%	0.1530	53.82%	-8.0150***	-9.7263***	
	(0.1830)	(58.17%, 74.02%)	(0.2411)	(42.08%, 65.15%)	(1.1629)	(1.7366)	
GRE Quantitative Percentile					0.1057***	0.1158***	
					(0.0134)	(0.0193)	
Gender					0.1987*	0.1142	
					(0.1015)	(0.1673)	
Demographic 1					-0.2471*	-0.3338	
					(0.1346)	(0.2403)	
Demographic 2					0.0406	-0.0047	
					(0.1504)	(0.2423)	
Advanced Math						0.6315***	
						(0.1766)	
Committee Score						0.0546	
						(0.0421)	
Year Transition More	-0.8466***	46.05%	0.7965**	72.10%	-0.0628	0.0285	
	(0.2184)	(39.60%, 52.64%)	(0.3769)	(58.90%, 82.34%)	(0.1667)	(0.2701)	
Program Rank					-0.0310***	-0.0283***	
					(0.0019)	(0.0027)	
Sigma							0.1913***
							(0.0252)
Obs | Param | LL | AIC	1000	21	-1376.5	2795.0			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Static Estimates (Suboptimal)

Table 21b: Simulated Restricted Static Estimates (Suboptimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.7340***	67.57%	0.1682	54.20%	-8.0582***	-6.3578***	
	(0.2062)	(58.17%, 75.73%)	(0.2593)	(41.58%, 66.29%)	(1.2803)	(1.7689)	
GRE Quantitative Percentile					0.1079***	0.0727***	
					(0.0148)	(0.0189)	
Gender					0.2185**	0.0409	
					(0.1042)	(0.1610)	
Demographic 1					-0.5505***	-0.3739	
					(0.1395)	(0.2380)	
Demographic 2					-0.0151	-0.0921	
					(0.1505)	(0.2356)	
Advanced Math						0.3897**	
						(0.1686)	
Committee Score						0.1754***	
						(0.0378)	
Year Transition More	-0.9965***	43.47%	0.6110	68.55%	-0.0772	0.0376	
	(0.2396)	(37.12%, 50.05%)	(0.3768)	(55.51%, 79.20%)	(0.1671)	(0.2807)	
Program Rank					-0.0305***	-0.0281***	
					(0.0019)	(0.0027)	
Sigma							0.2001***
							(0.0325)
Obs | Param | LL | AIC	1000	21	-1417.0	2875.9			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Unrestricted Myopic Estimates (Optimal)

Table 22a: Simulated Unrestricted Myopic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.6669***	-11.5730***	-2.2535***	-12.9989***	-2.3525***	-9.3999***	-1.4664***
	(0.0894)	(0.0224)	(0.1940)	(1.0222)	(0.1454)	(1.9897)	(0.2885)	(2.0583)	(0.2690)
GRE Quantitative Percentile				0.1234***	0.0240***	0.1355***	0.0245***	0.1145***	0.0179***
				(0.0117)	(0.0018)	(0.0221)	(0.0033)	(0.0226)	(0.0029)
Gender				0.2264	0.0441	0.1267	0.0229	0.1249	0.0195
				(0.1463)	(0.0283)	(0.2495)	(0.0451)	(0.2693)	(0.0419)
Demographic 1				-0.3011	-0.0586	-0.2603	-0.0471	-0.4657	-0.0727
				(0.1987)	(0.0386)	(0.3731)	(0.0672)	(0.3923)	(0.0608)
Demographic 2				0.1542	0.0300	0.1699	0.0307	-0.2420	-0.0378
				(0.2214)	(0.0431)	(0.3832)	(0.0693)	(0.4309)	(0.0672)
Advanced Math			0.6763***			0.7710***	0.1395***	0.6254**	0.0976**
			(0.2006)			(0.2527)	(0.0443)	(0.2743)	(0.0417)
Committee Score						0.0772	0.0140	0.0454	0.0071
						(0.0599)	(0.0108)	(0.0654)	(0.0102)
Year Transition More	0.0480	0.0120	0.1022	0.9845***	0.1917***	-0.9255***	-0.1675***	-0.0272	-0.0042
	(0.1265)	(0.0316)	(0.2069)	(0.1472)	(0.0263)	(0.2505)	(0.0427)	(0.2757)	(0.0430)
Program Rank								-0.0282***	-0.0044***
								(0.0027)	(0.0003)
$\sqrt{\text{MSE}}$			1.9512***						
			(0.0674)						
Obs | Param | LL | AIC	1000	29	-2419.4	4896.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Myopic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.7588***	-11.2089***	-2.1776***	-6.7003***	-1.3243***	-9.4427***	-1.4775***
	(0.0894)	(0.0224)	(0.1881)	(1.0127)	(0.1474)	(1.7354)	(0.3201)	(1.9470)	(0.2511)
GRE Quantitative Percentile				0.1224***	0.0238***	0.0547***	0.0108***	0.1144***	0.0179***
				(0.0117)	(0.0018)	(0.0190)	(0.0036)	(0.0214)	(0.0027)
Gender				0.2568*	0.0499*	0.0094	0.0019	0.0567	0.0089
				(0.1467)	(0.0283)	(0.2367)	(0.0468)	(0.2662)	(0.0416)
Demographic 1				-0.8281***	-0.1609***	-0.3330	-0.0658	-0.5034	-0.0788
				(0.1965)	(0.0371)	(0.3392)	(0.0669)	(0.3636)	(0.0564)
Demographic 2				0.0791	0.0154	0.1763	0.0348	-0.4606	-0.0721
				(0.2134)	(0.0414)	(0.3344)	(0.0659)	(0.3842)	(0.0598)
Advanced Math			0.7040***			0.3576	0.0707	0.5331**	0.0834**
			(0.1957)			(0.2374)	(0.0465)	(0.2696)	(0.0416)
Committee Score						0.2958***	0.0585***	0.0611	0.0096
						(0.0592)	(0.0106)	(0.0641)	(0.0100)
Year Transition More	0.0480	0.0120	0.0218	1.1069***	0.2151***	-0.7191***	-0.1421***	0.1276	0.0200
	(0.1265)	(0.0316)	(0.2022)	(0.1477)	(0.0257)	(0.2396)	(0.0455)	(0.2786)	(0.0434)
Program Rank								-0.0279***	-0.0044***
								(0.0028)	(0.0003)
$\sqrt{\text{MSE}}$			1.9289***						
			(0.0684)						
Obs | Param | LL | AIC	1000	29	-2464.5	4987.0					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Restricted Myopic Estimates (Optimal)

Table 23a: Simulated Restricted Myopic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0128	4.6811***	-0.3383***	0.7130	-0.0029	49.93%	-9.4033***	
	(0.0889)	(0.1930)	(0.0804)	(0.6090, 0.8348)	(0.2105)	(39.76%, 60.10%)	(1.9406)	
GRE Quantitative Percentile							0.1124***	
							(0.0214)	
Gender							0.2104*	
							(0.1205)	
Demographic 1							-0.3125*	
							(0.1789)	
Demographic 2							0.0218	
							(0.1733)	
Advanced Math		0.6793***					0.5373***	
		(0.2009)					(0.1692)	
Committee Score							0.0494	
							(0.0373)	
Year Transition More	0.0548	0.0997	-0.0233	0.6966	0.5674*	63.75%	0.0577	
	(0.1261)	(0.2067)	(0.0937)	(0.5914, 0.8204)	(0.3211)	(52.78%, 73.45%)	(0.2664)	
Program Rank							-0.0283***	
							(0.0027)	
$\sqrt{\text{MSE}}$		1.9515***						
		(0.0674)						
Sigma								0.1347***
								(0.0247)
Obs | Param | LL | AIC	1000	20	-2447.1	4934.2				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Myopic Estimates (Suboptimal)

Table 23b: Simulated Restricted Myopic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0122	4.8150***	-0.3290***	0.7196	-0.0983	47.55%	-8.1397***	
	(0.0891)	(0.1883)	(0.1039)	(0.5870, 0.8822)	(0.2266)	(36.76%, 58.56%)	(2.3601)	
GRE Quantitative Percentile							0.0922***	
							(0.0253)	
Gender							0.1605	
							(0.1277)	
Demographic 1							-0.5394**	
							(0.2132)	
Demographic 2							-0.0312	
							(0.1887)	
Advanced Math		0.7051***					0.3544**	
		(0.1963)					(0.1753)	
Committee Score							0.1744***	
							(0.0373)	
Year Transition More	0.0489	0.0008	-0.0616	0.6767	0.5779*	61.77%	0.1304	
	(0.1263)	(0.2020)	(0.0977)	(0.5578, 0.8209)	(0.3484)	(49.80%, 72.45%)	(0.2799)	
Program Rank							-0.0279***	
							(0.0028)	
$\sqrt{\text{MSE}}$		1.9300***						
		(0.0686)						
Sigma								0.1544***
								(0.0406)
Obs | Param | LL | AIC	1000	20	-2510.0	5059.9				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Unrestricted Mat-Dynamic Estimates (Optimal)

Table 24a: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.8946**	-0.4643**	4.8638***	-11.5888***	-2.2558***	-13.1535***	-2.3733***	-9.5541***	-1.4869***	-4.5425	-0.6043
	(0.7562)	(0.1830)	(1.3238)	(1.0233)	(0.1454)	(2.0077)	(0.2886)	(2.0721)	(0.2688)	(4.6438)	(0.6185)
GRE Quantitative Percentile	0.0207**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1161***	0.0181***	0.0587	0.0078
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0227)	(0.0029)	(0.0458)	(0.0061)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1290	0.0233	0.1263	0.0197	0.4869	0.0648
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.2699)	(0.0419)	(0.5061)	(0.0657)
Demographic 1	0.3693**	0.0905**	-0.3431	-0.3014	-0.0587	-0.2639	-0.0476	-0.4605	-0.0717	0.3568	0.0475
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3743)	(0.0672)	(0.3933)	(0.0608)	(0.7460)	(0.1001)
Demographic 2	0.0924	0.0226	-0.1806	0.1537	0.0299	0.1706	0.0308	-0.2340	-0.0364	0.1812	0.0241
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.4322)	(0.0672)	(0.7963)	(0.1069)
Advanced Math			0.6869***			0.7740***	0.1397***	0.6274**	0.0976**	0.8103	0.1078
			(0.2024)			(0.2535)	(0.0443)	(0.2749)	(0.0417)	(0.5770)	(0.0733)
Committee Score						0.0783	0.0141	0.0465	0.0072	-0.7409***	-0.0986***
						(0.0602)	(0.0108)	(0.0655)	(0.0102)	(0.2066)	(0.0235)
Year Transition/Matriculate More	0.0662	0.0162	0.0931	0.9841***	0.1916***	-0.9252***	-0.1669***	-0.0234	-0.0036	3.3262***	0.4425***
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.2763)	(0.0430)	(0.5667)	(0.0330)
Program Rank								-0.0282***	-0.0044***		
								(0.0027)	(0.0003)		
$\sqrt{\text{MSE}}$			1.9465***								
			(0.0673)								
Obs | Param | LL | AIC	1000	45	-2461.1	5012.2							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Unrestricted Mat-Dynamic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9060**	-0.4671**	5.2717***	-11.2295***	-2.1801***	-6.7297***	-1.3296***	-9.5961***	-1.4972***	-1.0400	-0.1351
	(0.7564)	(0.1830)	(1.2701)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.9602)	(0.2508)	(3.6082)	(0.4721)
GRE Quantitative Percentile	0.0208**	0.0051**	-0.0016	0.1226***	0.0238***	0.0550***	0.0109***	0.1162***	0.0181***	0.0129	0.0017
	(0.0090)	(0.0022)	(0.0144)	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0216)	(0.0027)	(0.0391)	(0.0051)
Gender	-0.2334*	-0.0572*	-0.1023	0.2575*	0.0500*	0.0098	0.0019	0.0583	0.0091	0.1297	0.0168
	(0.1309)	(0.0319)	(0.2051)	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.2669)	(0.0416)	(0.4717)	(0.0608)
Demographic 1	0.3690**	0.0904**	-0.4893*	-0.8292***	-0.1610***	-0.3340	-0.0660	-0.5154	-0.0804	0.6324	0.0821
	(0.1732)	(0.0421)	(0.2701)	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.3647)	(0.0563)	(0.6281)	(0.0823)
Demographic 2	0.0925	0.0227	-0.2744	0.0799	0.0155	0.1763	0.0348	-0.4742	-0.0740	-0.2225	-0.0289
	(0.1911)	(0.0468)	(0.2740)	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.3856)	(0.0598)	(0.6864)	(0.0880)
Advanced Math			0.7219***			0.3580	0.0707	0.5366**	0.0837**	0.5877	0.0763
			(0.1966)			(0.2375)	(0.0465)	(0.2703)	(0.0416)	(0.5321)	(0.0674)
Committee Score						0.2962***	0.0585***	0.0616	0.0096	-0.6092***	-0.0791***
						(0.0592)	(0.0106)	(0.0642)	(0.0100)	(0.1637)	(0.0176)
Year Transition/Matriculate More	0.0665	0.0163	0.0031	1.1107***	0.2156***	-0.7185***	-0.1420***	0.1314	0.0205	3.4846***	0.4526***
	(0.1281)	(0.0314)	(0.2010)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.2794)	(0.0434)	(0.6046)	(0.0349)
Program Rank								-0.0279***	-0.0044***		
								(0.0028)	(0.0003)		
$\sqrt{MSE}$			1.9186***								
			(0.0682)								
Obs | Param | LL | AIC	1000	45	-2506.5	5102.9							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Restricted Mat-Dynamic Estimates (Optimal)

Table 25a: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-2.1882***	5.5080***	-1.8385***	0.1591	-0.0309	49.23%	-9.5700***	-9.6624**	
	(0.7585)	(1.3416)	(0.2138)	(0.1046, 0.2419)	(0.2298)	(38.19%, 60.33%)	(1.9922)	(4.0874)	
GRE Quantitative Percentile	0.0243***	-0.0073					0.1149***	0.1166***	
	(0.0090)	(0.0153)					(0.0219)	(0.0397)	
Gender	-0.2211*	-0.0966					0.1535	0.7399	
	(0.1308)	(0.2097)					(0.1421)	(0.4538)	
Demographic 1	0.3601**	-0.3138					-0.3141	0.1062	
	(0.1729)	(0.2879)					(0.2124)	(0.7215)	
Demographic 2	0.0941	-0.1854					0.0161	0.0717	
	(0.1907)	(0.2982)					(0.2065)	(0.7746)	
Advanced Math		0.6491***					0.6118***	1.0516*	
		(0.2033)					(0.1753)	(0.5858)	
Committee Score							0.0257	-0.6841***	
							(0.0395)	(0.2011)	
Year Transition/Matriculate More	0.0704	0.1588	-1.1141***	0.0522	0.6267*	64.47%	0.0841	3.0506***	
	(0.1280)	(0.2057)	(0.2796)	(0.0280, 0.0972)	(0.3398)	(53.11%, 74.40%)	(0.2724)	(0.5403)	
Program Rank							-0.0281***		
							(0.0027)		
$\sqrt{\text{MSE}}$		1.9458***							
		(0.0672)							
Sigma									0.1438***
									(0.0288)
Obs | Param | LL | AIC	1000	36	-2508.0	5088.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Mat-Dynamic Estimates (Suboptimal)

Table 25b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-2.0969***	5.5146***	-1.5785***	0.2063	0.0521	51.30%	-8.0820***	-6.4822*	
	(0.7580)	(1.2768)	(0.3118)	(0.1120, 0.3801)	(0.2631)	(38.62%, 63.82%)	(2.4280)	(3.3848)	
GRE Quantitative Percentile	0.0232**	-0.0048					0.0924***	0.0774**	
	(0.0090)	(0.0145)					(0.0264)	(0.0364)	
Gender	-0.2275*	-0.1097					0.0806	0.3888	
	(0.1310)	(0.2034)					(0.1641)	(0.4605)	
Demographic 1	0.3529**	-0.4632*					-0.5186**	0.0455	
	(0.1730)	(0.2684)					(0.2595)	(0.6524)	
Demographic 2	0.0929	-0.2774					-0.1266	-0.4370	
	(0.1909)	(0.2721)					(0.2575)	(0.6992)	
Advanced Math		0.7098***					0.4414**	0.7348	
		(0.1960)					(0.1976)	(0.5514)	
Committee Score							0.1698***	-0.5422***	
							(0.0393)	(0.1604)	
Year Transition/Matriculate More	0.0723	0.0346	-1.3772***	0.0520	0.5873	65.46%	0.0893	3.1408***	
	(0.1281)	(0.2001)	(0.3779)	(0.0208, 0.1304)	(0.4121)	(50.68%, 77.76%)	(0.2917)	(0.5889)	
Program Rank							-0.0281***		
							(0.0027)		
$\sqrt{\text{MSE}}$		1.9142***							
		(0.0677)							
Sigma									0.1997***
									(0.0597)
Obs | Param | LL | AIC	1000	36	-2574.1	5220.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Unrestricted Wait-Dynamic Estimates (Optimal)

Table 26a: Simulated Unrestricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9049**	-0.4668**	4.8636***	-0.0252	-11.5884***	-2.2558***	-11.9628***	-1.6166***	-13.2707***	-1.6993***	-9.5534***	-1.4869***	-5.8313	-0.6741
	(0.7564)	(0.1830)	(1.3238)	(0.0518)	(1.0232)	(0.1454)	(2.5504)	(0.3154)	(2.4113)	(0.2845)	(2.0719)	(0.2688)	(6.4970)	(0.7706)
GRE Quantative Percentile	0.0208**	0.0051**	0.0008		0.1236***	0.0241***	0.1096***	0.0148***	0.1387***	0.0178***	0.1161***	0.0181***	0.0834	0.0096
	(0.0090)	(0.0022)	(0.0150)		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0264)	(0.0032)	(0.0227)	(0.0029)	(0.0644)	(0.0075)
Gender	-0.2335*	-0.0572*	-0.0783		0.2266	0.0441	0.0233	0.0032	0.3039	0.0389	0.1263	0.0197	0.4650	0.0538
	(0.1309)	(0.0319)	(0.2113)		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3287)	(0.0421)	(0.2699)	(0.0419)	(0.9003)	(0.1052)
Demographic 1	0.3690**	0.0904**	-0.3430		-0.3014	-0.0587	0.5534	0.0748	-0.8831*	-0.1131*	-0.4604	-0.0717	-0.2627	-0.0304
	(0.1732)	(0.0421)	(0.2885)		(0.1988)	(0.0386)	(0.4941)	(0.0665)	(0.4622)	(0.0578)	(0.3932)	(0.0608)	(0.7744)	(0.0880)
Demographic 2	0.0924	0.0226	-0.1805		0.1538	0.0299	0.5136	0.0694	-0.0525	-0.0067	-0.2341	-0.0364	-0.3966	-0.0459
	(0.1911)	(0.0468)	(0.3001)		(0.2215)	(0.0431)	(0.5204)	(0.0699)	(0.4528)	(0.0580)	(0.4322)	(0.0672)	(1.2105)	(0.1393)
Advanced Math			0.6870***				0.5066*	0.0685*	0.8622***	0.1104***	0.6275**	0.0977**	1.1399	0.1318
			(0.2024)				(0.2907)	(0.0390)	(0.3306)	(0.0421)	(0.2749)	(0.0417)	(0.9839)	(0.1080)
Committee Score							0.1061	0.0143	0.0378	0.0048	0.0464	0.0072	-0.9123**	-0.1055***
							(0.0689)	(0.0093)	(0.0672)	(0.0086)	(0.0655)	(0.0102)	(0.3787)	(0.0317)
Year Transition/Matriculate More	0.0664	0.0163	0.0932		0.9839***	0.1915***					-0.0235	-0.0037	3.3933***	0.3922***
	(0.1281)	(0.0314)	(0.2069)		(0.1472)	(0.0263)					(0.2763)	(0.0430)	(0.7682)	(0.0697)
Program Rank											-0.0282***	-0.0044***		
											(0.0027)	(0.0003)		
$\ddot{m}_2$				1.0892***			-1.5983**	-0.2160**						
				(0.0851)			(0.7869)	(0.1049)						
$\overline{m_{2A}}$									-2.0175***	-0.2583***				
									(0.7697)	(0.0964)				
$\sqrt{\text{MSE}}$			1.9465***	0.0015***										
			(0.0673)	(0.0005)										
Obs | Param | LL | AIC	1000	48	-594.3	1284.6										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

▸▸ Conclusion

A5: Simulated Unrestricted Wait-Dynamic Estimates (Suboptimal)

Table 26b: Simulated Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9050**	-0.4669**	5.2717***	-0.0498*	-11.2295***	-2.1801***	-6.7113***	-1.0637***	-7.1639***	-0.8002***	-9.5990***	-1.4976***	-1.4373	-0.2080
	(0.7564)	(0.1830)	(1.2701)	(0.0260)	(1.0144)	(0.1474)	(2.0007)	(0.3043)	(2.5069)	(0.2796)	(1.9604)	(0.2508)	(3.8091)	(0.5533)
GRE Quantative Percentile	0.0208**	0.0051**	-0.0016		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1162***	0.0181***	0.0355	0.0051
	(0.0090)	(0.0022)	(0.0144)		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0216)	(0.0027)	(0.0435)	(0.0063)
Gender	-0.2334*	-0.0572*	-0.1023		0.2575*	0.0500*	-0.2767	-0.0439	0.4103	0.0458	0.0583	0.0091	-0.3544	-0.0513
	(0.1309)	(0.0319)	(0.2051)		(0.1469)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.2669)	(0.0416)	(0.5436)	(0.0788)
Demographic 1	0.3690**	0.0904**	-0.4893*		-0.8293***	-0.1610***	0.2577	0.0408	-1.1549**	-0.1290**	-0.5155	-0.0804	0.3334	0.0482
	(0.1732)	(0.0421)	(0.2701)		(0.1966)	(0.0371)	(0.3774)	(0.0594)	(0.5110)	(0.0563)	(0.3647)	(0.0563)	(0.7088)	(0.1030)
Demographic 2	0.0925	0.0227	-0.2744		0.0797	0.0155	0.5369	0.0851	-0.3697	-0.0413	-0.4742	-0.0740	-0.7602	-0.1100
	(0.1911)	(0.0468)	(0.2740)		(0.2135)	(0.0414)	(0.3886)	(0.0605)	(0.4652)	(0.0521)	(0.3856)	(0.0598)	(0.7927)	(0.1089)
Advanced Math			0.7219***				0.2363	0.0374	0.4865	0.0543	0.5366**	0.0837**	0.3982	0.0576
			(0.1966)				(0.2639)	(0.0417)	(0.3566)	(0.0399)	(0.2704)	(0.0416)	(0.6445)	(0.0922)
Committee Score							0.3737***	0.0592***	0.0653	0.0073	0.0616	0.0096	-0.6725***	-0.0973***
							(0.0674)	(0.0096)	(0.0837)	(0.0093)	(0.0643)	(0.0100)	(0.1981)	(0.0217)
Year Transition/Matriculate More	0.0664	0.0163	0.0031		1.1108***	0.2156***					0.1314	0.0205	2.9410***	0.4256***
	(0.1281)	(0.0314)	(0.2010)		(0.1479)	(0.0257)					(0.2794)	(0.0434)	(0.6722)	(0.0578)
Program Rank											-0.0279***	-0.0044***		
											(0.0028)	(0.0003)		
$\ddot{m}_2$				0.8883***			-0.5496	-0.0871						
				(0.0489)			(0.5706)	(0.0900)						
$\overline{m_{2A}}$									-2.9213***	-0.3263***				
									(0.9934)	(0.1102)				
$\sqrt{\text{MSE}}$			1.9186***	-0.0010***										
			(0.0682)	(0.0002)										
Obs | Param | LL | AIC	1000	48	-430.6	957.3										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Restricted Wait-Dynamic Estimates (Optimal)

Table 27a: Simulated Restricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-2.1483***	4.7010***	-0.0252	-0.2209**	0.8018	0.1801	60.02%	-9.5045***	-5.8313		
	(0.7577)	(1.3237)	(0.0518)	(0.0865)	(0.6768, 0.9500)	(0.3070)	(49.17%, 69.97%)	(1.9667)	(6.4970)		
GRE Quantative Percentile	0.0238***	0.0028						0.1132***	0.0834		
	(0.0090)	(0.0150)						(0.0217)	(0.0644)		
Gender	-0.2140	-0.0645						0.2230*	0.4650		
	(0.1305)	(0.2113)						(0.1231)	(0.9003)		
Demographic 1	0.3579**	-0.3525						-0.3259*	-0.2627		
	(0.1725)	(0.2903)						(0.1842)	(0.7744)		
Demographic 2	0.0934	-0.1816						0.0202	-0.3966		
	(0.1906)	(0.3010)						(0.1773)	(1.2105)		
Advanced Math		0.6923***						0.5637***	1.1399		
		(0.2026)						(0.1717)	(0.9839)		
Committee Score								0.0501	-0.9123**		
								(0.0376)	(0.3787)		
Year Transition/Matriculate More	0.0734	0.0954		-0.0200	0.7859		74.43%	0.0701	3.3933***		
	(0.1278)	(0.2069)		(0.0820)	(0.6583, 0.9384)		(59.73%, 85.10%)	(0.2666)	(0.7682)		
Program Rank								-0.0282***			
								(0.0027)			
$\ddot{m}_2$			1.0892***								
			(0.0851)								
$\overline{m_{2A}}$						1.3960*					
						(0.8072)					
$\sqrt{\text{MSE}}$		1.9469***	0.0015***								
		(0.0673)	(0.0005)								
Beta										0.9751***	
										(0.0345)	
Sigma											0.1363***
											(0.0249)
Obs | Param | LL | AIC	1000	32	-625.7	1315.4							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Wait-Dynamic Estimates (Suboptimal)

Table 27b: Simulated Restricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	$\overline{m_{2A}}$	Cutoff 1	L(Cutoff)$_1$	Cutoff 3	P(Cutoff)$_3$	Success	Matriculate	Beta	Sigma
Intercept	-2.1670***	4.3821***	-0.0498*	-0.2877**	0.7500	0.2323	61.62%	-6.8847***	-1.4373		
	(0.7596)	(1.2671)	(0.0260)	(0.1209)	(0.5918, 0.9505)	(0.3539)	(47.35%, 74.14%)	(2.2787)	(3.8091)		
GRE Quantative Percentile	0.0241***	0.0091						0.0782***	0.0355		
	(0.0090)	(0.0144)						(0.0242)	(0.0435)		
Gender	-0.2191*	-0.0595						0.1414	-0.3545		
	(0.1307)	(0.2029)						(0.1168)	(0.5436)		
Demographic 1	0.3467**	-0.5657**						-0.4512**	0.3334		
	(0.1730)	(0.2717)						(0.1991)	(0.7088)		
Demographic 2	0.1001	-0.2626						0.0015	-0.7602		
	(0.1908)	(0.2740)						(0.1684)	(0.7927)		
Advanced Math		0.7355***						0.3288**	0.3982		
		(0.1973)						(0.1652)	(0.6445)		
Committee Score								0.1647***	-0.6725***		
								(0.0367)	(0.1981)		
Year Transition/Matriculate More	0.0686	0.0039		-0.0379	0.7221		77.13%	0.1685	2.9410***		
	(0.1278)	(0.2010)		(0.0870)	(0.5848, 0.8915)		(53.88%, 90.69%)	(0.2714)	(0.6722)		
Program Rank								-0.0281***			
								(0.0027)			
$\ddot{m}_2$			0.8883***								
			(0.0489)								
$\overline{m_{2A}}$						1.7371					
						(1.1521)					
$\sqrt{\text{MSE}}$		1.9217***	-0.0010***								
		(0.0689)	(0.0002)								
Beta										0.8711***	
										(0.0429)	
Sigma											0.1370***
											(0.0403)
Obs | Param | LL | AIC	1000	32	-486.0	1036.0							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Unrestricted Wait-Hybrid Estimates (Optimal)

Table 28a: Simulated Unrestricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0252	-11.5905***	-2.2561***	-11.9661***	-1.6170***	-13.2739***	-1.6996***	-8.1384***	-1.1679***	-9.5601***	-1.4878***	-5.8313	-0.6741
	(0.0518)	(1.0234)	(0.1454)	(2.5509)	(0.3154)	(2.4117)	(0.2845)	(1.1381)	(0.1344)	(2.0725)	(0.2688)	(6.4970)	(0.7706)
GRE Quantitative Percentile		0.1236***	0.0241***	0.1097***	0.0148***	0.1387***	0.0178***	0.1079***	0.0155***	0.1161***	0.0181***	0.0834	0.0096
		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0264)	(0.0032)	(0.0133)	(0.0015)	(0.0227)	(0.0029)	(0.0644)	(0.0075)
Gender		0.2266	0.0441	0.0233	0.0031	0.3039	0.0389	0.1399	0.0201	0.1264	0.0197	0.4650	0.0538
		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3287)	(0.0421)	(0.1739)	(0.0249)	(0.2699)	(0.0419)	(0.9003)	(0.1052)
Demographic 1		-0.3014	-0.0587	0.5530	0.0747	-0.8834*	-0.1131*	-0.2535	-0.0364	-0.4606	-0.0717	-0.2627	-0.0304
		(0.1988)	(0.0386)	(0.4940)	(0.0665)	(0.4622)	(0.0578)	(0.2359)	(0.0338)	(0.3933)	(0.0608)	(0.7744)	(0.0880)
Demographic 2		0.1538	0.0299	0.5128	0.0693	-0.0526	-0.0067	-0.1180	-0.0169	-0.2340	-0.0364	-0.3966	-0.0459
		(0.2215)	(0.0431)	(0.5204)	(0.0699)	(0.4528)	(0.0580)	(0.2581)	(0.0371)	(0.4322)	(0.0672)	(1.2105)	(0.1393)
Advanced Math				0.5065*	0.0684*	0.8621***	0.1104***			0.6273**	0.0976**	1.1399	0.1318
				(0.2907)	(0.0390)	(0.3306)	(0.0421)			(0.2749)	(0.0417)	(0.9839)	(0.1080)
Committee Score				0.1061	0.0143	0.0378	0.0048			0.0464	0.0072	-0.9123**	-0.1055***
				(0.0689)	(0.0093)	(0.0672)	(0.0086)			(0.0655)	(0.0102)	(0.3787)	(0.0317)
Year Transition/Matriculate More		0.9842***	0.1916***					-0.0586	-0.0084	-0.0233	-0.0036	3.3933***	0.3922***
		(0.1472)	(0.0263)					(0.1690)	(0.0242)	(0.2763)	(0.0430)	(0.7682)	(0.0697)
Program Rank								-0.0310***	-0.0045***	-0.0282***	-0.0044***		
								(0.0019)	(0.0002)	(0.0027)	(0.0003)		
$\ddot{m}_2$	1.0892***			-1.5983**	-0.2160**								
	(0.0851)			(0.7869)	(0.1049)								
$\overline{m_{2A}}$						-2.0170***	-0.2583***						
						(0.7696)	(0.0964)						
$\sqrt{\text{MSE}}$	0.0015***												
	(0.0005)												
Obs | Param | LL | AIC	1000	41	438.6	-795.1									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Unrestricted Wait-Hybrid Estimates (Suboptimal)

Table 28b: Simulated Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0498*	-11.2292***	-2.1801***	-6.7111***	-1.0638***	-7.1640***	-0.8002***	-8.2823***	-1.1912***	-9.6011***	-1.4979***	-1.4373	-0.2080
	(0.0260)	(1.0143)	(0.1474)	(2.0006)	(0.3043)	(2.5069)	(0.2796)	(1.1392)	(0.1341)	(1.9606)	(0.2508)	(3.8091)	(0.5533)
GRE Quantitative Percentile		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1097***	0.0158***	0.1162***	0.0181***	0.0355	0.0051
		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0133)	(0.0015)	(0.0216)	(0.0027)	(0.0435)	(0.0063)
Gender		0.2576*	0.0500*	-0.2763	-0.0438	0.4099	0.0458	0.1354	0.0195	0.0584	0.0091	-0.3544	-0.0513
		(0.1468)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.1741)	(0.0250)	(0.2669)	(0.0416)	(0.5436)	(0.0788)
Demographic 1		-0.8291***	-0.1610***	0.2570	0.0407	-1.1547**	-0.1290**	-0.2774	-0.0399	-0.5155	-0.0804	0.3334	0.0482
		(0.1966)	(0.0371)	(0.3773)	(0.0594)	(0.5110)	(0.0563)	(0.2344)	(0.0337)	(0.3648)	(0.0563)	(0.7088)	(0.1030)
Demographic 2		0.0799	0.0155	0.5360	0.0850	-0.3696	-0.0413	-0.1806	-0.0260	-0.4743	-0.0740	-0.7602	-0.1100
		(0.2135)	(0.0414)	(0.3885)	(0.0605)	(0.4652)	(0.0521)	(0.2566)	(0.0369)	(0.3856)	(0.0598)	(0.7927)	(0.1089)
Advanced Math				0.2361	0.0374	0.4867	0.0544			0.5367**	0.0837**	0.3982	0.0576
				(0.2639)	(0.0417)	(0.3566)	(0.0399)			(0.2704)	(0.0416)	(0.6445)	(0.0922)
Committee Score				0.3736***	0.0592***	0.0653	0.0073			0.0617	0.0096	-0.6725***	-0.0973***
				(0.0674)	(0.0096)	(0.0837)	(0.0093)			(0.0643)	(0.0100)	(0.1981)	(0.0217)
Year Transition/Matriculate More		1.1108***	0.2157***					-0.0767	-0.0110	0.1318	0.0206	2.9410***	0.4256***
		(0.1479)	(0.0257)					(0.1688)	(0.0243)	(0.2794)	(0.0434)	(0.6722)	(0.0578)
Program Rank								-0.0304***	-0.0044***	-0.0279***	-0.0044***		
								(0.0019)	(0.0002)	(0.0028)	(0.0003)		
$\ddot{m}_2$	0.8883***			-0.5486	-0.0870								
	(0.0489)			(0.5705)	(0.0900)								
$\overline{m_{2A}}$						-2.9216***	-0.3263***						
						(0.9934)	(0.1102)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	41	618.0	-1153.9									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A5: Simulated Restricted Wait-Hybrid Estimates (Optimal)

Table 29a: Simulated Restricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0252 (0.0518)	0.6891*** (0.1830)	66.58% (58.19%, 74.03%)	0.4185 (0.4184)	69.29% (54.94%, 80.68%)	-8.0491*** (1.1651)	-9.6599*** (1.7228)	-5.8313 (6.4970)		
GRE Quantitative Percentile						0.1060*** (0.0135)	0.1151*** (0.0191)	0.0834 (0.0644)		
Gender						0.1997** (0.1017)	0.1149 (0.1664)	0.4650 (0.9003)		
Demographic 1						-0.2477* (0.1349)	-0.3286 (0.2390)	-0.2627 (0.7744)		
Demographic 2						0.0405 (0.1507)	-0.0036 (0.2410)	-0.3966 (1.2105)		
Advanced Math							0.6284*** (0.1758)	1.1399 (0.9839)		
Committee Score							0.0548 (0.0419)	-0.9123** (0.3787)		
Year Transition/Matriculate More		-0.8464*** (0.2185)	46.08% (39.62%, 52.67%)		87.77% (63.97%, 96.67%)	-0.0609 (0.1668)	0.0204 (0.2685)	3.3933*** (0.7682)		
Program Rank						-0.0310*** (0.0019)	-0.0283*** (0.0027)			
$\ddot{m}_2$	1.0892*** (0.0851)									
$\overline{m_{2A}}$				2.4392* (1.4478)						
$\sqrt{\text{MSE}}$	0.0015*** (0.0005)									
Beta									0.9493*** (0.0396)	
Sigma										0.1917*** (0.0251)
Obs | Param | LL | AIC	1000	25	434.1	-818.1						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A5: Simulated Restricted Wait-Hybrid Estimates (Suboptimal)

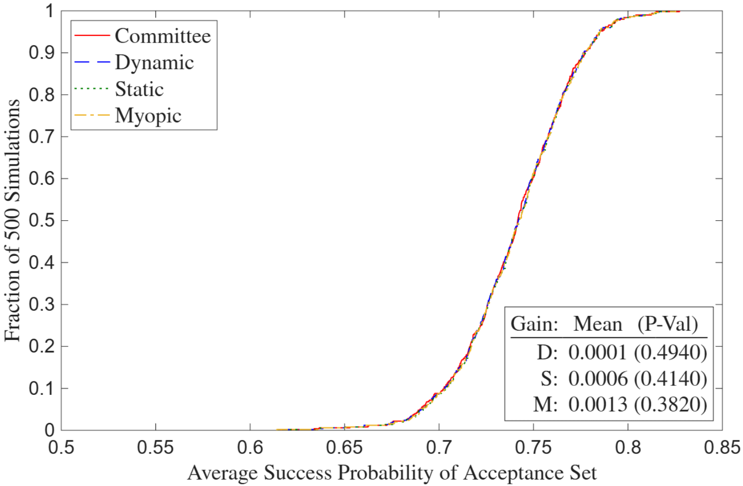
Table 29b: Simulated Restricted Waitlist-Hybrid Estimates (Suboptimal)

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0498*	0.7428***	67.76%	0.5736	74.61%	-8.0749***	-6.3299***	-1.4373		
	(0.0260)	(0.2082)	(58.29%, 75.97%)	(0.5852)	(55.80%, 87.24%)	(1.2691)	(1.7373)	(3.8091)		
GRE Quantitative Percentile						0.1082***	0.0718***	0.0355		
						(0.0147)	(0.0186)	(0.0435)		
Gender						0.2189**	0.0369	-0.3545		
						(0.1045)	(0.1594)	(0.5436)		
Demographic 1						-0.5509***	-0.3579	0.3334		
						(0.1397)	(0.2337)	(0.7088)		
Demographic 2						-0.0156	-0.0774	-0.7602		
						(0.1510)	(0.2321)	(0.7927)		
Advanced Math							0.3857**	0.3982		
							(0.1669)	(0.6445)		
Committee Score							0.1748***	-0.6725***		
							(0.0374)	(0.1981)		
Year Transition/Matriculate More		-1.0055***	43.47%		93.27%	-0.0797	0.1064	2.9410***		
		(0.2419)	(37.12%, 50.04%)		(41.58%, 99.63%)	(0.1668)	(0.2766)	(0.6722)		
Program Rank						-0.0306***	-0.0281***			
						(0.0019)	(0.0027)			
$\ddot{m}_2$	0.8883***									
	(0.0489)									
$\overline{m_{2A}}$				3.6299						
				(3.2728)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8280***	
									(0.0422)	
Sigma										0.2012***
										(0.0325)
Obs | Param | LL | AIC	1000	25	596.3	-1142.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

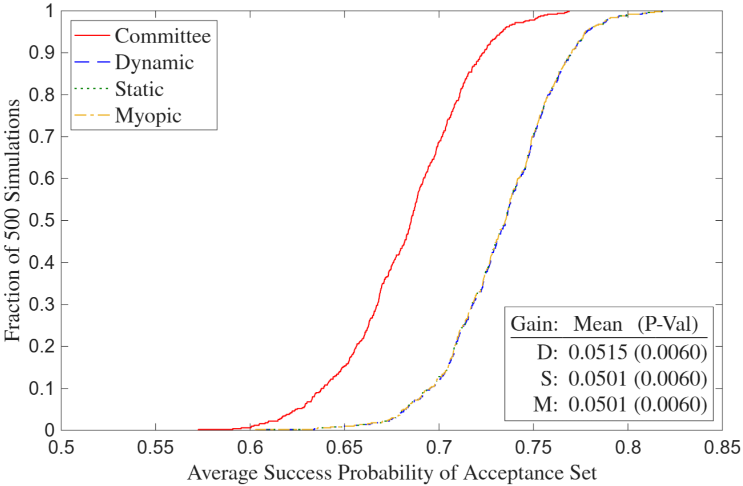
Conclusion

A5: Simulated Deviation Gains (Optimal)



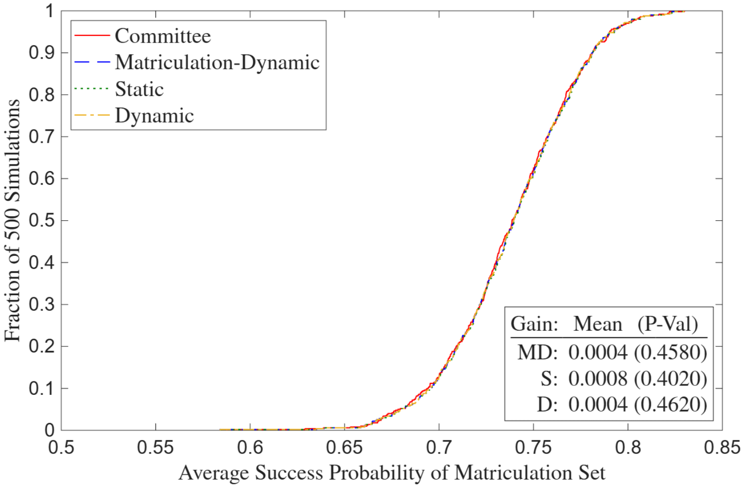
Conclusion

A5: Simulated Deviation Gains (Suboptimal)



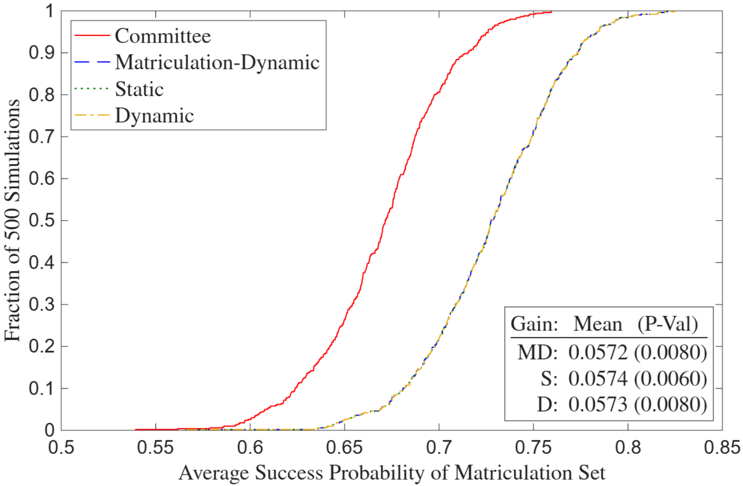
Conclusion

A5: Simulated Matriculation Deviation Gains (Optimal)



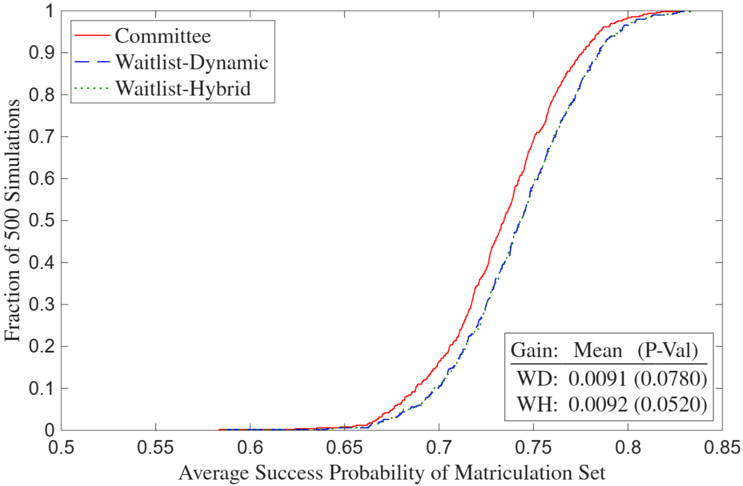
Conclusion

A5: Simulated Matriculation Deviation Gains (Suboptimal)



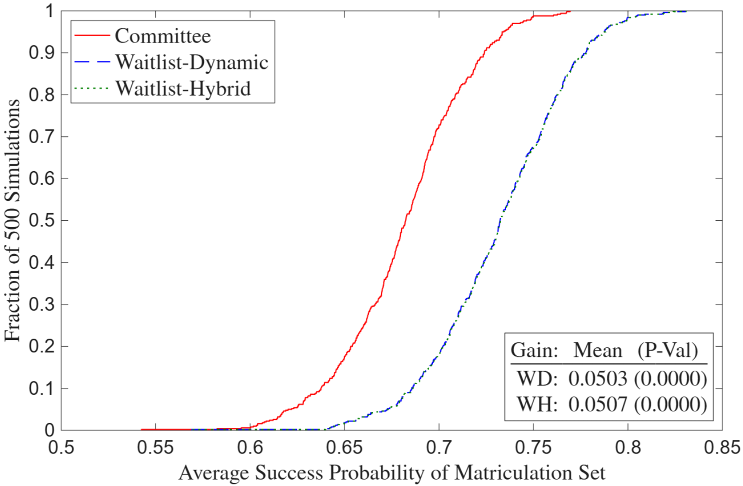
Conclusion

A5: Simulated Waitlist Deviation Gains (Optimal)



Conclusion

A5: Simulated Waitlist Deviation Gains (Suboptimal)



Conclusion

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